

# THE ROSEN FIBRATION

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**ABSTRACT.** Robert Rosen’s foundational work at the interface of mathematical biology and category theory combined unripe mathematical tools with considerable inventive insight. In this paper, we rekindle his programme using the resources offered by contemporary category theory.

In any monoidal closed category, we can cast the theory of Rosen’s  $(M,R)$ -systems: diagrams  $A \xrightarrow{f} B \xrightarrow{\Phi} [A, B]$  interpreted as process–repair pairs  $(f, \Phi)$ .

These *incoherent*  $(M,R)$ -systems assemble into a total category carrying a canonical span over  $C$ , fibred in the input variable but only *displayed* in the output variable. Examples arise in vector spaces, quantales, Chu spaces, and combinatorial species under convolution and substitution.

Repair becomes *polymorphic* once the repair map is required to be natural. We compare a hierarchy of naturality conditions, preferring to focus on those where the dependence on  $A, B$  is functorial; thus, we organise *coherent*  $(M,R)$ -systems into a profunctor, whose tabulator is the *Rosen fibration* of the title. Naturality is a severe constraint: in **Set**, and more generally in every localic topos, coherent  $(M,R)$ -systems are trivial. Nontrivial examples require the ambient category to carry isotropy, in a sense measured by a certain object of  $C$ , the end  $\int_X [X, X]$ , which classifies repair maps and which we call the *Bernstein centre*.

We then formalise Rosen’s *metabolic closure*. A universal closure point is unique when it exists, and an input object admits such a point for some natural repair family exactly when it is a retract of the monoidal unit; the possible inputs are therefore classified by split idempotent scalars. We extend the theory (both the incoherent and the coherent one) to diagrams  $A \rightarrow B \rightarrow [A, B] \rightarrow [B, [A, B]] \rightarrow \dots$  of arbitrary, even infinite, length. Finally, we analyse Arbib’s correspondence between  $(M,R)$ -systems and Mealy automata; we upgrade it to a fibred functor, and observe that metabolic closure translates into a fact of classical representation theory: a system is closed exactly when the initial state of the associated automaton has nontrivial isotropy, and a universal closure point is an element of its stabiliser. A substantial part of the mathematics below has been formalised and machine-checked in the Agda proof assistant.

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## 1. INTRODUCTION

Robert Rosen’s programme of *relational biology* sought to characterise living systems not by the physical and chemical substrate that realises them, but by the abstract pattern of relations among their constituent processes [Ros58a, Ros59]. Working at Chicago’s Committee on Mathematical Biology, Rosen sharpened this into the idea that a biological system should be modelled as a category of components together with functorial relations among them, adopting the then-recent theory of Eilenberg and Mac Lane [EML45] as its native formal language [Ros58b]; the programme culminated around 1964–1966 in the definition of what Rosen dubbed an  $(M,R)$ -system:

a diagram of shape

$$A \rightarrow B \rightarrow H(A, B) \tag{1.1}$$

where  $A, B$  are sets, and  $H(A, B)$  is an appropriately determined subset of the set  $B^A$  of functions from  $A$  to  $B$ ; this is meant to encode a ‘metabolic’ process  $A \rightarrow B$ , together with a ‘repair’ map  $B \rightarrow H(A, B)$  that reconstructs the very component responsible for carrying it out ( $H(A, B)$  is notation for the ‘hom’ set of functions  $A \rightarrow B$ , of which the process is a term).

Subsequent work recast (M,R)-systems as *sequential machines*, establishing their basic structural theory [Ros64a]: the state space of the associated machine is  $S = H(A, B)$ , of cardinality  $|B|^{|A|}$ , while the input alphabet is  $A$  itself, of size  $|A|$ , so that the alphabet is always negligible compared to the state space as soon as  $|B| \geq 2$ . Since reversibility of environmentally induced structural change was shown to require strong connectedness, one is led to restrict the ambient category of realisable maps, discarding enough set-theoretic functions for strong connectedness to become likely again [Ros64b]. The third paper in this line, in a terse remark whose consequences occupy an entire section of the present paper, observes that the totality of (M,R)-systems formed over a given category assembles, in a natural way, into a category [Ros66a]; it is precisely this remark that we unpack in full generality, in §3–4 below.

Demetrius characterised *behavioural reversibility* through the associated sequential machine [Dem66]. Arbib identified ‘fixed-genome’ (M,R)-systems (i.e. those with a fixed repair map) with a subcategory of automata whose state transitions depend on their outputs [Arb66]. Later on, Warner extended the correspondence to variable genomes and proved two comparison theorems [War82]. A relatively complete account of Arbib’s papers occupies our last section.

Băianu, alone and with Marinescu [Băi73, BM74], studied its algebraic and functorial properties, including the existence of projective limits. We generalise these completeness results in Remark 3.30 and Remark 3.33.

Several decades later, Palmer, Williams and Gatherer revisited the correspondence in the language of *X-machines*: automata equipped with a memory set, whose transitions are driven by a chosen type of partial functions on that memory rather than by raw input symbols. They show that ordinary finite-state machines, and even memory-equipped *stream X-machines*, are too weak to model the self-referential structure of an (M,R)-system, and that only *communicating X-machines* (networks of stream X-machines exchanging messages) can evade the obstruction [PWG16]. §6 below takes a different route back to Arbib’s own, older correspondence between (M,R)-systems and Mealy automata [Arb66]: Arbib relates the two only at the level of objects, and we show this assignment does not extend to a functor landing in the category of Mealy automata ordinarily considered in the literature; replacing the target by a category of ‘twisted’ Mealy automata, whose morphisms reindex input and output with opposite variance, repairs exactly this failure and turns Arbib’s correspondence into a genuine (indeed fibred) functor.

With the passing of time, Rosen’s emphasis gradually shifted from this automata-theoretic reading towards the notion of *closure to efficient causation*: informally, the requirement that every component responsible for carrying out one of its processes be itself produced by another process internal to the system, so that no efficient cause is left unentailed from outside [Ros12]. Rosen argued this closure to be incompatible with simulation by a Turing machine, a claim revisited and given precise categorical and  $\lambda$ -calculus formulations by Letelier and colleagues, who work it out explicitly in the compact closed category **Rel** and in complete lattices [LSAGA<sup>+</sup>04, LSAGA<sup>+</sup>06]; by Mossio, Longo and Stewart, who realise closure via a fixed-point combinator and argue that closure alone does not entail non-computability [MLS09]; and by Cárdenas and colleagues, who respond by insisting on Rosen’s own distinction between *simulating* and *modelling* a system [CLG<sup>+</sup>10].

These works seem aware of, but generally fail to make explicit use of, the fact that closure to efficient causation is essentially the property, of certain models of the  $\lambda$ -calculus (i.e. Cartesian closed categories), of admitting a so-called *reflexive object*: an object  $X$  equipped with an isomorphism  $X \cong X^X$ , exhibiting  $X$  itself as a space of ( $X$ -valued) functions on  $X$ . Nontrivial such models of  $\lambda$ -calculus were first constructed by Dana Scott [Sco75], and accordingly the categories that they form go under the name ‘Scott domains’. In Lemma 4.2 we show that a rather general notion of ‘coherent’ (M,R)-system (one whose repair map is natural, rather than merely a bare

arrow) is forced to be trivial in the category **Set** of sets and functions; in Proposition 4.23, by contrast, we exhibit a category that is still, in the relevant sense, a model of the  $\lambda$ -calculus, but admits genuinely nontrivial coherent (M,R)-systems. The obstruction responsible for triviality in **Set**, isolated in Remark 4.6 below, is exactly that the terminal object of **Set** happens to be a *generator* (or *separator*) of the whole category [AR94]. This property fails, and so no longer obstructs nontriviality, once one leaves **Set** for a suitable non-well-pointed topos.

The interpretation of (M,R)-systems as self-application, loosely written ‘ $f(f) = f$ ’ in [SAJGL11], is related to this notion of ‘reflexive object’ for untyped  $\lambda$ -calculus. Those authors trace it to Lawvere’s fixed-point theorem, which unifies the diagonal arguments of Cantor, Russell, Gödel and Tarski [Law69], and to the extension by Soto-Andrade and Varela [SAV84]. Our uncurrying–recurring operation (Definition 2.19) makes the change of type explicit: under symmetry of the tensor,  $\Phi : B \rightarrow [A, B]$  corresponds to  $\Phi^\vee : A \rightarrow [B, B]$ , a map into an internal endomorphism object.

The notion of closure was, in the following decades, generalised well beyond (M,R)-systems proper into an account of *biological organisation* as closure of constraints, by Montévil and Mossio, who reformulate closure in terms of constraints (structures that canalise a process without themselves being altered by it) mutually enabling one another’s maintenance [MM15]. Louie’s two monographs remain the most extensive syntheses of the whole programme available to date, and we lean on their terminology and conventions throughout: the earlier [Lou09] a direct continuation of Rosen’s own *Life Itself*, the later [Lou17] a still more systematic functorial treatment. Rosen’s own, older notion of a *modelling relation*—the bidirectional encoding/decoding correspondence between a natural system and a formal system representing it—has been recast in explicitly adjunction-theoretic terms by Zafiris, who exhibits encoding and decoding as a pair of adjoint functors between categories of observables [Zaf12]; and a Cartesian fragment of Maturana and Varela’s autopoiesis has been cast categorically by Nomura [Nom01].

Another chapter of the story involves Rosen’s disciple, A. Louie, who in the early 2000s strengthened Rosen’s non-simulability claim into an explicit noncomputability result for any material realisation of an (M,R)-system: his [Lou05] gives the general impredicativity argument and rebuts several published attempts at explicit computable counterexamples, while a sharper 2007 note targets one specific critique [CH06] directly, arguing it rests on an incorrect definition of ‘mechanism’ and on a mishandling of categorical products and coproducts [Lou07]. The claim has not gone unchallenged: Chu and Ho have argued at length, reconstructing Rosen’s argument from *Life Itself* in careful categorical detail, that several of his central notions are ill-defined and that his central proof does not, in fact, establish non-simulability [CH06].<sup>1</sup>

With essentially no exception, the literature surveyed above, foundational or contemporary, casts (M,R)-systems in the category **Set** of sets and functions, or at best in a Cartesian category, improperly treated in all respects as if it were **Set**; Remark 3.2 and Remark 3.4 below record a suspicion, seemingly not articulated before, that this is a rather degenerate special case of a much richer phenomenon.

**Structure of the paper.** We start with reformulating Rosen’s (M,R)-system (3.1) in an arbitrary monoidal closed category  $C$ , i.e. a category equipped with a tensor product  $- \otimes -$  and an internal hom  $[-, -]$  satisfying  $C(X \otimes A, B) \cong C(X, [A, B])$ , so that  $[A, B]$  plays the role of ‘the object of maps from  $A$  to  $B$ ’ even when  $C$  is not **Set**. The reformulation is carried out with a deliberately minimal set of standing hypotheses: we do not assume  $C$  symmetric (i.e. we assume no identification  $A \otimes B \cong B \otimes A$ ), and throughout the paper we track exactly where symmetry, braiding, *biclosure* (the existence of a second, independent internal hom allowing one to ‘curry in the other direction’, needed precisely because symmetry is not assumed), or the existence of limits and colimits becomes necessary, rather than assuming them wholesale from the outset (Remark 3.3).

<sup>1</sup>At the moment we take no side in this debate, which concerns the physical and computational instantiation of (M,R)-systems rather than their mathematical structure, the latter being the sole object of the present paper.

The preliminaries (section 2) recall the Yoneda lemma, comma categories, inserters, coends, and closed structures. They also describe uncurrying–recurring (Definition 2.19), which is available in a symmetric or biclosed category, and which we use in composing systems and in the end construction of Remark 4.8.

In section 3, *incoherent* systems have a bare repair arrow rather than a natural family. They form a total category **iMRS** with a canonical span over  $C$ , giving a common framework for Rosen’s categorical construction [Ros66a] and Băianu’s completeness results. We also identify them with lenses  $[A/B] \rightarrow [B/B]$ : process and uncurried repair supply the ‘get’ and ‘put’ maps familiar from functional programming [FGM<sup>+</sup>07, JRW12, PGW17, Ril18, CEG<sup>+</sup>24]. The comparison explains why their sequential composition is associative but only left-unital (Proposition 3.38).

The examples cover vector spaces, relations, join-semilattices, quantales [Yet90, Ros90], pointed sets, endofunctors, and categories under the ‘funny’ tensor product. We also treat combinatorial species under convolution and substitution [Joy86, BLL98], and Chu spaces, carrying linear-logical and physical interpretations [Bar91, Pra99, Abr10], in relative detail. The section closes with the Cartesian case, studying how  $(M,R)$ -systems can be described as dynamical systems in a certain coKleisli category (the coKleisli category of the ‘reader’ comonad).

In §4 we reach the core of the paper, considering the problem of upgrading the repair map  $\Phi$  of an incoherent  $(M,R)$ -system to a natural transformation, so that repair becomes *polymorphic* and can vary together with the process it is supposed to repair. We survey a hierarchy of increasingly abstract naturality conditions, distinguished by exactly what is required to vary naturally: an *id-natural*  $(M,R)$ -system (§4.1) asks  $\Phi_X : X \rightarrow [A, X]$  to vary naturally over *every* object  $X$  of  $C$ , with no reference to any process at all; a *cod-natural* (§4.3.1) one instead asks  $\Phi$  to vary naturally over every *process*  $h : X \rightarrow Y$  (formally, over the arrow category of  $C$ ), so that repair is tied to how a system was actually produced, which is arguably the more biologically faithful reading; an *A/C-natural* (§4.3.2) one weakens this further, demanding naturality only among processes sharing a fixed domain  $A$ ; and a fully *profunctorial*  $(M,R)$ -system (§4.3.5) drops the reference to a fixed  $A$  altogether.

These conditions reveal a tension between two opposing desiderata of the theory, tractability and nontriviality. More naturality can force a great many almost-trivial models:

- First, as a consequence of the Yoneda lemma, in the category **Set** no nontrivial id-natural  $(M,R)$ -system exists (Lemma 4.2); this mirrors, in a sense, the Yoneda-flavoured argument Rosen himself sketches informally in [Ros12, §3.5]; we observe that outside well-pointed toposes, by contrast, nontrivial Cartesian coherent  $(M,R)$ -systems genuinely exist, as we exhibit concretely in the topos of  $C_2$ -sets (Proposition 4.23). Toposes of monoid actions, toposes of graphs, sheaves over nontrivial topological spaces. . . are all non-well-pointed.
- Second, more naturality does not automatically yield a *profunctor* of  $(M,R)$ -systems, which is the structure needed to reindex an  $(M,R)$ -system along a change of  $A$  and  $B$  (reindexing along  $B$  was not possible for incoherent  $(M,R)$ -systems): we prove that id-natural and cod-natural  $(M,R)$ -systems are in fact in bijection, because the codomain functor of the arrow category is a *lax epimorphism* (Lemma 4.33), and that this shared notion of coherent  $(M,R)$ -system does assemble into a profunctor; but *A/C-natural*  $(M,R)$ -systems, despite being arguably the most semantically natural of the four, provably do not give a profunctor (subsubsection 4.3.2).

When this has been made precise, we assemble the profunctor of cod/id-natural  $(M,R)$ -systems into a fibration taking its tabulator, the *Rosen fibration* of the title (Definition 4.10): the category whose objects are  $(M,R)$ -systems and whose fibres over  $C^{\text{op}} \times C$  record, at each pair  $(A, B)$ , exactly the coherent  $(M,R)$ -systems  $\langle\langle f, \Phi \rangle\rangle$  based there.

In section 5, metabolic closure is the property of a system that evaluating repair at a certain point of  $B$  recovers  $f : A \rightarrow B$ . A *universal closure point* instead recovers the identity process on  $A$ , and hence supplies closure for every process with that input. Metabolic closure is preserved by reindexing. A point  $u : I \rightarrow A$  is universal for  $\Phi$  exactly when  $u \circ p(\Phi) = 1_A$ , and it is then unique (Proposition 5.5). Thus the input objects admitting some universally closed repair family are precisely the retracts of  $I$ , classified by split idempotent scalars (Corollary 5.6 and Proposition 5.9).

Armed with this theoretical result, we classify universal closure points in categories of modules, families of vector spaces, representations, quasi-coherent sheaves, bimodules, and modules over a quantale.

We extend the whole theory to  $(M, R)$ -systems of arbitrary finite length  $A \rightarrow B \rightarrow [A, B] \rightarrow [B, [A, B]] \rightarrow \dots$ , the categorical avatar of the construction Rosen sketches informally, as equations (3.80)–(3.81) of *Anticipatory Systems*, under the heading ‘Another Relational Metaphor’ [Ros12, §3.5]. That extension is not merely a matter of drawing longer diagrams. A system of length  $n$  carries  $n - 1$  natural transformations rather than one, all of whose components are available at once (Construction 5.16). Taking the limit of the truncation categories gives the category  $\mathbf{MRS}_\infty$  of *infinitely extended*  $(M, R)$ -systems (Definition 5.17). We isolate an additional compatibility across levels, *tower-coherence* (Definition 5.22). In a symmetric monoidal closed category, when the underlying chain has a colimit, tower-coherence is equivalent to the existence of a unique repair transformation on that colimit (Proposition 5.23); when the unit is a generator, this is equivalently a single global label. Universal closure supplies retractions of the connecting maps, and compatible universal points characterise towers whose connecting maps are all isomorphisms (Proposition 5.26 and Proposition 5.27). Explicit examples separate tower-coherence from levelwise closure, and the one-dimensional linear case admits a complete classification by zero labels or a six-periodic sequence of nonzero labels (Proposition 5.29).

Finally, section 6 develops Arbib’s correspondence with Mealy automata [Arb66]. Building on the fibred theory of automata in [Lor25b, Lor25a], we extend the object-level correspondence to a fibred functor. Its target is a category of ‘twisted’ automata, whose morphisms reindex input and output with opposite variance. Neither the remark that the correspondence on objects underlies a functor, nor the remark that its codomain is a category of twisted automata, appears in Arbib’s paper.

We close by translating metabolic closure into the vocabulary of classical representation theory [Ste16], reading  $[A, B]$  as the carrier of a representation of the monoid  $A^*$  of iterated inputs: closure turns out to be exactly the condition that the initial state of the associated Mealy automaton has nontrivial *isotropy* (a nonidentity element fixing it), and a universal closure point is exactly an element of the *stabiliser* of that state. In this way a diagram-chasing biological notion is translated into a single, classical fact of algebra.

*A remark on method.* The literature surveyed here spans nearly seventy years and close to three thousand pages. Much of it relies on intuition without a shared formal standard; some claims are too loosely stated to admit a decisive proof or refutation. Many insights came from biology, cybernetics, or philosophy, before the necessary categorical tools were available to phrase them unequivocally. Chu and Ho’s contention, recounted above, that several of Rosen’s own central notions are ill-defined is not an isolated skirmish but a symptom of exactly this state of affairs: a tradition that has, after decades, still not agreed on whether its own foundational proof is even correct, because no version of that proof was ever pinned down precisely enough to admit a definitive answer. That so many different communities sensed there was something worth pursuing is itself a testament to how good the intuition was. But insight, of which Rosen had no shortage, needs precision to harden into knowledge, and precision is what twenty-first-century mathematics can now supply.

This is why a substantial part of the mathematics below has been formalised and machine-checked in the Agda proof assistant (v2.8.0); the codebase is maintained as a subfolder (Rosen ) in a fork of the `agda-categories`  library.

We state precisely what that claim covers, since a formalisation is worth exactly as much as the accuracy of the claims made on its behalf. Each statement that has been formalised carries a marker linking to the corresponding line of the codebase, and the marker comes in two colours:

- a black ‘’ means the statement is proved outright: its proof term is complete and appeals to no postulate. See for example Definition 3.5;
- a red ‘’ means the opposite, and is a warning: the formalisation is complete in *shape* but rests on an explicitly postulated step, so *nothing* about that statement has been verified. See Proposition 3.51.

A statement carrying no marker at all has not been formalised: this is the case for the whole of section 2, which recalls standard material, and for a number of statements elsewhere. As of this writing 34 statements are black, none is red, and no statement carries an unresolved placeholder — the red marker is kept because it is the honest thing to have available, not because we currently need it.

Two qualifications, which we prefer to state than to have discovered. First, of the 37 modules in the codebase 29 pass Agda’s `--safe` check, which mechanically refuses every postulate; most of the 34 black statements live in one of those. The rest are proved outright as well, but sit in modules that are not `--safe` — either because they are instantiated over **Set** and so assume function extensionality, or because they share a file with a postulate they do not themselves use. Second, function extensionality is an ordinary axiom rather than a gap, and it is the *only* postulate in the whole development, isolated in a single file, `Axioms.agda`, so that its use can be audited. Where we have found a claim of ours to be false we have removed it and documented the obstruction in its place, rather than assuming it: a postulate asserting a falsehood is worse than an absence, because it reads as a result. Both lists are recoverable from the codebase in one `grep`, and we would rather the reader run it than take our word.

We believe *verified truth* is an appropriate tribute to a mind as brilliant and imaginative as Rosen’s, decades ahead of the category theory that could substantiate his ideas.

**Remark 1.1.** The task of building this codebase has been greatly assisted by various generative AI models, primarily ChatGPT 5.4, ChatGPT 5.6 Luna and Sol, Claude Opus 5, and Claude Sonnet 5. The same models were also used to assemble, retrieve and organise the literature survey above, and to help with proofreading and streamlining the text. No human-readable mathematical argument has been AI-generated.

## 2. PRELIMINARIES

This section fixes notation and recalls the categorical tools used below. Brief remarks provide references and explain how each notion enters the paper. We write  $1_X$  for the identity of an object  $X$ ,  $\text{id}_C$  or simply  $\text{id}$  for the identity functor of  $C$ ,  $I$  for the monoidal unit, and  $1$  for a terminal object when one exists. In arguments involving bifunctors, an object may stand for its identity morphism; for example,  $[A, f]$  means  $[1_A, f]$ .

**Definition 2.1** (Yoneda embedding). Every small category  $C$  admits a fully faithful functor into the category  $[C^{\text{op}}, \mathbf{Set}]$  (denoted  $\widehat{C}$  for short) of *presheaves* on  $C$ , i.e. contravariant functors from  $C$  into sets. This functor is called the *Yoneda embedding*

$$\mathbb{Y} : C \longrightarrow \widehat{C} \quad (2.1)$$

of  $C$  and it sends each object  $C$  to the *representable* hom-functor  $\mathbb{Y}C = C(-, C)$ .

**Theorem 2.2** (Yoneda lemma). For every  $F \in \widehat{C}$  and every object  $C \in C$ , there is an isomorphism

$$\varepsilon_{F,C} : \widehat{C}(\mathbb{Y}C, F) \xrightarrow{\cong} FC \quad (2.2)$$

induced by evaluation at  $1_C$ , namely the function

$$\varepsilon_{F,C}(\alpha) := \alpha_C(1_C) \in FC \quad (2.3)$$

is bijective and natural in  $C$  (and in  $F$ ).

Equivalently, there is a natural isomorphism between the *evaluation functor* [Rie16, §2.2],

$$\text{ev}_C : C^{\text{op}} \times \widehat{C} \rightarrow \mathbf{Set}, \quad (C, F) \mapsto FC, \quad (2.4)$$

and the composite functor  $\widehat{C}(\mathbb{Y}, \text{id}_{\widehat{C}})$ , filling the triangle

$$\begin{array}{ccc} & (\widehat{C})^{\text{op}} \times \widehat{C} & \\ \mathbb{Y}^{\text{op}} \times \text{id}_{\widehat{C}} \nearrow & \Downarrow \varepsilon & \searrow \widehat{C}(-, -) \\ C^{\text{op}} \times \widehat{C} & \xrightarrow{\text{ev}_C} & \mathbf{Set} \end{array} \quad (2.5)$$

**Remark 2.3.** The Yoneda lemma originates in [Yon60], and the reader can consult [ML98] for the classical textbook statement; the ‘fully-natural’ formulation used above is contained in [Rie16]; the Yoneda lemma is the main obstruction in the passage from *incoherent*  $(M, R)$ -systems (Definition 3.5) to *coherent* ones (Lemma 4.2).

**Definition 2.4** (Comma category). Given categories  $\mathcal{A}, \mathcal{B}, \mathcal{C}$  and functors  $F : \mathcal{A} \rightarrow \mathcal{C}$  and  $G : \mathcal{B} \rightarrow \mathcal{C}$  with the same codomain, the *comma category*  $(F/G)$  is the category whose

- objects are triples  $(A, B, p)$ , where  $p \in \mathcal{C}(FA, GB)$  is an arrow of  $\mathcal{C}$ ;
- arrows  $(a, b) : (A, B, p) \rightarrow (A', B', q)$  are pairs of arrows  $a : A \rightarrow A'$  and  $b : B \rightarrow B'$  such that the following square commutes:

$$\begin{array}{ccc} FA & \xrightarrow{p} & GB \\ Fa \downarrow & & \downarrow Gb \\ FA' & \xrightarrow{q} & GB'. \end{array} \quad (2.6)$$

Some useful properties of comma categories are collected in the following proposition.

**Proposition 2.5** (Some properties of  $(F/G)$ ).

- The comma category comes equipped with two *projection* functors:

$$\mathcal{A} \xleftarrow{\pi_F} (F/G) \xrightarrow{\pi_G} \mathcal{B}. \quad (2.7)$$

The left projection  $\pi_F : (F/G) \rightarrow \mathcal{A}$  is a fibration, while the right projection  $\pi_G : (F/G) \rightarrow \mathcal{B}$  is an opfibration (these are in a suitable sense the free fibration on  $F$  and the free opfibration on  $G$ ; however, this remark is inessential for our purposes).

- The comma category  $(F/G)$  enjoys the following universal property. The square

$$\begin{array}{ccc} (F/G) & \xrightarrow{\pi_F} & \mathcal{A} \\ \pi_G \downarrow & \not\parallel_{\theta_{FG}} & \downarrow F \\ \mathcal{B} & \xrightarrow{G} & \mathcal{C} \end{array} \quad (2.8)$$

together with the 2-cell filling it, whose component at  $(A, B, p)$  is  $p : FA \rightarrow GB$ , is terminal. Concretely, every other span  $\mathcal{B} \leftarrow X \rightarrow \mathcal{A}$ , equipped with a 2-cell

$$\chi : F \circ l \Rightarrow G \circ r, \quad (2.9)$$

factors uniquely through  $\theta_{FG}$ ; i.e. there is a unique functor  $\langle l, r \rangle : X \rightarrow (F/G)$  such that  $\chi = \theta_{FG} * \langle l, r \rangle$  (the ‘whiskering’ of the natural transformation  $\theta_{FG}$  and the functor  $\langle l, r \rangle$ ):

$$\begin{array}{ccccc} X & & \xrightarrow{l} & & \mathcal{A} \\ & \searrow \langle l, r \rangle & & \searrow \pi_F & \\ & (F/G) & \xrightarrow{\pi_F} & & \mathcal{A} \\ & \pi_G \downarrow & \not\parallel_{\theta_{FG}} & \downarrow F & \\ & \mathcal{B} & \xrightarrow{G} & & \mathcal{C}. \end{array} \quad (2.10)$$

- The comma category can be computed as a limit in **Cat**. More precisely, there is a pullback square of functors

$$\begin{array}{ccc} (F/G) & \longrightarrow & \mathcal{C}^{\rightarrow} \\ \langle \pi_F, \pi_G \rangle \downarrow & \lrcorner & \downarrow \langle d_0, d_1 \rangle \\ \mathcal{A} \times \mathcal{B} & \xrightarrow{F \times G} & \mathcal{C} \times \mathcal{C}. \end{array} \quad (2.11)$$

**Remark 2.6.** Comma categories have been a standard tool of category theory since Lawvere’s thesis [Law63]; the reader can consult [ML98, Jac99] for a systematic treatment; comma constructions are ubiquitous in describing categories of ‘machines’ and will appear in §6 relating (M,R)-systems and ‘Mealy automata’.

The comma category  $(F/G)$  is an important example of a 2-categorical limit. Another important example is the *inserter* of two functors.

**Definition 2.7** (Inserter of two functors). Given two parallel functors  $F, G : \mathcal{B} \rightarrow \mathcal{C}$ , the *inserter*  $(F \multimap G)$  is the category whose

- objects are pairs  $(B, h)$ , where  $h : FB \rightarrow GB$ ;
- arrows  $(B, h) \rightarrow (B', k)$  are arrows  $u : B \rightarrow B'$  such that

$$Gu \circ h = k \circ Fu. \quad (2.12)$$

The inserter  $(F \multimap G)$  enjoys a universal property analogous to that of  $(F/G)$ . The square

$$\begin{array}{ccc} (F \multimap G) & \xrightarrow{U} & \mathcal{B} \\ U \downarrow & \not\cong_{\theta_{FG}} & \downarrow F \\ \mathcal{B} & \xrightarrow{G} & \mathcal{C} \end{array} \quad (2.13)$$

where  $U : (F \multimap G) \rightarrow \mathcal{B}$  is the evident forgetful functor, comes equipped with a 2-cell

$$\iota : F \circ U \Rightarrow G \circ U \quad (2.14)$$

and is terminal among squares of this form. Thus, although there need not be a natural transformation  $F \Rightarrow G$  on the whole of  $\mathcal{B}$ , there is a canonical one after restriction along  $U$ .

The difference between a comma category and an inserter is worth making explicit. In an inserter, the two functors are parallel and there is a single projection  $U : (F \multimap G) \rightarrow \mathcal{B}$ . By contrast, a comma category has two generally different projections. In particular, when  $F$  and  $G$  are parallel, both  $(F/G)$  and  $(F \multimap G)$  are defined, but they are different categories.

An important example of an inserter is the category of algebras, or coalgebras, for an endofunctor.

**Definition 2.8** (Algebras and coalgebras as inserters). Let  $F : \mathcal{C} \rightarrow \mathcal{C}$  be an endofunctor. The category  $\mathbf{Alg} F$  of  $F$ -algebras is defined as the inserter

$$\begin{array}{ccc} \mathbf{Alg} F & \xrightarrow{U} & \mathcal{C} \\ U \downarrow & \not\cong & \downarrow F \\ \mathcal{C} & \xrightarrow{F} & \mathcal{C} \end{array} \quad (2.15)$$

Dually, the category  $\mathbf{coAlg} F$  of  $F$ -coalgebras is defined as the inserter

$$\begin{array}{ccc} \mathbf{coAlg} F & \xrightarrow{U} & \mathcal{C} \\ U \downarrow & \not\cong & \parallel \\ \mathcal{C} & \xrightarrow{F} & \mathcal{C} \end{array} \quad (2.16)$$

**Remark 2.9.** The presentation of algebras and coalgebras for an endofunctor as inserters is standard; the reader can consult [Jac99, ML98]; we will use the notion of endofunctor algebra for the characterisation in Proposition 3.29.

**Definition 2.10** (Module for an endofunctor). Let  $F : \mathcal{C} \rightarrow \mathcal{C}$  be an endofunctor; a *module* for  $F$  (see [Dub70]) consists of a functor  $X : \mathcal{X} \rightarrow \mathcal{C}$  and a natural transformation  $\xi : F \circ X \Rightarrow X$ .

Thus, the category of algebras for  $F$  can be characterised as the universal module for  $F$ , via the structure given by the forgetful functor  $U : \mathbf{Alg}(F) \rightarrow \mathcal{C}$ .



**Remark 2.11.** The notion of module for an endofunctor, in the enriched setting, is contained in [Dub70]; we will observe in Remark 4.13 that the repair map of a coherent  $(M, R)$ -system can be seen as a module in this sense.

The next technical tool we need is the *dinatural equivalence relation* used in the concrete construction of a coend, a particular kind of colimit associated with functors of type  $C^{\text{op}} \times C \rightarrow \mathcal{D}$ . We shall only need the case in which  $\mathcal{D} = \mathbf{Set}$ .

**Definition 2.12** (Dinatural equivalence relation). Let  $T : C^{\text{op}} \times C \rightarrow \mathbf{Set}$  be a functor. The relation of dinatural equivalence is the equivalence relation  $\sim$  on the disjoint union  $\sum_X T(X, X)$  generated by declaring

$$(X, \xi) \sim (Y, \theta) \quad (2.17)$$

whenever  $\xi \in T(X, X)$ ,  $\theta \in T(Y, Y)$ , and there exists an arrow  $f : X \rightarrow Y$  such that

$$T(X, f)(\xi) = T(f, Y)(\theta). \quad (2.18)$$

**Definition 2.13.** The *coend* of  $T$  is, when it exists, the quotient of the coproduct  $\sum_X T(X, X)$  by the dinatural equivalence relation,

$$\int^X T(X, X) := \left( \sum_X T(X, X) \right) / \sim, \quad (2.19)$$

denoted simply by  $\int T$  when there is no risk of confusion. It comes equipped with a canonical family of functions

$$\iota_A : T(A, A) \longrightarrow \int^X T(X, X) \quad (2.20)$$

satisfying the following *cowedge condition*:

For every arrow  $u : A \rightarrow B$  in  $C$ , the square

$$\begin{array}{ccc} T(B, A) & \xrightarrow{T(u, A)} & T(A, A) \\ T(B, u) \downarrow & & \downarrow \iota_A \\ T(B, B) & \xrightarrow{\iota_B} & \int^X T(X, X) \end{array} \quad (2.21)$$

commutes.

The family  $(\iota_A \mid A \in C)$  is the ‘initial cowedge’, i.e. it is initial among families satisfying this condition.

**Remark 2.14.** Ends, coends, and dinatural transformations are treated systematically in [Lor21], with the classical account given in [ML98]; we will use this notion in Remark 4.8, where we reduce an id-coherent  $(M, R)$ -system to a single map into an appropriate *end*, and in subsection 5.5, where we characterise a ‘procomonad’ structure on the profunctor of id-coherent  $(M, R)$ -systems.

**Definition 2.15** (Closed category). A category is *closed* if it is equipped with the following data:

- a functor  $[-, -] : C^{\text{op}} \times C \rightarrow C$ , called the *internal hom* or simply the hom of  $C$ ;
- an object  $I \in C$ , called the *unit object*;
- a natural isomorphism  $i : \text{id}_C \xrightarrow{\cong} [I, -]$ , exhibiting each object as its internal hom from the unit;
- a family of morphisms  $j_X : I \rightarrow [X, X]$  forming an extranatural family in the variable  $X$ , meaning that for every  $h : X \rightarrow Y$  in  $C$ , the square

$$\begin{array}{ccc} I & \xrightarrow{j_X} & [X, X] \\ j_Y \downarrow & & \downarrow [X, h] \\ [Y, Y] & \xrightarrow{[h, Y]} & [X, Y] \end{array} \quad (2.22)$$

commutes. Note that this is a *wedge* condition, dual to the one in (2.21) above. It models how one chooses an identity for each object  $X$ , compatibly with the morphisms of  $C$ .

These data are subject to certain coherence conditions, expressing the expected interaction among the hom,  $I$ ,  $i$ , and  $j$ , and generalising the usual Yoneda-type identities. However, we do not dwell on the precise definition, as the notion of closed category is a relatively impoverished one.

By contrast, the categories one encounters most often are *monoidal* closed, among which the *Cartesian closed* categories form a particularly important subclass; see [Lam80].

**Definition 2.16.** A monoidal category is equipped with a bifunctor

$$- \otimes - : C \times C \longrightarrow C \quad (2.23)$$

satisfying the axioms of associativity, left and right unitality (with respect to a *monoidal unit* object  $I \in C_0$ ), as well as the coherence conditions of [ML98]. We denote the unitors by  $\lambda_X : I \otimes X \xrightarrow{\cong} X$  and  $\rho_X : X \otimes I \xrightarrow{\cong} X$ .

For the full unraveling of the axioms, see [ML98, EGNO15, FS19]. We focus here on monoidal closure.

**Definition 2.17** (Monoidal closed category). A closed category  $(\mathcal{V}, [-, -])$  is *monoidal closed* if it is equipped with a monoidal product  $- \otimes -$ , with unit  $I$ , and a natural isomorphism

$$\mathcal{V}(X, [Y, Z]) \cong \mathcal{V}(X \otimes Y, Z). \quad (2.24)$$

**Remark 2.18.** The original definition of closed category is due to [EK66]; its monoidal refinement, treated above, is discussed in [Day70b, Koc70]; it is a ubiquitous assumption of the present paper to work in an ambient (monoidal) closed category.

**Definition 2.19** (Uncurrying-recurrying). In a monoidal closed category  $C$  as in Definition 2.17, every object  $Y \in C$  induces a natural isomorphism

$$C(X \otimes Y, Z) \cong C(X, [Y, Z]) \quad (2.25)$$

exhibiting  $- \otimes Y$  as left adjoint to  $[Y, -]$ ; we call the passage from left to right *currying*, and its inverse *uncurrying*.

Given a map  $\Phi : B \rightarrow [A, B]$ , *uncurrying*  $\Phi$  (i.e. applying the isomorphism above right-to-left, with  $X = B$ ,  $Y = A$ ,  $Z = B$ ) yields

$$\varphi := \text{ev}_{A,B} \circ (\Phi \otimes 1_A) : B \otimes A \rightarrow B, \quad (2.26)$$

where  $\text{ev}_{A,B} : [A, B] \otimes A \rightarrow B$  is the counit of  $- \otimes A \dashv [A, -]$ .

*Recurrying in the other tensor variable* turns  $\varphi : B \otimes A \rightarrow B$  into a map from  $A$  to an internal endomorphism object of  $B$ . This requires one of the following additional structures.

- $C$  is *biclosed*: besides the internal hom  $[-, -]$  above, right adjoint to  $- \otimes Y$  for every  $Y$ , there is a second internal hom  $\{-, -\} : C^{\text{op}} \times C \rightarrow C$ , right adjoint instead to  $Y \otimes -$ , i.e. a natural isomorphism

$$C(Y \otimes X, Z) \cong C(X, \{Y, Z\}). \quad (2.27)$$

Taking  $Y = B$ ,  $X = A$ ,  $Z = B$  in this second adjunction,  $\varphi : B \otimes A \rightarrow B$  curries directly into a map  $A \rightarrow \{B, B\}$ , with no need for a symmetry.

- $- \otimes -$  is *symmetric*, with symmetry isomorphism  $\sigma_{B,A} : B \otimes A \xrightarrow{\cong} A \otimes B$ . Precomposing  $\varphi$  with  $\sigma_{B,A}^{-1}$  gives a map  $\varphi \circ \sigma_{B,A}^{-1} : A \otimes B \rightarrow B$ , which currying along the original adjunction (with  $X = A$ ,  $Y = Z = B$ ) turns into a map  $A \rightarrow [B, B]$ .

When  $C$  is symmetric and biclosed, the two homs  $[-, -]$  and  $\{-, -\}$  are canonically isomorphic, and the two recipes above agree up to this isomorphism; we do not distinguish between them notationally, writing  $\Phi^\vee : A \rightarrow [B, B]$  for the map obtained by either route, and calling it the *uncurrying-recurrying* of  $\Phi$ .

In short: starting from  $\Phi : B \rightarrow [A, B]$ , one uncurries into  $\varphi : B \otimes A \rightarrow B$ ; assuming either that  $C$  is biclosed, or that  $- \otimes -$  is symmetric, one then recurries  $\varphi$ , along the other tensor slot, into  $\Phi^\vee : A \rightarrow [B, B]$ .

**Remark 2.20.** Mere closedness provides the minimal structure needed for the curried formulation of an  $(M, R)$ -system, whereas a monoidal closed category allows one to pass freely between the curried map  $\Phi : B \rightarrow [A, B]$  and its uncurried counterpart  $\Phi' : B \otimes A \rightarrow B$ . Neither formulation is indispensable in complete generality. In a monoidal category  $C$  that is not closed, one may take the uncurried map  $\Phi'$  as primitive; conversely, in a closed category whose closed structure does not arise from a monoidal product, one may work directly with the curried map  $\Phi$ . The most convenient formulation therefore depends on the ambient category.

A full review of the theory of fibrations would take us too far afield. Standard references are [Jac99, Chapter 1] and [LR20]. We shall nevertheless make repeated use of one of that theory's fundamental results: the *Grothendieck–Bénabou equivalence*.

**Theorem 2.21.** There is an equivalence of categories

$$\mathbf{Fib}/C \cong \mathbf{Cat}_{ps}(C^{\text{op}}, \mathbf{Cat}) \quad (2.28)$$

between the category of *fibrations* over  $C$  and the category of *pseudofunctors*  $C^{\text{op}} \rightarrow \mathbf{Cat}$  ([Jac99, Definition 1.4.4]), induced by the correspondence of [Jac99, Proposition 1.4.5]. This equivalence restricts to one between the category of *presheaves*  $C^{\text{op}} \rightarrow \mathbf{Set}$  and the category of *discrete fibrations* over  $C$ .

We will also make somewhat implicit use of the most general version of this correspondence, which in recent years has generally been dubbed a ‘straightening’ construction (because it allows one to straighten the fibres of a fibration into a presheaf).

**Theorem 2.22.** There is an equivalence of categories between  $\mathbf{Cat}/\mathcal{B}$ , the slice category of functors  $F : X \rightarrow \mathcal{B}$ , whose arrows  $H : (X, F) \rightarrow (Y, G)$  are commutative triangles  $G \circ H = F$ , and the category of *normal lax functors*

$$\mathcal{B} \longrightarrow \mathbf{Prof} \quad (2.29)$$

sending an object  $B$  of  $\mathcal{B}$  to a category  $F_{\natural}B$  (the fibre of  $F$  over  $B$ ) and a morphism  $u : B \rightarrow B'$  to a *profunctor* (see below for the definition)  $F_{\natural}u : (F_{\natural}B)^{\text{op}} \times (F_{\natural}B') \rightarrow \mathbf{Set}$ .

**Remark 2.23.** Fibrations and the Grothendieck construction are covered extensively in [Bén85, Str23, Gra66].

We now recall the notion of profunctor used in this statement.

A *profunctor*  $P$  between categories  $\mathcal{A}, \mathcal{B}$  consists of a functor of type

$$P : \mathcal{A} \times \mathcal{B}^{\text{op}} \longrightarrow \mathbf{Set}. \quad (2.30)$$

**Definition 2.24** (Profunctors and tabulators). The *tabulator* of  $P$  (also called the *category of elements* or simply the *total category* of  $P$ ; but we reserve such names for different constructions here) is defined as follows:

- objects are triples  $(A, B; \xi)$  where  $\xi \in P(A, B)$  is an element;
- arrows  $(A, B; \xi) \rightarrow (A', B'; \theta)$  are pairs of arrows  $u \in \mathcal{A}(A, A'), v \in \mathcal{B}(B, B')$  such that

$$P(A, v)(\xi) = P(u, B')(\theta) \quad \text{as elements of } P(A, B'). \quad (2.31)$$

The verification that this defines a category  $\circledast P$  is easy; note that the category  $\circledast P$  comes equipped with two *projection* functors

$$\mathcal{A} \xleftarrow{\pi^l} \circledast P \xrightarrow{\pi^r} \mathcal{B} \quad (2.32)$$

and a natural transformation of type

$$\circledast P(-, -) \quad P(\pi^l -, \pi^r -) \quad (2.33)$$

with components given by sending  $(u, v)$  as above to the common value  $t = P(A, v)(\xi) = P(u, B')(\theta)$ , and satisfying the following universal property: every cell of shape

$$\begin{array}{ccc}
 \begin{array}{c} \mathcal{X} \\ \swarrow p \quad \searrow q \\ \mathcal{A} \xrightarrow{P} \mathcal{B} \end{array} & \text{factors uniquely as} & \begin{array}{c} \mathcal{X} \\ \swarrow p \quad \searrow q \\ \mathcal{A} \xrightarrow{P} \mathcal{B} \\ \uparrow \pi^l \quad \uparrow \pi^r \\ \mathcal{P} \end{array}
 \end{array} \quad (2.34)$$

**Remark 2.25.** The tabulator of a distributor is discussed at length in [Béner], and its relationship with (co)ends is treated in the literature on ‘double categories’ [GP99b, Gra19]; tabulators will allow us to construct, in Definition 4.10, a two-sided discrete fibration associated to the profunctor of coherent (M,R)-systems.

The span in (2.32) is a *two-sided* (discrete) *fibration* canonically associated to  $P$ : one leg is a discrete opfibration, the other is a discrete fibration, and they are suitably compatible in the sense of [?].

A similar construction, which however reverses the direction of  $u : A \rightarrow A'$  into  $u : A' \rightarrow A$ , is the following.

**Definition 2.26** (Category of twisted elements of a profunctor). Let  $P$  be a profunctor, in the notation of (2.30); the *category of twisted elements* of  $P$  is defined as follows:

- the objects are triples  $(A, B; \xi)$  where  $\xi \in P(A, B)$  is an element;
- a morphism  $(A, B; \xi) \rightarrow (A', B'; \theta)$  is a pair of arrows  $u \in \mathcal{A}(A', A)$ ,  $v \in \mathcal{B}(B, B')$  such that the function  $P(u, v) : P(A, B) \rightarrow P(A', B')$  sends  $\xi$  to  $\theta$ .

It is clear that this gives rise to a category  $\tau\mathcal{EL}[P]$ ; the category comes equipped with a forgetful functor  $V : \tau\mathcal{EL}[P] \rightarrow C^{\text{op}} \times C$ .

**Remark 2.27.** The twisted arrow category is a standard construction attached to profunctors, essential to the theory of ends and coends, as well as to some applications to categorical logic; see [ML98, Lor21] for more details; this construction is essential to the development of Construction 5.16, where we build categories of ‘long’ (and possibly infinitely extended) (M,R)-systems. (Strictly speaking the category of Proposition 3.42 is not an example of a twisted arrow category, since a profunctor of incoherent (M,R)-systems does not exist; but the notation comes in handy.)

### 3. INCOHERENT (M,R)-SYSTEMS

**Notation 3.1.** Informally speaking, an ‘(M,R)-system’ consists of a diagram of shape

$$A \xrightarrow{f} B \xrightarrow{\Phi} [A, B] \quad (3.1)$$

Following Rosen, we call  $f$  the *process* and  $\Phi$  the *repair map*; Remark 3.6 discusses this terminology. It is diagrams of this shape, or suitable variations thereof, that we want to study.

Throughout,  $C$  denotes a monoidal closed category unless stated otherwise. Symmetry is not assumed. Our convention in the meaning of ‘closed’ is that we can uncurry  $B \rightarrow [A, B]$  into  $B \otimes A \rightarrow B$ .

Additional assumptions on limits and colimits are stated where needed, as in Proposition 3.29.

Our choice of a general monoidal rather than a Cartesian setting requires explanation. To our knowledge, this perspective has not been developed in Rosen’s programme [Ros64a, Ros64b, Ros66a] or its later categorical treatments [LSAGA<sup>+</sup>06, Nom01, Lou17].

The separation between these research communities may partly explain the omission. Resource-sensitive logics are familiar in mathematical logic and computer science [Gir87, LS86]; resource sensitivity is the organising principle of linear logic [Gir87, See89].

Rosen gestures towards this distinction when he suggests working with relations rather than functions [Ros64a, Ros12]. His discussion anticipates the role of **Rel** as a model of linear logic under the tensor product described in [See89]. Its relevance to mathematical biology appears to have received little attention.

**Remark 3.2** (On Cartesian versus purely monoidal  $(M, R)$ -systems). The internal logic of a Cartesian category is a setting in which assumptions may be duplicated and discarded without restriction, via the ‘diagonal’ and ‘terminal’ maps,

$$A \xrightarrow{\Delta} A \times A \quad A \xrightarrow{!} 1 \quad (3.2)$$

In proof-theoretic terms, contraction and weakening are built into the logic, so that resources are treated as freely reusable.<sup>2</sup> From this perspective, Cartesian logic may be regarded as a logic of unrestricted reproducibility [LS86, See89].

By contrast, the internal logic of a *symmetric monoidal* category is, in general, resource-sensitive. *Linear* logic makes this principle explicit by treating assumptions as resources that cannot, in general, be duplicated or discarded (unless what is called an ‘exponential modality’ is present). The logical formalism therefore keeps track of the consumption, preservation, and transformation of resources [Gir87, See89].

Resource scarcity constrains both biological systems and the processes they can realise. A resource-sensitive internal logic therefore seems a natural setting for their study. We nevertheless take a certain interest in the Cartesian case in subsection 3.2, both to connect with the existing literature and to establish results specific to that setting, such as Proposition 3.53.

**Remark 3.3.** For the basic definition, the assumptions can be weakened further:

It is sufficient that  $C$  is either a monoidal category (where in (3.1) the repair map  $\Phi$  is assumed to be uncurried), or a closed category (where  $\Phi$  is curried) in the sense of Eilenberg–Kelly [EK66].

Thus, ‘bare  $(M, R)$ -systems’ require neither symmetry nor completeness. However, the theory of ‘purely closed’ categories is a bit unwieldy; a monoidal closed structure allows us to use both the curried and uncurried presentations for a repair map; braidings, symmetry, and limits or colimits are every now and then needed to substantiate the further characterisations developed below.

Working in the Cartesian setting, naturality introduces a pretty rigid restriction on how many  $(M, R)$ -systems exist.

**Remark 3.4.** In section 4, we require repair to extend to a natural transformation. This yields *coherent*  $(M, R)$ -systems, but naturality can sharply restrict the available examples. In the set-theoretic setting of [Ros64a, Ros64b, Ros66a, LSAGA<sup>+</sup>04, LSAGA<sup>+</sup>06, Lou17], the only natural transformation of type

$$\Phi^\natural : C \begin{array}{c} \xrightarrow{\text{id}} \\ \Downarrow \\ \xrightarrow{[A, -]} \end{array} C \quad (3.3)$$

has components that uncurry to the projections  $\pi_X : X \times A \rightarrow X$ , by Lemma 4.2. The repair map is recovered as  $\Phi_B^\natural$ . Nontrivial Cartesian examples instead arise in non-well-pointed Grothendieck toposes, as Proposition 4.23 shows.

So far, naturality and its consequences have not been examined thoroughly; the vast majority of references we could consult work in a category of sets, interpreting  $A, B$  as sets and  $B \otimes A = B \times A$  as a Cartesian product. In such a situation, there is just *one* natural transformation  $\Phi^\natural$  of the above type, the components of which must all uncurry to projections  $\pi_X : X \times A \rightarrow X$ ; see Lemma 4.2.

We begin with incoherent systems, whose repair maps carry no naturality requirement.

<sup>2</sup>From a more algebraic perspective, we refer the reader to the so-called *Fox theorem* [Fox76]: a monoidal category  $(C, \otimes, I)$  is Cartesian if and only if every object  $A \in C$  carries a *comonoid* structure

$$\Delta_A : A \rightarrow A \otimes A, \quad e_A : A \rightarrow I,$$

such that, moreover, every arrow  $f : A \rightarrow B$  is a comonoid homomorphism (i.e.  $\Delta_B \circ f = (f \otimes f) \circ \Delta_A$  and  $e_B \circ f = e_A$ );  $\Delta$  and  $e$  are then respectively the diagonal and the unique map to the monoidal unit (which can then be proved to be a terminal object of  $C$ ).

**Definition 3.5** (Incoherent  $(M,R)$ -system,  $\mathcal{U}$ ). Let  $A, B$  be objects of  $C$ . An *incoherent  $(M,R)$ -system* (of type  $A, B$ ) in  $C$  consists of a diagram of shape

$$A \xrightarrow{f} B \xrightarrow{\Phi} [A, B]. \quad (3.4)$$

We write  $\mathcal{M} = \langle f, \Phi \rangle$ . A *morphism*  $(\ell, r) : \langle f, \Phi \rangle \rightarrow \langle g, \Psi \rangle$  consists of arrows

$$\ell : A \rightarrow A' \quad r : B \rightarrow B', \quad (3.5)$$

such that the equations

$$r \circ f = g \circ \ell \quad [\ell, B'] \circ \Psi \circ r = [A, r] \circ \Phi \quad (3.6)$$

hold; in diagrammatic terms, both diagrams in

$$\begin{array}{ccccc} A & \xrightarrow{f} & B & \xrightarrow{\Phi} & [A, B] \\ \downarrow \ell & & \downarrow r & & \downarrow [A, r] \\ & (i) & & (ii) & [A, B'] \\ & & & & \uparrow [\ell, B'] \\ A' & \xrightarrow{g} & B' & \xrightarrow{\Psi} & [A', B'] \end{array} \quad (3.7)$$

are commutative.

We denote by  $\mathbf{iMRS}(A, B)$  (or  $\mathbf{iMRS}_C(A, B)$  when there is risk of ambiguity) the collection of incoherent  $(M,R)$ -systems of type  $(A, B)$ .

**Remark 3.6** ( $(M,R)$ -systems as process-repair pairs). An incoherent  $(M,R)$ -system can be thought of as consisting, first, of a process (or ‘metabolism function’)  $f : A \rightarrow B$ , carrying an input of type  $A$  to an output of type  $B$ , and, second, of a repair map  $\Phi : B \rightarrow [A, B]$ . The latter assigns to each state or product  $b$  a possible process  $A \rightarrow B$ : in the terminology developed from Rosen’s original  $(M, R)$  systems,  $\Phi$  is a ‘selector’ or ‘replacement map’, rather than necessarily a material repair operation [Ros64a, Ros64b, Ros66a, LSAGA<sup>+</sup>04, LSAGA<sup>+</sup>06]. In this sense, in a more mathematical dialect,  $\Phi$  provides a term of type  $f$ , internalised in  $C$  via the closed structure.

In the Cartesian setting used in Rosen’s early treatments, this is the familiar state-transition reading; the uncurried representation makes it evident that one can think of  $f$  as a generalised element of  $B$ , and  $\Phi$  as an action on  $B$ , from which we can access the ‘iterates’

$$A \xrightarrow{f} B, \quad A \rightarrow A \times A \xrightarrow{f \times A} B \times A \xrightarrow{\Phi} B, \quad A \xrightarrow{(f \times A \times A) \circ \Delta} B \times A \times A \xrightarrow{\Phi \times A} B \times A \xrightarrow{\Phi} B, \dots \quad (3.8)$$

In a general monoidal category the tensor instead records that state and input are resources whose combination need not admit copying or deletion [Ros64a, Ros12, Gir87, See89].

The analogy was subsequently developed in automata-theoretic directions. Rosen recast  $(M, R)$  systems as sequential machines, making the output-dependent state transition explicit [Ros64a, Ros64b, Ros66a]; Arbib related systems with a fixed repair component to Mealy automata [Arb66], and Warner represented more general  $(M, R)$  systems through categories of automata [War82]. Later work revisited the same correspondence using communicating  $X$ -machines, arguing that ordinary finite-state machines are too weak to capture the self-referential structure [PWG16]. The present term “process-repair datatype” keeps the common typed interface in view while postponing stronger requirements: in the next section, we ask when the repair assignment can itself vary coherently, rather than remaining a single, unconstrained map attached to one system.

**Definition 3.7** (Total category of incoherent  $(M,R)$ -systems,  $\mathcal{U}$ ). Incoherent  $(M,R)$ -systems and their morphisms form the *total category*  $\mathbf{iMRS}$ , or  $\mathbf{iMRS}_C$  when the base category needs to be specified.<sup>3</sup> Explicitly,

<sup>3</sup>The subscript is useful when varying the base: the assignment  $C \mapsto \mathbf{iMRS}_C$  defines a pseudofunctor  $\mathbf{CMonCat}_{\text{lax}} \rightarrow \mathbf{Cat}$ , where  $\mathbf{CMonCat}_{\text{lax}}$  denotes the 2-category of closed monoidal categories and lax monoidal functors; see [ML98, §XI.2]. At the moment of writing, we do not have any practical use for this observation in what follows.

- the composition of arrows is performed componentwise and uses the bifactoriality property of  $[-, -]$ : given a composable pair of arrows  $\mathcal{M} \rightarrow \mathcal{N} \rightarrow \mathcal{P}$ , destructured as

$$\langle\langle f, \Phi \rangle\rangle \xrightarrow{(\ell, r)} \langle\langle g, \Psi \rangle\rangle \xrightarrow{(\ell', r')} \langle\langle h, \Theta \rangle\rangle, \quad (3.9)$$

the composite  $(\ell', r') \circ (\ell, r)$  is  $(\ell' \circ \ell, r' \circ r)$  and it is such that

$$\begin{aligned} [\ell' \circ \ell, 1] \circ \Theta \circ r' \circ r &= [\ell, 1] \circ [\ell', 1] \circ \Theta \circ r' \circ r && \text{(hom-functor)} \\ &= [\ell, 1] \circ [1, r'] \circ \Psi \circ r && \text{((ii)-comm)} \\ &= ([1, r'] \circ [\ell, 1]) \circ \Psi \circ r && \text{(hom-bifunctor)} \\ &= [1, r'] \circ [1, r] \circ \Phi && \text{((ii)-comm)} \\ &= [1, r' \circ r] \circ \Phi. && \text{(hom-functor)} \end{aligned}$$

- The identity of  $\langle\langle f, \Phi \rangle\rangle$  is  $(1_A, 1_B)$ .

**Remark 3.8.** Uncurrying gives an equivalent presentation of **iMRS**, with repair maps  $\Phi' : B \otimes A \rightarrow B$ . We use whichever presentation suits the argument; Proposition 3.36, for example, is easier to prove in uncurried form.

The second condition in (3.6) then becomes the commutative square

$$\begin{array}{ccc} B \otimes A & \xrightarrow{\Phi'} & B \\ r \otimes A \downarrow & \searrow r \otimes \ell & \downarrow r \\ B' \otimes A & \xrightarrow{B' \otimes \ell} B' \otimes A' \xrightarrow{\Psi'} & B' \end{array} \quad (3.10)$$

(or in equational form,  $r \circ \Phi' = \Psi' \circ r \otimes \ell$ , where clearly  $r \otimes \ell$  is the common value  $(B' \otimes \ell) \circ (r \otimes A) = (r \otimes A') \circ (B \otimes \ell)$  given by bifactoriality of the tensor operation).

We collect several examples of incoherent (M,R)-systems in various categories, to illustrate the generality of the notion.

**Example 3.9.** In the category of vector spaces, one can consider an (M,R)-system where  $B = k$  is the ground field, and find that a diagram

$$A \xrightarrow{\alpha} k \xrightarrow{\beta} [A, k] \quad (3.11)$$

consists of a pair of  $k$ -linear maps  $\alpha : A \rightarrow k, \beta : k \rightarrow [A, k]$ ; clearly, the second map amounts to another  $\beta : A \rightarrow k$ , due to the isomorphism  $\mathbf{Vect}(k, [A, k]) \cong \mathbf{Vect}(A, [k, k]) \cong \mathbf{Vect}(A, k)$ , thus giving a pair  $\alpha, \beta : A \rightarrow k$  of covectors of  $A$ ; the tensor product  $\alpha \otimes \beta : A \otimes A \rightarrow k \otimes k \cong k$ , which coincides with the uncurrying of  $\beta \circ \alpha$ , now amounts to a bilinear pairing on  $A$ , and one can recover all (M,R)-systems of type (3.11) as all and only the pairings  $A \otimes A \rightarrow k$  that are pure tensors in that they arise as Kronecker products of the form  $\alpha \otimes \beta$ .

**Example 3.10.** A similar computation can be performed in the categories **Rel** of relations and  $\vee \mathbf{Lat}$  of join semilattices; take  $B = 1$ , the singleton set that is the monoidal unit of **Rel**; then an (M,R)-system  $\langle\langle \alpha, \beta \rangle\rangle$  consists of a pair of relations of type  $\alpha : A \rightarrow 1, \beta : 1 \rightarrow [A, 1]$  i.e., since **Rel** is compact closed with  $A^* \cong A$  and hence  $[A, 1] \cong A$ , a pair of subsets  $\alpha, \beta \subseteq A$ ; the tensor  $\alpha \otimes \beta \subseteq A \times A$ , which coincides with the uncurrying of  $\beta \circ \alpha$ , is exactly the product relation  $\alpha \times \beta$ , so, as in 3.3.9, every such (M,R)-system is a ‘pure tensor’, now in the sense of a Cartesian product of two subsets of  $A$ . The identical computation, with 1 replaced by the two-element lattice  $\mathbf{2} = \{\perp < \top\}$  (the monoidal unit of  $\vee \mathbf{Lat}$ ), applies verbatim in  $\vee \mathbf{Lat}$ .

More generally, we can reproduce this reasoning in every compact closed category; we refrain from doing so in full detail, but the reader will be able to fill in the argument by abstracting from the examples of (finite-dimensional) vector spaces, **Rel** and  $\vee \mathbf{Lat}$ .

**Example 3.11.** Recall that a *quantale* [Yet90, Ros90] is a suplattice that, *qua* category, is monoidal (bi)closed. It is then equipped with a monotone monoid operation  $*$  :  $Q \times Q \rightarrow Q$ , so that there is a Galois connection

$$x * y \leq z \iff x \leq y \multimap z \iff y \leq x \multimap z, \quad (3.12)$$

where the existence of arbitrary suprema ensures that the adjoint functor theorem holds in the form

$$y \multimap z := \bigvee \{t \mid t * y \leq z\} \quad x \multimap z := \bigvee \{t \mid x * t \leq z\} \quad (3.13)$$

In a quantale  $Q$ , an  $(M, R)$ -system consists of a chain of inequalities  $a \leq b \leq a \multimap b$ , which uncurries into the simpler ‘right action’ condition  $b * a \leq b$ . In all integral quantales (where the top element and the monoid unit for  $*$  coincide, cf. [Ros90]) like the Booleans, more generally all Heyting algebras, the suplattice of ideals of a commutative ring  $R$ ,  $\mathbf{Cost} := ([0, \infty], \geq, +, 0)$ , or  $\mathbf{Ult} := ([0, \infty], \geq, \max, 0)$ , the inclusion  $b \leq a \multimap b$  is always satisfied, so an  $(M, R)$ -system consists simply of an inequality  $a \leq b$ . In non-integral quantales like the free quantale  $2^{S^*}$  (the powerset of the free monoid on a set  $S$  of ‘letters’), the condition  $b \leq a \multimap b$  is not vacuous: the inequality  $b \leq a \multimap b$  says precisely that  $b$  is closed under right concatenation by any word of  $a$ , i.e.  $b \cdot a \subseteq b$ , recovering the right-action condition  $b * a \leq b$  directly rather than as an automatic consequence of integrality. For instance, fixing a single letter  $s \in S$  and taking  $a = \{s\}$ , any language  $b$  closed under appending  $s$ , such as  $b = S^*s$  (the words ending in  $s$ ) or  $b = \{s\}^*$  (the words consisting of repeated copies of  $s$ ), gives a genuine  $(M, R)$ -system  $a \leq b \leq a \multimap b$  that is not forced by the ambient order alone, in contrast with the automatic inclusion holding for every pair in the integral case.

**Example 3.12.** In any monoidal closed category  $C$  admitting countably infinite coproducts, one can uncurry the repair arrow  $\Phi$  of an  $(M, R)$ -system  $\langle f, \Phi \rangle = (A \rightarrow B, B \rightarrow [A, B])$  into an arrow  $B \otimes A \rightarrow B$ , which extends into a unique right action  $B \otimes A^* \rightarrow B$  of the free monoid  $A^* = \sum_{n \in \mathbb{N}} A^n$  on  $B$ . In particular, in the category of vector spaces (or more generally in  $R$ -modules over a ring  $R$ , or more generally in any abelian category), it is a reasonable question to ask what an  $(M, R)$ -system that is in addition a *complex* of maps looks like, i.e. such that the composite  $\Phi \circ f : A \rightarrow B \rightarrow [A, B]$  is the zero map. In such a situation,  $\Phi$  reduces to a right action  $B \otimes A^* \rightarrow B : (b, \underline{a}) \mapsto b \cdot \underline{a}$ , with the property that the image  $f(A) \subset B$  is contained in the annihilator

$$\text{Ann}(\Phi) := \{b \in B \mid \forall \underline{a} \in A^*, b \cdot \underline{a} = 0\}. \quad (3.14)$$

We are now ready to collect more lengthy examples.

**Remark 3.13** ( $(M, R)$ -systems in species, with convolution). A particularly rich example of  $(M, R)$ -system in a monoidally cocomplete category is the following; consider the category  $\mathbf{Spc}$  of Joyal’s *combinatorial species* [Joy86, BLL98], i.e. the functor category  $\mathbf{Bij} \rightarrow \mathbf{Set}$  where  $\mathbf{Bij}$  is the groupoid core of finite sets; among all species, the *representable* ones, denoted  $y(n)$ , are ‘concentrated on  $n$ -sets’ in the sense that  $y(n)[S] \neq \emptyset$  if and only if  $S$  has  $n$  elements, and in that case it is a singleton.

Equipped with *Day convolution*, see [Day70a, Day70b],

$$(F \otimes G)[S] = \bigsqcup_{S=S_1 \sqcup S_2} F[S_1] \times G[S_2], \quad (3.15)$$

the category  $\mathbf{Spc}$  is symmetric monoidal closed, with unit the species  $y(0)$  concentrated on the empty set; the internal hom  $\{F, G\}$  is defined by  $\{F, G\}[S] = \mathbf{Spc}(F, G[S \sqcup \_])$ , in such a way as to represent the functor  $\mathbf{Spc}(\_ \otimes F, G)$ .

For a species  $A$ , we write  $A^*$  to denote the free monoid on  $A$  in  $(\mathbf{Spc}, \otimes)$ , i.e.  $A^* := \sum_{n \geq 0} A^{\otimes n}$  is the countable coproduct of iterated Day convolutions of  $A$  with itself.

By the universal property of free monoids, for any species  $B$  an action of  $A^*$  on  $B$ , i.e. a right module structure  $B \otimes A^* \rightarrow B$ , is equivalent to a single map  $\varphi : B \otimes A \rightarrow B$ , resulting from uncurrying the repair function  $\Phi$  of an incoherent  $(M, R)$ -system  $\langle f, \Phi \rangle$ .

For example, if  $A = y(1)$  is the representable at 1, an  $(M, R)$ -system of the form

$$b_0 : y(1) \rightarrow B, \quad B \otimes y(1)^* \rightarrow B \quad (3.16)$$

carries no more information than the pair

$$b_0 \in B[1], \quad \varphi : B \otimes L \rightarrow B, \quad (3.17)$$

where  $L = \sum_{n \geq 0} y(1)^{\otimes n} = \sum_{n \geq 0} y(n)$  is the species of *linear orders*, see [BLL98, p. 10].



Thus, an incoherent  $(M, R)$ -system  $\langle f, \Phi \rangle$  based at  $(y(1), B)$  in the category **Spc** consists of a pointed  $L$ -module  $B$ , and more generally, an incoherent  $(M, R)$ -system  $\langle f, \Phi \rangle$  based at  $(A, B)$  consists of a pointed  $A^*$ -module  $B$ .

More can be said for an incoherent  $(M, R)$ -system  $\langle f, \Phi \rangle$  based at  $(y(1), B)$ , given that the repair function  $\Phi : B \rightarrow \{y(1), B\}$  has a quite direct combinatorial interpretation: recall from [BLL98, §1.4] that  $\{y(1), B\}$  is the *species derivative* of  $B$ , i.e. the species  $\partial B$  acting on a set  $S$  as  $B[S \sqcup 1]$ .<sup>4</sup>

Given all this,  $\Phi$  is equivalent to the datum of a  $\partial$ -coalgebra having components  $\Phi_{[n]} : B[n] \rightarrow B[n \sqcup 1]$  where  $[n]$  is a shorthand for an  $n$ -element set; as such, taking also into account that  $B$  is ‘pointed’ by  $b_0 : y(1) \rightarrow B$ , we can map  $b_0$  to  $b_1 := \Phi_{[1]}(b_0) \in B[2]$ , which then can be mapped to  $b_2 := \Phi_{[2]}(b_1) = \Phi_{[2]}(\Phi_{[1]}(b_0))$ , etc.

This is a ‘dynamical system’ in the ‘space’  $B$ , in the sense that follows. To each species  $B$  one can associate its *category of elements*  $\nabla B$ , having objects the pairs  $(S, x)$ , where  $S$  is a finite set and  $x \in B[S]$ , and morphisms  $\sigma : (S, x) \rightarrow (T, y)$  the bijections  $\sigma : S \rightarrow T$  such that  $B[\sigma](x) = y$ ; there is a canonical forgetful functor  $\pi_B : \nabla B \rightarrow \mathbf{Bij}$  which is a discrete opfibration, with fibre  $\pi_B^{-1}([n]) = B[n]$ , and allows one to recover the species  $B$  under this identification; thus, in light of the equivalence between functors  $\mathbf{Bij} \rightarrow \mathbf{Set}$  and discrete opfibrations over  $\mathbf{Bij}$ , the total space  $\nabla B$  is precisely the ‘bundle’ obtained by gluing the fibres  $B[n]$  along the bijections that act on them.

Now, given an incoherent  $(M, R)$ -system  $\langle f, \Phi \rangle$ , iterating  $\Phi$  from  $b_0$  traces out a sequence

$$\begin{array}{ccccccccccc} B[1] & & B[2] & & B[3] & & B[4] & & \dots & & B[n+1] & & \dots \\ b_0 \longmapsto & b_1 \longmapsto & b_2 \longmapsto & b_3 \longmapsto & \dots \longmapsto & b_n & \dots \end{array} \quad (3.18)$$

that can be seen as an orbit in the ‘total space’  $\nabla B$ : the generator of  $L = y(1)^*$  acts on the ‘seed’  $b_0$ , driving it stepwise through the successive fibres of  $\nabla B$ .

Clearly, the two descriptions of an incoherent  $(M, R)$ -system based at  $(y(1), B)$  given above, as a module  $\varphi : B \otimes L \rightarrow B$  and as a coalgebra  $\Phi : B \rightarrow \partial B$ , are Curry-dual to one another, and the orbit  $(b_n)_{n \geq 0}$  obtained by iterating  $\Phi$  coincides with the trajectory of the semiautomaton  $\varphi$  started at  $b_0$ , as well as with the essentially unique factorisation of  $b_0$  through the free construction on  $A$ .

The same ‘trace semantics’ for an  $(M, R)$ -system holds for arbitrary generating species  $A$ , and it once again becomes cleanest in the language of fibrations.

The Day convolution of species  $B \otimes A$  has a total category  $\nabla(B \otimes A)$ , such that an object over  $S$  is a decomposition  $S = S' \sqcup S''$  into disjoint subsets, together with an object of  $\nabla B$  over  $S'$  and one of  $\nabla A$  over  $S''$ , so  $\varphi$  becomes a fibred extension map, attaching an  $A$ -structured piece to a  $B$ -structured piece along a disjoint union of supports. Similarly, an object of  $\nabla(A^*)$  over a finite set  $U$  is an ordered decomposition  $U = S_1 \sqcup \dots \sqcup S_k$  together with elements  $a_i \in A[S_i]$ : a finite word of disjoint  $A$ -blocks covering  $U$ . The pointed module map  $\psi : A^* \otimes y(1) \rightarrow B$  extending  $b_0$  is then, under the Grothendieck construction, a morphism of discrete opfibrations over  $\mathbf{Bij}$ ,

$$\nabla \psi : \nabla(A^* \otimes y(1)) \longrightarrow \nabla B, \quad (3.19)$$

sending an object over  $U \sqcup \{1\}$ , i.e. a word of  $A$ -blocks anchored at the seed 1, to the point of  $B[U \sqcup \{1\}]$  obtained by consuming the blocks of the word in order, starting from  $b_0$ .

In this sense every generator  $a$  of  $A^*$  leaves a *trace* in the total space  $\nabla B$ , obtained by looking at the orbit of  $b_0$ : the action  $\varphi$  is nothing but a fibrewise map from the total space of words in  $A$  anchored at the seed into the total space of  $B$ , and the fibre of  $\nabla(A^* \otimes y(1))$  over  $U \sqcup \{1\}$  indexes precisely the paths through  $\nabla B$ , starting at the point over  $\{1\}$  determined by  $b_0$ , that this map records.

**Example 3.14** ( $(M, R)$ -systems in species, with substitution). The category **Spc** carries a second monoidal structure, the *substitution* product of [BLL98, §1.4],

$$(F \circ G)[S] = \coprod_{\pi \vdash S} F[\pi] \times \prod_{P \in \pi} G[P] \cong \int^{T \in \mathbf{Bij}} F[T] \times G^{\otimes T}[S], \quad (3.20)$$

where  $\pi$  ranges over the partitions of  $S$  and  $G^{\otimes T}$  is the  $T$ -indexed convolution power of Remark 3.13 (the two descriptions agree as soon as  $G[\emptyset] = \emptyset$ , which we assume throughout). Its unit

<sup>4</sup>Adjoint representations of this kind, for algebras and coalgebras arising in **Spc**, have been studied to some extent in [Lor25a].

is  $y(1)$ , and not the  $y(0)$  of the convolution structure. Substitution is not symmetric, nor braided, and it is closed on one side only:  $- \circ G$  is cocontinuous, being a coend of copies of its argument, whereas  $F \circ -$  preserves no colimit in general. The internal hom is defined as

$$[A, B][T] \cong \mathbf{Spc}(A^{\otimes T}, B), \quad (3.21)$$

so that an incoherent  $(M, R)$ -system  $\langle f, \Phi \rangle$  based at  $(A, B)$  uncurries to a pair  $f : A \rightarrow B$ ,  $\varphi : B \circ A \rightarrow B$ .

In particular,  $\varphi$  exhibits  $B$  as a *right module over an operad* in the sense of [Fre09]: an  $(M, R)$ -system based at  $(A, B)$  is a right module over the free operad  $\mathcal{F}(A)$ , pointed by  $f$ . The free monoid is no longer  $\sum_{n \geq 0} A^{\circ n}$ , since that formula needs the tensor to preserve countable coproducts in *both* variables and substitution does not; it is instead the free operad, whose value on  $S$  is the set of rooted trees with leaves labelled by  $S$  and each node of arity  $k$  decorated by an element of  $A[k]$ ; words are replaced by trees.

The comparison with Remark 3.13 is sharpest at  $A = y(1)$ , which was there the interesting case and is here the degenerate one:  $y(1)$  has become the monoidal unit, so  $[y(1), B] \cong B$ , the repair map is a bare endomorphism of the species  $B$ , and the free monoid on it is  $\sum_{n \geq 0} y(1)$ , a single unary operation iterated. Nothing leaves the fibre it starts in, whereas under convolution the same choice produced the derivative  $\partial B$  and a seed travelling through  $B[1], B[2], B[3], \dots$ . The first nondegenerate case is  $A = y(2)$ , for which  $A^{\otimes T}$  is the species of perfect matchings indexed by  $T$ , whence

$$[y(2), B][T] \cong B[T \times 2]^{(\mathbb{Z}/2)^T} : \quad (3.22)$$

the  $B$ -structures on a doubled support that are invariant under the swap inside each pair. Iterating  $\Phi$  from the seed by  $c_1 := f$  and  $c_{n+1} := \varphi \circ (c_n \circ A) : A^{\circ(n+1)} \rightarrow B$  then doubles the support at every step,  $2, 4, 8, \dots$ , against the  $+1$  of the derivative; the trace of a seed is an  $A$ -decorated tree in  $\nabla B$ , not a path. In fibred terms, an object of  $\nabla(B \circ A)$  over  $S$  is a partition of  $S$  together with a  $B$ -structure on the set of its blocks and an  $A$ -structure on each block, and  $\varphi$  is the fibred contraction of such a configuration, where  $\nabla(B \otimes A)$  recorded only a two-block decomposition.

**Example 3.15** ( $(M, R)$ -systems in  $\mathbf{Chu}(\mathbf{Set}, k)$ ). Let  $k$  be a set, interpreted as a space of truth values for propositions; a *Chu space* [Bar91, Pra99] over a set  $k$  is a triple  $A = (|A|, A^\bullet, e_A)$  consisting of a set of *points*  $a \in |A|$ , a set of *tests*  $q \in A^\bullet$ , and a pairing  $e_A : |A| \times A^\bullet \rightarrow k$  that assigns to each pair  $(a, q)$  an ‘answer’ to the question ‘is  $a$  such that  $q$ ?’.

A morphism  $A \rightarrow B$  of Chu spaces is a pair  $(r_! : |A| \rightarrow |B|, r^\bullet : B^\bullet \rightarrow A^\bullet)$  such that  $e_B(r_!(a), \beta) = e_A(a, r^\bullet(\beta))$  for all  $a \in |A|, \beta \in B^\bullet$ ; these compose in the evident way, yielding the category  $\mathbf{Chu}(\mathbf{Set}, k)$  (or more briefly,  $\mathbf{Chu}_k$ ).

The category  $\mathbf{Chu}_k$  is symmetric monoidal closed:  $|A \otimes B| = |A| \times |B|$ , and the internal hom  $A \multimap B$  has  $|A \multimap B| = \mathbf{Chu}_k(A, B)$ , with test set  $(A \multimap B)^\bullet = |A| \times B^\bullet$  and pairing  $\langle t, (a, \beta) \rangle = e_B(t_!(a), \beta)$ , representing  $\mathbf{Chu}_k(- \otimes A, B)$ .

Unlike the examples in Example 3.10,  $\mathbf{Chu}_k$  is not compact closed for  $|k| > 2$ : the internal hom  $A \multimap B$  is in general not isomorphic to  $A^\perp \otimes B$ . The operation  $\multimap$  is the ‘cofree constructor for linear implication’; see [Bar91, See89, Gir87].

We now embark on the task of studying  $(M, R)$ -systems  $\langle f, \Phi \rangle$  based at  $(A, B)$  in  $\mathbf{Chu}_k$ , i.e. a diagram of shape

$$A \xrightarrow{(f, f^\bullet)} B \xrightarrow{(\Phi_!, \Phi^\bullet)} (A \multimap B). \quad (3.23)$$

Unfolding the definition of the repair arrow  $\Phi$ , one sees that it assigns to every point  $b \in |B|$  a Chu-transform  $\Phi_!(b) : A \rightarrow B$  (in turn splitting as a pair  $(\Phi_!(b)_!, \Phi_!(b)^\bullet)$ ), together with a backward map  $\Phi^\bullet : |A| \times B^\bullet \rightarrow B^\bullet$  satisfying  $e_B(b, \Phi^\bullet(a, \beta)) = e_B(\Phi_!(b)_!(a), \beta)$ .

This condition says that testing  $b$  against the composite test built from a resource  $a \in |A|$  and a  $B$ -test  $\beta$  is the same as running  $b$ ’s repair transform on  $a$  and testing the result with  $\beta$ : the observable behaviour of  $b$  already fixes what  $\Phi_!(b)$  does to every input, with no further datum available.

To see these data in a fully worked instance, let  $k = \{0 < \frac{1}{2} < 1\}$ ; this is an algebraic lattice where conjunction gives a pairing  $k \times k \rightarrow k$  and thus a Chu space  $\mathbf{K} = (k, k, \min)$ , modelling Łukasiewicz/Kleene three-valued logic, see [CDM00]. Let  $A = B = \mathbf{K}$ . For each  $c \in k$ , we can

define an  $(M, R)$ -system based at  $(K, K)$  taking as components of the metabolism  $\varphi : K \rightarrow K$  the maps  $(\varphi_c)_! = \varphi_c^\bullet = \min(-, c)$ ; evidently

$$\min((\varphi_c)_!(x), y) = \min(x, c, y) = \min(x, (\varphi_c)^\bullet(y)), \quad (3.24)$$

by associativity and commutativity of  $\min$ , so we have really defined a Chu transformation. Note also that  $\varphi_c \circ \varphi_{c'} = \varphi_{\min(c, c')}$ .

We thus fix  $c \in k$  and take the metabolism map to be  $f = \varphi_c$ ; we define the repair map  $\Phi_! : k \rightarrow \mathbf{Chu}_k(K, K)$  by

$$\Phi_!(b) = \varphi_b \circ \varphi_c = \varphi_{\min(b, c)}, \quad \Phi^\bullet(a, \beta) = \min(a, c, \beta). \quad (3.25)$$

This is a valid Chu-transform  $K \rightarrow (K \multimap K)$ : for every  $b, a, \beta \in k$ ,

$$e_B(b, \Phi^\bullet(a, \beta)) = \min(b, \min(a, c, \beta)) = \min(\varphi_{\min(b, c)}(a), \beta) = e_B(\Phi_!(b)_!(a), \beta), \quad (3.26)$$

so  $\langle f, \Phi \rangle$  is a well-defined incoherent  $(M, R)$ -system based at  $(K, K)$ .

Uncurrying the composite  $A \xrightarrow{f} B \xrightarrow{\Phi} (A \multimap B)$  gives, for  $a_1, a_2 \in k$ ,

$$(a_1, a_2) \mapsto \Phi_!(f_!(a_1))_!(a_2) = \varphi_{\min(a_1, c)}(a_2) = \min(a_1, a_2, c). \quad (3.27)$$

The composite combines two resource readings with the threshold  $c$  by taking  $\min(a_1, a_2, c)$ . The two inputs remain distinct, making explicit the bilinear structure that a Cartesian diagonal would conceal. The same construction works for any complete lattice  $Q$ : put  $K = (Q, Q, \wedge)$  and  $\varphi_c = - \wedge c$ . Associativity, commutativity, and idempotency of meet give the uncurried composite  $(a_1, a_2) \mapsto a_1 \wedge a_2 \wedge c$ . The metabolism tests against  $c$ , while repair at  $b$  replaces the threshold by  $b \wedge c$ .

Hilbert spaces give a related example. For a complex separable Hilbert space  $\mathcal{H}$ , Abramsky [Abr10] represents the quantum system by  $K_{\mathcal{H}} = (\mathcal{H}_o, L(\mathcal{H}), e_{\mathcal{H}})$ , where  $\mathcal{H}_o$  consists of non-zero vectors,  $L(\mathcal{H})$  is the complete ortholattice of closed subspaces, and  $e_{\mathcal{H}}(\psi, S) = \|P_S \psi\|^2 / \|\psi\|^2$  is the Born-rule probability. The Chu-transforms  $K_{\mathcal{H}} \rightarrow K_{\mathcal{H}}$  are the unitary and antiunitary operators, recovering Wigner's theorem. For our repair construction we use a different pairing:  $K = (L(\mathcal{H}), L(\mathcal{H}), \wedge)$ . This is the complete-lattice example with  $Q = L(\mathcal{H})$ . A fixed closed subspace  $C$  gives metabolism  $f = \varphi_C = - \wedge C$  and repair  $\Phi_!(B) = \varphi_{B \wedge C}$ ; the uncurried self-application map is  $(S_1, S_2) \mapsto S_1 \wedge S_2 \wedge C$ . Thus two tested subspaces are intersected with a fixed reference subspace. The lattice is not distributive, so this instance lies outside the Heyting-algebra setting.

**Example 3.16.** Let  $\mathbf{N}$  denote the set of natural numbers; we regard  $\mathbf{N}$  as a discrete category, equipped with a monoidal structure given by addition.

Recall that the category  $\mathbf{Set}^{\mathbf{N}}$  of  $\mathbf{N}$ -graded sets can be defined as the functor category  $\mathbf{Cat}(\mathbf{N}, \mathbf{Set})$ , equipped with the monoidal closed structure coming from Day convolution; it is a natural question whether an intuition similar to the one in Remark 3.13 holds in  $\mathbf{Set}^{\mathbf{N}}$  as well. Indeed, this is the case, as the category has a similar ‘tensor-by-[1]’ and ‘derivative’ adjunction; one can easily compute that an incoherent  $(M, R)$ -system in  $\mathbf{Set}^{\mathbf{N}}$  consists of the following data:

- a ‘process’ given by  $\{f_n : A_n \rightarrow B_n \mid n \in \mathbf{N}\}$ ;
- a ‘dynamics’ given by  $\{\Phi_n : (B \otimes A)_n \rightarrow B_n\}$ .

The latter is in particular equivalent to a point  $b \in B_1$  and a family of functions  $B_1 \rightarrow B_2 \rightarrow \dots$  when  $A = y(1)$  is the family given by  $y(1)_n = \{\bullet\}$  if  $n = 1$  and empty otherwise.

All in all, a similar line of reasoning works whenever one considers categories of presheaves over bases that are ‘graded’ in a suitable sense; among many, the case of single-sorted PROPs [ML65] (symmetric monoidal categories where the monoid of objects is isomorphic to  $(\mathbf{N}, +, 0)$ ) is emblematic.

**Example 3.17.** Recall that the category of *pointed sets* consists of the coslice  $1/\mathbf{Set}$ : objects are pairs  $(X, x_0)$  made of a set  $X$  and a distinguished point  $x_0 \in X$ ; arrows are basepoint-preserving functions, in the obvious sense.

Much as for the category of pointed sets, we can speak of the category of pointed topological spaces, or more generally of pointed objects in each category  $C$  that admits a terminal object  $1$ ; as long as  $C$  admits at least the pushouts that are necessary to give this definition, the category of such pointed objects is monoidal closed, with respect to a structure where

- the tensor is called the *smash product* of pointed objects: a pair  $(X, x_0), (Y, y_0)$  of pointed objects is tensored into the object appearing in the pushout

$$\begin{array}{ccc} (X \times \{y_0\}) \cup (\{x_0\} \times Y) & \longrightarrow & X \times Y \\ \downarrow & & \downarrow \\ 1 & \longrightarrow & X \wedge Y, \end{array} \quad (3.28)$$

where the left vertical map collapses the *wedge*  $X \vee Y := (X \times \{y_0\}) \cup (\{x_0\} \times Y) \subseteq X \times Y$  to a point, and the induced basepoint of  $X \wedge Y$  is the image of 1.

- Each functor  $- \wedge Y$  (and thus each  $X \wedge -$ ) admits a right adjoint  $[Y, -]$ , realising the isomorphism

$$C_*((X \wedge Y, x_0 \wedge y_0), (Z, z_0)) \cong C_*((X, x_0), ([Y, Z], [y_0, z_0])); \quad (3.29)$$

the set of functions  $Y \rightarrow Z$  is indeed pointed, by the function that is constant at the basepoint of  $Z$ , and this definition is enough to ensure that there is a bijection with the set of basepoint-preserving functions  $(X \wedge Y, x_0 \wedge y_0) \rightarrow (Z, z_0)$ .

It is natural to ask what an incoherent  $(M, R)$ -system in the category of pointed sets looks like; the category of pointed sets is exactly the Kleisli category of the maybe monad on **Set**: a basepoint-preserving map  $A \rightarrow B$  is the same datum as a partial function  $A \rightarrow B$  (which is undefined if  $f(t) = b_0$  is the basepoint). Consequently, an incoherent  $(M, R)$ -system  $f : A \rightarrow B, \Phi : B \rightarrow [A, B]$  in the category of pointed sets is literally metabolism and repair implemented as partial functions allowed to fail:  $f$  may send a resource to ‘depleted’ (the basepoint), and  $\Phi$  assigns to every state  $b \in B$  a partial repair recipe that need not be defined on every input. This is a genuinely different flavour of resource-sensitivity from the one exhibited by the Chu-space examples above: there, every process comes with a dual test that is always defined (an *observational* form of accountability), whereas here a repair process can simply fail to fire, rather than being required to answer every test, and fail if it doesn’t pass one.

More generally, by a theorem of Kock [Koc70, Koc72], the Kleisli category of any *commutative* strong monad  $T$  on a Cartesian closed category inherits a *symmetric monoidal* structure, with  $\mathbf{Set}_T$  recovering the maybe-monad case above as the simplest instance; commutativity of  $T$  is exactly the hypothesis needed to make this induced tensor well defined and symmetric. Kock’s theorem stops there, however: it does not, by itself, make  $\mathbf{Set}_T$  *closed*. Whether the Kleisli category of a commutative strong monad is monoidal closed is a considerably more delicate question in general, addressed only under further hypotheses on  $T$  by Keigher [Kei78] and, more recently and in the specific setting of algebraic  $\lambda$ -calculi, by Valiron [Val13]; it does not follow from commutativity alone. The two instances below are closed for an independent reason, namely that each resulting Kleisli category happens to be a well-known compact closed category in its own right, quite apart from any general Kleisli-closure theorem. Taking  $T = \mathcal{P}$ , the (covariant) powerset monad,  $\mathbf{Set}_{\mathcal{P}}$  is exactly the category **Rel** of 3.3.10, compact closed with  $[A, B] \cong A \times B$ : an incoherent  $(M, R)$ -system there is precisely the pair of subsets described above, now read as a ‘resource is present/absent’ pairing. Taking instead  $T = \mathcal{M}$ , the finite-multiset (free commutative monoid) monad,  $\mathbf{Set}_{\mathcal{M}}$  restricted to finite sets is the likewise compact closed category  $\mathbf{Mat}_{\mathbb{N}}$  of finite sets and  $\mathbb{N}$ -valued matrices under composition by matrix multiplication over the semiring  $(\mathbb{N}, +, \times)$ , again with  $[A, B] \cong A \times B$ . Taking  $B = 1$  as in 3.3.10, an incoherent  $(M, R)$ -system there consists of a pair  $\alpha, \beta : A \rightarrow \mathbb{N}$  of finitely-supported multiplicity functions, and the tensor  $\alpha \otimes \beta : A \times A \rightarrow \mathbb{N}, (a, a') \mapsto \alpha(a)\beta(a')$ , records the number of ways of jointly witnessing one copy from  $\alpha$  and one from  $\beta$ ; this refines the merely Boolean, existence-only reading of **Rel** to an honest count of resource copies, closer in spirit to counting distinct molecules of a metabolite than to recording their mere presence.

**Example 3.18** ( $(M, R)$ -systems in ‘funny’ categories). Recall that, for categories  $A, B$ , the *funny tensor product*  $A \boxtimes B$  [FLK80] has the same objects as the Cartesian product  $A \times B$ , whereas its arrows are freely generated by pairs  $(f, 1_b) : (a, b) \rightarrow (a', b)$  for  $f : a \rightarrow a'$  in  $A$ , and  $(1_a, g) : (a, b) \rightarrow (a, b')$  for  $g : b \rightarrow b'$  in  $B$ , subject only to the relations that make each ‘horizontal’ slice

$A \rightarrow A \boxtimes B$  (at a fixed object of  $B$ ) and each ‘vertical’ slice  $B \rightarrow A \boxtimes B$  (at a fixed object of  $A$ ) into functors, crucially without imposing the interchange relation  $(1, g) \circ (f, 1) = (f, 1) \circ (1, g)$  that holds automatically in the Cartesian product  $A \times B$ . The category **Cat**, equipped with  $\boxtimes$ , is monoidal closed, with internal hom  $[B, C]_u$  having as objects the functors  $B \rightarrow C$ , and as morphisms  $F \Rightarrow G$  arbitrary object-indexed families of morphisms  $F(b) \rightarrow G(b)$  in  $C$ : an ‘unnatural transformation’.

An incoherent (M,R)-system  $f : A \rightarrow B, \Phi : B \rightarrow [A, B]_u$  in  $(\mathbf{Cat}, \boxtimes)$  assigns to every object  $b \in B$  a whole functor  $\Phi_b : A \rightarrow B$  (a full metabolic subsystem indexed by the state  $b$ ), but as  $b$  varies along a morphism  $b \rightarrow b'$  of  $B$ , the induced comparison between  $\Phi_b$  and  $\Phi_{b'}$  is not required to satisfy any coherence condition. From the point of view of the semantics, such a system corresponds to a  $\mathcal{B}_0$ -indexed family of arrows  $\Phi_X : X \rightarrow [A, X]$  able to reconstruct a process of given any type; in particular, one of type  $f : A \rightarrow B$ , if given one.

**Example 3.19.** In this example we take  $C$  to be a small-complete category (i.e. a—large—category admitting all limits of diagrams with small domain), and consider the—even larger—category  $[C, C]$  of endofunctors of  $C$ , (strict) monoidal under composition  $F \otimes G := G \circ F$ , with (strict) unit the identity functor  $\text{id}_C$ , and (right) internal hom  $[F, G] := \text{Ran}_F G$ , the right Kan extension of  $G$  along  $F$ , which exists because  $C$  is complete.

This is precisely the monoidal category in which monads on  $C$  are the internal monoids, see [ML98, p. 138]: we now make the point that an incoherent (M,R)-system here is one step short of the classical theory of monads and their modules.

Concretely, an incoherent (M,R)-system consists of a natural transformation  $f : A \Rightarrow B$  of endofunctors of  $C$ , together with  $\Phi : B \Rightarrow \text{Ran}_A B$ ; uncurrying gives an isomorphism  $\text{Nat}(B, \text{Ran}_A B) \cong \text{Nat}(B \circ A, B)$ , so  $\Phi$  transposes to a natural transformation  $\mu : B \circ A \Rightarrow B$ : this is the structure exhibiting  $B$  as a right module (algebra) for the endofunctor  $A$ , acting by precomposition. Currying the self-application composite  $A \circ A \xRightarrow{f \circ A} B \circ A \xRightarrow{\mu} B$  recovers a natural transformation  $A \circ A \Rightarrow B$ , the ‘two-resource’ action one would expect of an iterated repair.

The absence-of-a-diagonal phenomenon already discussed in Remark 3.2 recurs here in a sharp form: for a ‘duplication’  $A \Rightarrow A \circ A$  to exist,  $A$  must be a *comonad* on  $C$ .

**Remark 3.20.** The total category **iMRS** comes equipped with two functors

$$C \xleftarrow{[-]_A} \mathbf{iMRS} \xrightarrow{[-]_B} C \quad (3.30)$$

sending, respectively, the typical object  $(A, B; \langle f, \Phi \rangle)$  to  $A$  and to  $B$ , and the typical arrow  $(\ell, r) : (A, B; \langle f, \Phi \rangle) \rightarrow (A', B'; \langle g, \Psi \rangle)$  to  $\ell$  and  $r$ .

**Definition 3.21** (Fibres of **iMRS**,  $\mathcal{U}$  and  $\mathcal{V}$ ). Given objects  $A, B \in C_0$ , we can access the fibres over  $A$  and over  $B$  of the total category **iMRS**:

- The fibre  $\mathbf{iMRS}^A$  has objects the (M,R)-systems with fixed input  $A$  and variable output  $B$ ; this means that the typical arrow  $v : (B; \langle f, \Phi \rangle) \rightarrow (B'; \langle g, \Psi \rangle)$  consists of an arrow  $v : B \rightarrow B'$  in  $C$  such that  $v \circ f = g$  and  $[A, v] \circ \Phi = \Psi \circ v$ , i.e. such that the diagram

$$\begin{array}{ccc} A & \xrightarrow{f} & B \xrightarrow{\Phi} [A, B] \\ & \searrow g & \downarrow v \quad \downarrow [A, v] \\ & & B' \xrightarrow{\Psi} [A, B'] \end{array} \quad (3.31)$$

commutes;

- The fibre  $\mathbf{iMRS}_B$  has objects the (M,R)-systems with fixed output  $B$  and variable input  $A$ ; this means that the typical arrow  $u : (A; \langle f, \Phi \rangle) \rightarrow (A'; \langle g, \Psi \rangle)$  consists of an arrow  $u : A \rightarrow A'$  in  $C$  such that  $g \circ u = f$  and  $[u, B] \circ \Psi = \Phi$ , i.e. such that the diagram

$$\begin{array}{ccc} A & \xrightarrow{f} & B \xrightarrow{\Phi} [A, B] \\ u \downarrow & \searrow g & \downarrow \Psi \quad \downarrow [u, B] \\ A' & \xrightarrow{g} & B \xrightarrow{\Psi} [A', B] \end{array} \quad (3.32)$$

commutes.

**Remark 3.22.** These are the fibres of  $[-]_A$  and  $[-]_B$ .<sup>5</sup> The discrete category  $\mathbf{iMRS}(A, B)$  of Definition 3.5 is the fibre over  $(A, B)$  of

$$\langle [-]_A, [-]_B \rangle : \mathbf{iMRS} \longrightarrow C \times C \quad (3.33)$$

(in order to avoid confusion with different meanings for the angled brackets, it is probably better to rename this functor to  $[-]_{(A,B)}$ ).

**Proposition 3.23.** The functor  $[-]_A$  of (3.30) is a Grothendieck fibration.

*Proof.* It is sufficient to exhibit the reindexing functors

$$u^* : \mathbf{iMRS}^{A'} \longrightarrow \mathbf{iMRS}^A \quad (3.34)$$

induced by an arrow  $u : A \rightarrow A'$  in the base  $C$ ;  $u^*$  acts on an object  $\langle f, \Phi \rangle$ , based at  $(A', B)$ , sending it to the incoherent  $(M, R)$ -system  $\langle f \circ u, [u, B] \circ \Phi \rangle$  based at  $(A, B)$ ; that this assignment is functorial on morphisms  $v : B \rightarrow B'$  follows from the fact that the diagram

$$\begin{array}{ccccccc} A & \xrightarrow{f \circ u} & B & \xrightarrow{\Phi} & [A', B] & \xrightarrow{[u, B]} & [A, B] \\ & \searrow & \downarrow & \ddots & \downarrow [A', v] & & \downarrow [A, v] \\ & A' & & & & & \\ & \swarrow & \downarrow & \ddots & \downarrow [A', v] & & \downarrow [A, v] \\ & B' & \xrightarrow{\Phi'} & [A', B'] & \xrightarrow{[u, B']} & [A, B'] \end{array} \quad (3.35)$$

commutes in all its parts, using bifunctionality of  $[-, -]$  and the assumption that  $\dagger$  and  $\ddagger$  are commutative (since  $v$  is an arrow in  $\mathbf{iMRS}^{A'}$ ).  $\square$

**Remark 3.24.** On the other hand, due to the covariant-contravariant dependence on the parameter  $B$  in the type of an incoherent  $(M, R)$ -system, the functor  $[-]_B$  is *neither* a fibration nor an opfibration; it only exhibits  $\mathbf{iMRS}$  as a *displayed* category over  $C$ —which amounts to the fact that the dependence on  $B$  of the fibres  $\mathbf{iMRS}_B$  is *profunctorial*, and not functorial; moreover, it is only functorial on the groupoid core of  $C$ .

*Proof.* There is nothing new to prove, since the (inverse) Grothendieck-Bénabou construction applied to the functor  $[-]_B$  yields a lax normal functor  $C \rightarrow \mathbf{Prof}$  sending  $B$  to  $\mathbf{iMRS}_B$  and an arrow  $v : B \rightarrow B'$  to a reindexing profunctor  $v_* : \mathbf{iMRS}_B \rightarrow \mathbf{iMRS}_{B'}$ ; but we define the profunctor in detail for completeness. Let  $v : B \rightarrow B'$  be an arrow in the base category  $C$ ; then, there is a profunctor  $v_* : \mathbf{iMRS}_B^{\text{op}} \times \mathbf{iMRS}_{B'} \rightarrow \mathbf{Set}$  sending

- a pair of objects  $(\langle f, \Phi \rangle, \langle f', \Phi' \rangle)$  to the set

$$v_*(\langle f, \Phi \rangle, \langle f', \Phi' \rangle) := \left\{ u : A \rightarrow A' \mid \begin{array}{c} v \circ f = f' \circ u \\ [1, v] \circ \Phi = [u, 1] \circ \Phi' \circ v \end{array} \right\} \quad (3.36)$$

- a pair of morphisms  $(s : \langle g, \Psi \rangle \rightarrow \langle f, \Phi \rangle, t : \langle f', \Phi' \rangle \rightarrow \langle g', \Psi' \rangle)$  in  $\mathbf{iMRS}_B^{\text{op}} \times \mathbf{iMRS}_{B'}$  to the function

$$v_*(\langle f, \Phi \rangle, \langle f', \Phi' \rangle) \longrightarrow v_*(\langle g, \Psi \rangle, \langle g', \Psi' \rangle) \quad (3.37)$$

defined by sending the element  $u : A \rightarrow A'$  in the domain to  $t \circ u \circ s$ ; clearly  $v \circ g = g' \circ t \circ u \circ s$ , because

$$\begin{aligned} v \circ g &= (v \circ f) \circ s \\ &= f' \circ (u \circ s) \\ &= g' \circ (t \circ (u \circ s)) \end{aligned}$$

<sup>5</sup>The fibre  $p^{-1}(B)$  of a functor  $p : \mathcal{E} \rightarrow \mathcal{B}$  consists of the objects sent to  $B$  and the arrows sent to  $1_B$ .

and equally clearly  $[1, v] \circ \Psi = [t \circ u \circ s, 1] \circ \Psi' \circ v$ , because

$$\begin{aligned}
 [1, v] \circ \Psi &= ([1, v] \circ [s, 1]) \circ \Phi \\
 &= [s, 1] \circ ([1, v] \circ \Phi) \\
 &= ([s, 1] \circ [u, 1]) \circ (\Phi' \circ v) \\
 &= [u \circ s, 1] \circ (\Phi' \circ v) \\
 &= ([u \circ s, 1] \circ [t, 1]) \circ (\Psi' \circ v) \\
 &= [t \circ u \circ s, 1] \circ (\Psi' \circ v)
 \end{aligned}$$

The identity and functoriality of this assignment are evident once unfolded, and this concludes the proof.  $\square$

The discussion so far has established the following theorem:

**Theorem 3.25.** The total category of incoherent  $(M, R)$ -systems forms, with the functors of Remark 3.20, a *discrete span* (by which we mean a span  $\mathcal{A} \xleftarrow{p} \mathcal{E} \xrightarrow{q} \mathcal{B}$  such that the induced functor  $\langle p, q \rangle$  has discrete fibres), whose left leg is a Grothendieck fibration.

**Remark 3.26** (On the iterated composites of **iMRS**). Considering the diagram of Remark 3.20 as an object of the monoidal category of spans  $(\mathbf{Span}(C, C), \circ)$ , one can construct the iterated composites

$$\mathbf{iMRS}^{(0)} := C \quad \mathbf{iMRS}^{(n+1)} := \mathbf{iMRS}^{(n)} \circ \mathbf{iMRS}; \quad (3.38)$$

we refer to  $\mathbf{iMRS}^{(n)}$  as *n-ary incoherent  $(M, R)$ -systems*.

In particular  $\mathbf{iMRS}^{(2)}$  is obtained as the pullback of  $[-]_B$  and  $[-]_A$  yielding the category having

- as objects the pairs  $(\langle f, \Phi \rangle, \langle g, \Psi \rangle)$  of incoherent  $(M, R)$ -systems

$$\begin{array}{ccccc}
 A & \xrightarrow{f} & B & \xrightarrow{\Phi} & [A, B] \\
 & & \parallel & & \\
 & & X & \xrightarrow{g} & Y \xrightarrow{\Psi} [X, Y]
 \end{array} \quad (3.39)$$

such that  $B = [f, \Phi]_B = [g, \Psi]_A = X$ ;

- as morphisms the pairs of morphisms  $(\ell, r) : \langle f, \Phi \rangle \rightarrow \langle f', \Phi' \rangle$  and  $(\ell', r') : \langle g, \Psi \rangle \rightarrow \langle g', \Psi' \rangle$  of incoherent  $(M, R)$ -systems such that  $r = \ell'$  as arrows  $B \rightarrow B'$  in the appropriate commutative diagram;

compositions and identities are inherited from those of the product category  $\mathbf{iMRS} \times \mathbf{iMRS}$ , of which  $\mathbf{iMRS}^{(2)}$  is evidently a regular subobject.

**Construction 3.27** (Sequential composition of incoherent  $(M, R)$ -systems,  $\circ$ ). There exists a functor

$$[-] \circ [-] : \mathbf{iMRS}^{(2)} \longrightarrow \mathbf{iMRS} \quad (3.40)$$

defined as follows:

- on objects, the pair  $(\langle f, \Phi \rangle, \langle g, \Psi \rangle)$  goes to the incoherent  $(M, R)$ -system based at  $(A, Y)$  and obtained as the composition

$$\langle f, \Phi \rangle \circ \langle g, \Psi \rangle := \langle g \circ f, f^* \Psi \rangle : A \xrightarrow{f} B \xrightarrow{g} Y \xrightarrow{\Psi} [B, Y] \xrightarrow{[f, Y]} [A, Y] \quad (3.41)$$

- on morphisms, one acts accordingly; a morphism of incoherent  $(M, R)$ -systems is such that the diagram marked in red here

$$\begin{array}{ccccc}
 & & X' & \xrightarrow{\Psi'} & [B', X'] \\
 & \nearrow w & & & \searrow [f', X'] \\
 X & & & & [A', X'] \\
 \downarrow \Psi & & & & \nearrow [v, X'] \\
 [B, X] & \xrightarrow{[B, w]} & [B, X'] & \xrightarrow{[f, X']} & [A, X'] \\
 \searrow [f, X] & & \searrow [f, X'] & & \nearrow [u, X'] \\
 & [A, X] & \xrightarrow{[A, w]} & [A, X'] &
 \end{array} \tag{3.42}$$

commutes, using only the bifunctionality of  $[-, -]$ .

Two structural properties of  $-\circlearrowright-$  are worth recording explicitly. First,  $-\circlearrowright-$  is strictly associative: for composable incoherent  $(M, R)$ -systems  $\langle\langle f, \Phi \rangle\rangle : A \rightarrow B$ ,  $\langle\langle g, \Psi \rangle\rangle : B \rightarrow Y$ ,  $\langle\langle h, \Xi \rangle\rangle : Y \rightarrow Z$ , both

$$(\langle\langle f, \Phi \rangle\rangle \circlearrowright \langle\langle g, \Psi \rangle\rangle) \circlearrowright \langle\langle h, \Xi \rangle\rangle \quad \text{and} \quad \langle\langle f, \Phi \rangle\rangle \circlearrowright (\langle\langle g, \Psi \rangle\rangle \circlearrowright \langle\langle h, \Xi \rangle\rangle) \tag{3.43}$$

compute to  $\langle\langle h \circ g \circ f, [f, Z] \circ [g, Z] \circ \Xi \rangle\rangle$ , using associativity of composition in  $C$  together with the identity  $[g \circ f, Z] = [f, Z] \circ [g, Z]$  furnished by bifunctionality of  $[-, -]$ .

Second, and this line of reasoning will be expanded further in Proposition 3.51,  $-\circlearrowright-$  is only *one-sidedly* unital. For any object  $A$  and *any* choice of repair map  $\eta$ , the system  $\langle\langle 1_A, \eta \rangle\rangle$  based at  $(A, A)$  is a left unit:  $\langle\langle 1_A, \eta \rangle\rangle \circlearrowright \langle\langle g, \Psi \rangle\rangle = \langle\langle g \circ 1_A, [1_A, Y] \circ \Psi \rangle\rangle = \langle\langle g, \Psi \rangle\rangle$ , since the formula for  $-\circlearrowright-$  never references the repair map of its left argument. By contrast,  $\langle\langle f, \Phi \rangle\rangle \circlearrowright \langle\langle 1_B, \eta \rangle\rangle = \langle\langle f, [f, B] \circ \eta \rangle\rangle$ , which recovers  $\langle\langle f, \Phi \rangle\rangle$  only for those systems whose repair map happens to factor as  $\Phi = [f, B] \circ \eta$ ; no single  $\eta$  achieves this simultaneously for every  $\langle\langle f, \Phi \rangle\rangle$  based at  $B$ . Consequently,  $(C_0, \mathbf{iMRS}(-, -), -\circlearrowright-)$  falls short of being an ordinary category: it is associative but merely left-unital, a semigroupoid rather than a category. This is the same phenomenon, seen from a different angle, as the failure of  $[-]_B$  to be a fibration recorded in Remark 3.24: the dependence of an incoherent  $(M, R)$ -system's repair map on  $B$  is profunctorial rather than functorial, and it is exactly this profunctoriality that  $-\circlearrowright-$  cannot repair when composing on the right.

**Remark 3.28.** Continuing the induction, one can see that a ternary incoherent  $(M, R)$ -system consists of three incoherent  $(M, R)$ -systems  $\langle\langle f, \Phi \rangle\rangle, \langle\langle g, \Psi \rangle\rangle, \langle\langle h, \Xi \rangle\rangle$  arranged in a diagram of the following type:

$$\begin{array}{ccccc}
 A & \xrightarrow{f} & B & \xrightarrow{\Phi} & [A, B] \\
 & & \downarrow g & & \\
 & & Y & \xrightarrow{h} & Z \xrightarrow{\Xi} [Y, Z] \\
 & & \downarrow \Psi & & \\
 & & [B, Y] & &
 \end{array} \tag{3.44}$$

It is evident that from this diagram one can extract a 3-simplex  $(f, g, h)$  in the nerve of  $C$ , and that there is a canonical way to complete the composition  $h \circ g \circ f$  to an incoherent  $(M, R)$ -system with repair function

$$Z \xrightarrow{\Xi} [Y, Z] \xrightarrow{[g \circ f, Z]} [A, Z]. \tag{3.45}$$

This sets up a functor  $\mathbf{iMRS}^{(3)} \rightarrow \mathbf{iMRS}$  and the induction proceeds.

We now collect a number of equivalent presentations for incoherent  $(M, R)$ -systems; from time to time we will assume that the base category admits some limits and colimits. Note that some of these abstract characterisations will shed light on the results found in the above examples, in particular Remark 3.13 and Example 3.17.



**Proposition 3.29** ((M,R)-systems at  $A$  as algebras,  $\mathcal{C}$ ). For this example, assume that  $C$  admits binary coproducts; then (since each tensoring functor  $B \otimes -$  commutes with all colimits), one can uncurry  $\Phi$  and pair the arrows  $(f, \hat{\Phi})$  in an incoherent (M,R)-system into a single map  $[f, \hat{\Phi}] : A + B \otimes A \rightarrow B$ , which in turn corresponds to

$$(I + B) \otimes A \xrightarrow{\sim} I \otimes A + B \otimes A \xrightarrow[\text{unit}^t + B \otimes A]{\sim} A + B \otimes A \xrightarrow{[f, \hat{\Phi}]} B. \quad (3.46)$$

This is the correspondence on objects of an equivalence of categories between  $\mathbf{iMRS}^A$  and the category of algebras for the endofunctor

$$K_A = \lambda X.((I + X) \otimes A) : C \longrightarrow C \quad (3.47)$$

*Proof.* The discussion above outlines the action on objects of a candidate functor

$$(-)^a : \mathbf{iMRS}^A \longrightarrow \mathbf{Alg}(K_A). \quad (3.48)$$

Clearly this is a bijection on objects whose inverse is defined by precomposition with the coproduct injections, and currying. It is now straightforward to check that a morphism  $v : \langle f, \Phi \rangle \rightarrow \langle g, \Psi \rangle$  of incoherent (M,R)-systems based at  $A$  corresponds, under this assignment, to a  $K_A$ -algebra homomorphism  $v^a : \langle f, \Phi \rangle^a \rightarrow \langle g, \Psi \rangle^a$ , and bijectively so. The correspondence is built so that the two conditions in (3.6) coincide with the fact that the map  $[f, \hat{\Phi}] : K_A B = A + B \otimes A \rightarrow B$  determined in (3.47) makes the square

$$\begin{array}{ccc} K_A B & \xrightarrow{(I+v) \otimes A} & K_A B' \\ [f, \hat{\Phi}] \downarrow & & \downarrow [g, \hat{\Psi}] \\ B & \xrightarrow{v} & B' \end{array} \quad (3.49)$$

commute. The key adjunction identity says that the left square below commutes if and only if the right square does:

$$\begin{array}{ccc} B & \xrightarrow{\Phi} & [A, B] \\ v \downarrow & & \downarrow [A, v] \\ B' & \xrightarrow{\Psi} & [A, B'] \end{array} \quad \begin{array}{ccc} B \otimes A & \xrightarrow{\hat{\Phi}} & B \\ v \otimes A \downarrow & & \downarrow v \\ B' \otimes A & \xrightarrow{\hat{\Psi}} & B' \end{array} \quad (3.50)$$

Thus, this comparison functor  $(-)^a$  is an equivalence.  $\square$

**Remark 3.30** (Limits and colimits of  $A$ -based (M,R)-systems). The previous result comes in handy because the theory of endofunctor algebras provides powerful tools for establishing completeness and accessibility properties of  $\mathbf{iMRS}^A$ : assuming  $C$  is a locally presentable category [AR94, GU71] (thus accessible, complete, and cocomplete) we have that

- the category  $\mathbf{iMRS}^A$  is complete, and limits are created in  $C$  by the forgetful functor  $U = [-]_B|_{\mathbf{iMRS}^A}$ ;
- the functor  $U$  has a left adjoint yielding the free  $K_A$ -algebra on an object  $X$ , obtained by sending an object  $X$  of  $C$  to the initial algebra of the endofunctor  $Y \mapsto X + K_A Y$ , i.e. to the colimit of the free-algebra sequence of [Adá74],

$$X + K_A \emptyset \longrightarrow X + K_A(X + K_A \emptyset) \longrightarrow \dots \quad (3.51)$$

which is  $A^* \otimes (1 + X)$ ;

- the category  $\mathbf{iMRS}^A$  is accessible, because  $K_A$  is an accessible functor and accessible categories are closed under bilimits [MP89, §5.1]; thus,  $\mathbf{iMRS}^A$  is locally presentable, hence also cocomplete; the initial  $K_A$ -algebra is then the colimit of the sequence

$$\emptyset \longrightarrow K_A \emptyset \longrightarrow K_A K_A \emptyset \longrightarrow K_A K_A K_A \emptyset \longrightarrow \dots \quad (3.52)$$

which is easily seen to be the free monoid on  $A$ , with structure maps

$$A \xrightarrow{(-)} A^* \quad A^* \xrightarrow{a \mapsto \lambda x.(a \cdot x)} [A, A^*] \quad (3.53)$$

where we use the notation  $\underline{a} \mapsto \lambda x.(\underline{a} : x)$ , thinking of a  $\lambda$ -term, to denote the arrow  $\sum_{n \geq 0} A^{\otimes(n+1)} \rightarrow \sum_{m \geq 0} A^{\otimes m}$  induced by  $\iota_{n+1} : A^{\otimes(n+1)} \rightarrow A^*$ .

**Proposition 3.31** ((M,R)-systems at  $B$  as a slice category). In a very similar vein, we can characterise the fibres  $\mathbf{iMRS}_B$  via a universal property. Assume that  $C$  is symmetric monoidal closed and that it admits binary products; then one can uncurry, swap the domain objects, and re-curry the repair arrow  $\Phi$  of an incoherent (M,R)-system into a single map

$$A \longrightarrow B \times [B, B]. \quad (3.54)$$

If in addition  $C$  admits coproducts with the monoidal unit  $I$ , we can rewrite this as a map

$$A \longrightarrow [I, B] \times [B, B] \stackrel{\sim}{=} [I + B, B]; \quad (3.55)$$

sending  $\langle f, \Phi \rangle$  to this map constitutes the action on objects of an equivalence between  $\mathbf{iMRS}_B$  and the slice category  $C/[I + B, B]$ .

*Proof.* The argument is a straightforward check that the correspondence just defined on objects extends to morphisms, sending a diagram (3.7) to a homomorphism in the slice category; by way of example, we show part of the argument.

Denote by  $\tilde{p}$  the mate of an arrow  $p : X \otimes Y \rightarrow Z$  according to the monoidal closed structure on  $C$ , i.e. the arrow

$$Y \xrightarrow{\eta_Y} [X, X \otimes Y] \xrightarrow{[X, p]} [X, Z] \quad (3.56)$$

Then the claim of functoriality amounts to the fact that the triangle

$$\begin{array}{ccc} A & \xrightarrow{u} & A' \\ & \searrow \langle \widetilde{f \circ \rho_A}, \tilde{\Phi} \rangle & \swarrow \langle \widetilde{g \circ \rho_{A'}}, \tilde{\Psi} \rangle \\ & [I, B] \times [B, B] & \end{array} \quad (3.57)$$

is commutative, which can be checked ‘componentwise’ as  $\langle \widetilde{g \circ \rho_{A'}}, \tilde{\Psi} \rangle \circ u = \langle \widetilde{g \circ \rho_{A'}}, \tilde{\Psi} \rangle \circ u = \langle \widetilde{g \circ \rho_{A'}}, \tilde{\Psi} \rangle \circ u$  and

$$\begin{aligned} [I, g \circ \rho_{A'}] \circ \eta_{A'} \circ u &= [I, g \circ \rho_{A'}] \circ [I, u \otimes I] \circ \eta_A \\ &= [I, g \circ \rho_{A'} \circ u \otimes I] \circ \eta_A \\ &= [I, f \circ \rho_A] \circ \eta_A \\ &= \eta_{A'} \circ f \circ \rho_A \\ &= \widetilde{f \circ \rho_A}. \end{aligned}$$

One reasons similarly, if more tediously, for  $\tilde{\Psi} \circ u$  and this concludes the proof.  $\square$

**Remark 3.32** (On the symmetry hypothesis). This is the only result that seems to require symmetry, or at least biclosure of  $C$ , because it is the only one that seems to rely on the uncurrying-recurrying operation (Definition 2.19).

We nonetheless assume symmetry, in the statement and in the formalisation, because in a biclosed setting the two internal homs must be carried, related and reasoned about separately throughout, and the codebase would not have benefited from this extra generality. The Agda development makes the choice visible: `Incoherent/Slice.agda` takes a `Symmetric` structure as an explicit module parameter, and the braiding occurs there exactly where the argument above predicts.

**Remark 3.33.** As a consequence of the equivalence just outlined, a set of corollaries similar to the ones in Remark 3.30 applies to  $\mathbf{iMRS}_B$  and the existence of co/limits therein, as well as its accessibility.

The existence of co/limits in the fibres  $\mathbf{iMRS}^A$  and  $\mathbf{iMRS}_B$  is a first step towards establishing completeness and cocompleteness of the total category  $\mathbf{iMRS}$ . We now embark on providing a modern, explicit argument for the existence of co/limits in the  $\mathbf{iMRS}$  of Definition 3.5 using the language of accessible categories.

The treatment here settles a matter the existing literature leaves genuinely obscure. Several authors—among them Rosen, Arbib and Warner [Ros66a, Arb66, War82]—assert the existence of co/limits in subcategories of **iMRS**, but none proves completeness or cocompleteness directly, and no two of them work with quite the same category.

Băianu states a completeness theorem, in the form of a claim that ‘projective systems’ of general  $(M, R)$ -systems admit ‘projective limits’, and that discrete families admit ‘direct sums’ and ‘cokernels’ [Băi73, Theorems 1–2]; the latter clearly echoes the classical results reducing cocompleteness to the existence of arbitrary coproducts and coequalisers. The proof, however, is attributed to an older paper of Căzănescu couched in the language of automata [Căz67], which is effectively unfindable; the Băianu–Marinescu paper cited alongside it is likewise unavailable for verification [BM74].

Louie, for his part, exhibits a neighbouring category of systems [Lou09, §7.10, 13.23], only to observe that it fails to admit all equalisers.

The net effect is a tower of Babel: definitions differing in details seldom made explicit, carrying theorems that variously affirm and deny the existence of the same co/limits, and no dictionary for passing between them. There is great disorder under heaven; the situation is excellent, in that the disagreements turn out to concern not mathematics but which category one is standing in. We therefore begin by rectifying names, in the sense of *Analects* XIII.3; Proposition 3.36 below then establishes completeness and cocompleteness for the category **iMRS** that we defined in Definition 3.5.

**3.1. Limits and colimits in iMRS.** The cocompleteness theorem makes heavy use of the following construction, that we recall from [ACC<sup>+</sup>24, Example 2.24].

**Definition 3.34** (The fibration of parametric algebras). Define a category  $\int_A \mathbf{Alg}(- \otimes A)$  of *parametric algebras*, having

- objects the tuples  $(X, \xi)^A$  consisting of an object  $A$  of  $C$ , an object  $X$  of  $C$ , and a map  $\xi : X \otimes A \rightarrow X$  in  $C$ ;
- arrows  $(X, \xi)^A \rightarrow (Y, \theta)^B$  the pairs  $(u, v)$  of arrows  $u : A \rightarrow B$  and  $v : X \rightarrow Y$  in  $C$  such that the diagram

$$\begin{array}{ccc} X \otimes A & \xrightarrow{v \otimes u} & Y \otimes B \\ \xi \downarrow & & \downarrow \theta \\ X & \xrightarrow{v} & Y \end{array} \quad (3.58)$$

commutes.

Evidently, all the category axioms are satisfied, and the assignment  $(X, \xi)^A \mapsto A$  defines a functor  $p_\otimes : \int_A \mathbf{Alg}(- \otimes A) \rightarrow C$ .

In [ACC<sup>+</sup>24] it is argued that this category deserves the name of *total category of semiautomata*, given that a semiautomaton is precisely an algebra for the endofunctor  $- \otimes A$  on  $C$ , and that the typical object of  $\int_A \mathbf{Alg}(- \otimes A)$  is a pair (parameter, algebra for the functor defined at that parameter).

There we also argue, among other things, that the category  $\int_A \mathbf{Alg}(- \otimes A)$  can be seen as a ‘semidirect product’ of  $C$  acting on itself by the ‘regular’ representation  $(X, Y) \mapsto X \otimes Y$ . We thus adopt the shorthand  $C \ltimes_\otimes C := \int_A \mathbf{Alg}(- \otimes A)$  from now on.

**Remark 3.35.** The category  $C \ltimes_\otimes C$  comes equipped with two functors

$$C \xleftarrow{p_\otimes} C \ltimes_\otimes C \xrightarrow{V} C \quad (3.59)$$

defined respectively by sending the typical object  $(X, \xi)^A$  to  $X = V((X, \xi)^A)$  and to  $A = p_\otimes((X, \xi)^A)$ . The functor  $p_\otimes$  is moreover a *Grothendieck fibration*, and the fibre over an object  $A$  is precisely the category  $\mathbf{Alg}(- \otimes A)$  of algebras for the endofunctor  $- \otimes A$ ; this explains in a nutshell the notation and the relation between  $\int_A \mathbf{Alg}(- \otimes A)$  and  $p_\otimes$ : the latter is the Grothendieck fibration associated, under the equivalence of Theorem 2.21, to the pseudofunctor

$$C^{\text{op}} \longrightarrow \mathbf{Cat} : A \longmapsto \mathbf{Alg}(- \otimes A) \quad (3.60)$$

**Proposition 3.36.** Let  $C$  be a closed monoidal category, which in addition is *locally presentable* in the sense of [AR94, Theorem 1.20];<sup>6</sup> then the category  $\mathbf{iMRS}_C$  of Definition 3.7 is itself locally presentable, and thus (complete and) cocomplete.

The proof of this proposition revolves around the following lemma, which we record separately since it is of independent interest: it provides a highly conceptual characterisation of the total category  $\mathbf{iMRS}$  of incoherent  $(M, R)$ -systems.

**Lemma 3.37.** Let  $C$  be a closed monoidal category, and let  $C \ltimes_{\otimes} C$  be the total category of semi-automata as above. Then the category  $\mathbf{iMRS}$  of incoherent  $(M, R)$ -systems (in uncurried form, see Remark 3.8) fits in a pullback of categories

$$\begin{array}{ccccc}
 \mathbf{iMRS} & \longrightarrow & \mathcal{Y} & \longrightarrow & C^{[0 \rightarrow 1]} \\
 \downarrow & \lrcorner & \downarrow & \lrcorner & \downarrow \text{dom} \\
 \mathcal{X} & \longrightarrow & C \ltimes_{\otimes} C & \xrightarrow{p_{\otimes}} & C \\
 \downarrow & \lrcorner & \downarrow v & & \\
 C^{[0 \rightarrow 1]} & \xrightarrow{\text{cod}} & C & & 
 \end{array} \tag{3.61}$$

*Proof.* The proof is straightforward: an object of  $\mathcal{X}$ , the pullback of the pair  $(\text{cod}, V)$ , is a pair  $(f : X \rightarrow B, \Phi : B \otimes A \rightarrow B)$  for certain objects  $X, A \in C_0$ ; at the same time, an object of  $\mathcal{Y}$ , the pullback of the pair  $(\text{dom}, p_{\otimes})$ , is a pair  $(g : A \rightarrow Y, \Psi : Z \otimes A \rightarrow Z)$  for certain objects  $Y, Z \in C_0$ . It is clear how the functors in the cospan  $\mathcal{X} \rightarrow C \ltimes_{\otimes} C \leftarrow \mathcal{Y}$  act, and that their pullback is exactly an incoherent  $(M, R)$ -system based at  $(A, B)$ . The same reasoning applies to morphisms, and this concludes the proof.

It is worth observing that the destructor functors  $[-]_A, [-]_B$  defined in Remark 3.20 are ‘hidden’ in (3.61), and thus admit a more fundamental (and universal) characterisation in terms of this limit diagram.  $\square$

*Proof of Proposition 3.36.* We combine standard results on accessible categories in three steps.

- Apply the *completeness theorem for accessible categories* [MP89, Theorem 5.1.6] to a pullback square

$$\begin{array}{ccc}
 \mathcal{E} & \xrightarrow{P} & \mathcal{X} \\
 \downarrow Q & \lrcorner & \downarrow G \\
 \mathcal{Y} & \xrightarrow{F} & \mathcal{A}
 \end{array} \tag{3.62}$$

of categories, if  $F$  and  $G$  are accessible functors between accessible categories, then the pullback  $\mathcal{E}$  is itself an accessible category, and the projection functors  $P, Q$  are accessible.

- In particular, we would like to deduce that if  $\mathcal{A}$  is locally presentable, then so is  $\mathcal{E}$ , in the same notation and under the same assumptions. This follows from an application of [ML98, V.6, Exercise 3], where it is asserted that in any strict pullback square such as (3.62), if  $G$  preserves and  $F$  creates limits, then  $P$  creates limits. Thus  $\mathcal{E}$  is complete if  $\mathcal{X}$  is. An accessible category is locally presentable if and only if it is complete, so  $\mathcal{E}$  is locally presentable.
- Now, we can apply the above result once we have characterised  $\mathbf{iMRS}$  via an (iterated) pullback of categories; this is exactly what the presentation of  $\mathbf{iMRS}$  outlined in (3.61) attains. The functors involved in the pullback are easily seen to be accessible, provided  $C$  is, and the base category  $C$  is cocomplete (thus locally presentable) by assumption; thus, repeatedly applying [MP89]’s completeness theorem, we conclude that  $\mathbf{iMRS}$  is locally presentable.

We conclude that  $\mathbf{iMRS}$  is accessible and complete, thus locally presentable.  $\square$

<sup>6</sup>Recall that this means that  $C$  is cocomplete, and it is generated under filtered colimits by the subcategory of its  $\kappa$ -compact objects.

A similar line of reasoning works for the category of twisted elements of incoherent  $(M, R)$ -systems, that we define below.

Now for something different. The categorical theory of lenses, and more generally of *optics*, has by now an extensive literature; the reader can consult [FGM<sup>+</sup>07] for the original definition of a lens, and its motivating application to the view-update problem for bidirectional transformations, [JRW12] for the identification of lenses with certain fibrational data, [Ril18] for the unifying account of optics (lenses among them) as morphisms in a category built from a monoidal action, and [CEG<sup>+</sup>24] for the profunctor representation of optics that we tacitly use below; for applications of lenses across computer science, from modular data accessors [PGW17, BG18] to open games [GHWZ18], supervised learning [FST19] and dependent lenses [Spi19], the literature is by now vast.

**Proposition 3.38** ((M,R)-systems as lenses). The type of an incoherent  $(M, R)$ -system is that of a lens  $\begin{bmatrix} A \\ B \end{bmatrix} \rightarrow \begin{bmatrix} B \\ B \end{bmatrix}$ .

*Proof.* Recall from [FGM<sup>+</sup>07, JRW12] that a (bimorphic) lens  $\begin{bmatrix} S \\ S' \end{bmatrix} \rightarrow \begin{bmatrix} V \\ V' \end{bmatrix}$  in a Cartesian category consists of a pair ( $\text{get} : S \rightarrow V$ ,  $\text{put} : S \times V' \rightarrow S'$ ). In the Cartesian setting the order of the two factors of  $\text{put}$  carries no information, since  $S \times V' \cong V' \times S$  naturally; in the monoidal, not necessarily symmetric setting of the present paper it does, and we fix once and for all the convention

$$\text{get} : S \rightarrow V, \quad \text{put} : V' \otimes S \rightarrow S', \quad (3.63)$$

placing the residual to the *right* of the update. With this convention the correspondence is immediate and appeals to no braiding: an incoherent  $(M, R)$ -system based at  $(A, B)$  is the lens  $\begin{bmatrix} A \\ B \end{bmatrix} \rightarrow \begin{bmatrix} B \\ B \end{bmatrix}$  whose  $\text{get}$  is the process map  $f : A \rightarrow B$ , and whose  $\text{put}$  is the uncurried repair map  $\hat{\Phi} : B \otimes A \rightarrow B$  of Notation 3.1. This identification is a bijection on the nose, not merely up to isomorphism.

The choice of side is forced, rather than conventional. Since  $\text{get} = f$  pins the source object to  $S = A$ , the opposite convention  $\text{put} : S \otimes V' \rightarrow S'$  would demand an isomorphism  $A \otimes V' \cong B \otimes A$ , which is unavailable for *any* choice of target pair  $\begin{bmatrix} V \\ V' \end{bmatrix}$  unless  $A \cong B$ ; in particular, retargeting the lens to  $\begin{bmatrix} B \\ A \end{bmatrix}$  does not repair the mismatch.  $\square$

**Remark 3.39** (Lenses versus optics). The word ‘lens’ is used above for the bare ( $\text{get}, \text{put}$ ) pair, which is its original meaning in [FGM<sup>+</sup>07] and the one that the  $(M, R)$ -system diagram instantiates. It should not be conflated with the *optic* for the  $\otimes$ -action, i.e. the coend

$$\int^{M \in C} C(S, M \otimes V) \times C(M \otimes V', S') \quad (3.64)$$

of [Ril18, CEG<sup>+</sup>24]. The two agree when  $C$  is Cartesian, where the co-Yoneda lemma collapses the coend at  $M = S$  and returns exactly a ( $\text{get}, \text{put}$ ) pair; but in a general monoidal closed category the coend instead computes the *linear lens*  $C(S, [V', S'] \otimes V)$ , which is a different notion. It is the pair, not the optic, that Proposition 3.38 identifies with an incoherent  $(M, R)$ -system.

**Remark 3.40.** At first sight this proposition does not appear to be of any use beyond restating the definition in different notation: the general theory of (polymorphic) lenses composes morphisms  $\begin{bmatrix} S \\ S' \end{bmatrix} \rightarrow \begin{bmatrix} V \\ V' \end{bmatrix}$  and  $\begin{bmatrix} V \\ V' \end{bmatrix} \rightarrow \begin{bmatrix} U \\ U' \end{bmatrix}$  only when the target pair  $\begin{bmatrix} V \\ V' \end{bmatrix}$  of the first matches the source pair of the second *exactly* in both coordinates. An incoherent  $(M, R)$ -system based at  $(A, B)$ , read as a lens  $\begin{bmatrix} A \\ B \end{bmatrix} \rightarrow \begin{bmatrix} B \\ B \end{bmatrix}$ , has target pair  $\begin{bmatrix} B \\ B \end{bmatrix}$ ; a subsequent incoherent  $(M, R)$ -system based at  $(B, Y)$  is a lens  $\begin{bmatrix} B \\ Y \end{bmatrix} \rightarrow \begin{bmatrix} Y \\ Y \end{bmatrix}$ , with *source* pair  $\begin{bmatrix} B \\ Y \end{bmatrix} \neq \begin{bmatrix} B \\ B \end{bmatrix}$  unless  $Y = B$ . Consequently the general lens-composition formula (which threads the first  $\text{put}$  through the second,  $\text{put}(s, u') := \text{put}_1(s, \text{put}_2(\text{get}_1(s), u'))$ ), which duplicates  $s$  and therefore presupposes a diagonal, so that it is in any case unavailable in a non-Cartesian  $C$ ) is not even type-correct here, and cannot be invoked to compose incoherent  $(M, R)$ -systems. This explains, from the optics perspective, why the sequential composition  $- \circ -$  of Construction 3.27 had to be defined by hand rather than inherited from the general theory of lenses, and why it turned out to be associative but only left-unital rather than assembling **iMRS** into a genuine category of lenses.

In addition to the functors of Remark 3.20, it would be desirable to have functors  $[-]_f$  and  $[-]_\Phi$ , selecting the process and the repair map of an  $(M, R)$ -system, respectively, i.e. some correspondence of type  $[\mathcal{M}]_f = f$  and  $[\mathcal{M}]_\Phi = \Phi$  if  $\mathcal{M} = \langle\langle f, \Phi \rangle\rangle$ ; however, due to the variance of the diagrams in (3.7), the best one can do starting from the category **iMRS** is to define a span

$$C^{[0 \rightarrow 1]} \xleftarrow{[-]_f} \mathbf{iMRS} \quad \mathbf{iMRS}^A \xrightarrow{[-]_\Phi} C^{[0 \rightarrow 1]} \quad (3.65)$$

In particular, one can destructure an  $(M, R)$ -system into its repair map, functorially, only if the domain of the  $(M, R)$ -system is fixed.

This is undesirable, and a possible adjustment is to resort to a modified version of the category **iMRS**.

**Definition 3.41** (The category of incoherent twisted elements,  $\mathcal{E}(\mathbf{iMRS})$ ). The category of *twisted elements* for incoherent  $(M, R)$ -systems, denoted  $\mathcal{E}(\mathbf{iMRS})$ , is defined having

- objects the  $(M, R)$ -systems as in Definition 3.5;
- arrows the *twisted* morphisms of  $(M, R)$ -systems:  $\langle\langle f, \Phi \rangle\rangle \rightarrow \langle\langle g, \Psi \rangle\rangle$  consists of a pair  $\ell : A' \rightarrow A, r : B \rightarrow B'$  such that (equationally)

$$r \circ f \circ \ell = g \quad \Psi \circ r = [\ell, r] \circ \Phi \quad (3.66)$$

or diagrammatically, the following square and twisted square both commute.

$$\begin{array}{ccccc} A & \xrightarrow{f} & B & \xrightarrow{\Phi} & [A, B] \\ \uparrow \ell & \text{(i)} & \downarrow r & \text{(ii)} & \downarrow [\ell, r] \\ A' & \xrightarrow{g} & B' & \xrightarrow{\Psi} & [A', B'] \end{array} \quad (3.67)$$

It is immediate to verify that

**Proposition 3.42** ( $\mathcal{E}(\mathbf{iMRS})$ ).  $\mathcal{E}(\mathbf{iMRS})$  is a category.

**Remark 3.43.** The category  $\mathcal{E}(\mathbf{iMRS})$  fits in a pullback diagram similar to (3.61),

$$\begin{array}{ccccc} \mathcal{E}(\mathbf{iMRS}) & \longrightarrow & \mathcal{Y} & \longrightarrow & C^{[0 \rightarrow 1]} \\ \downarrow & \lrcorner & \downarrow & \lrcorner & \downarrow \text{dom} \\ X & \longrightarrow & C \times_{\otimes} C & \xrightarrow{p_{\otimes}} & C \\ \downarrow & \lrcorner & \downarrow V & & \\ \mathbf{Tw}(C) & \xrightarrow{\text{cod}} & C & & \end{array} \quad (3.68)$$

(where  $\text{cod}$  is a small abuse of notation for the composite functor  $\mathbf{Tw}(C) \rightarrow C^{\text{op}} \times C \xrightarrow{\pi'} C$ ) but we have to be content with a weaker statement: the twisted arrow category of  $C$  might not inherit colimits from  $C$  (a blatant example is  $C = \mathbf{Set}$ , where  $\mathbf{Tw}(\mathbf{Set})$  lacks an initial object).

**Proposition 3.44** ( $\mathcal{E}(\mathbf{iMRS})$ ). There is a span of “projection” functors

$$\mathbf{Tw}(C) \xleftarrow{\lambda - \mathfrak{f}_f} \mathcal{E}(\mathbf{iMRS}) \xrightarrow{\lambda - \mathfrak{f}_\Phi} C^{[0 \rightarrow 1]} \quad (3.69)$$

defined on objects respectively as  $\lambda f, \Phi \mathfrak{f}_f = f$  and  $\lambda f, \Phi \mathfrak{f}_\Phi = \Phi$ , and mapping morphisms to the squares (i) and (ii) in (3.67).

The theory of coherent  $(M, R)$ -systems will yield another ‘ $[-]_\Phi$  accessor’, defined only on a category where the typical object is not just an arrow  $u : X \rightarrow Y$ , but a pair  $(A, \Phi : \text{id} \Rightarrow [A, -])$  (the resulting category **Rep** is the *fibration of repair maps*). In Definition 4.28 we will construct a coreflective subcategory

$$J : \mathbf{Rep} \hookrightarrow \mathbf{iMRS} \quad (3.70)$$

of the fibration of repairs into the total category of incoherent  $(M, R)$ -systems, whose right adjoint is a functor in the opposite direction, defined on objects by sending  $\langle\langle f, \Phi \rangle\rangle$  to  $\Phi$  (the definition of **Rep** is engineered to make  $K$  functorial).

**Remark 3.45** (Functorial abstraction of incoherent  $(M, R)$ -systems). We can devise the following very general abstraction of  $(M, R)$ -system; let now  $C$  be any category, and  $F : C \rightarrow C$  an endofunctor which admits a right adjoint  $R$ ; clearly, this abstracts the situation where  $C$  is monoidal closed and  $(F \dashv R)$  is the parametric adjunction  $- \otimes A \dashv [A, -]$ . An  $F$ -( $M, R$ )-system now consists of a pair

$$A \xrightarrow{f} B \quad FB \xrightarrow{\Phi} B \quad (3.71)$$

where  $f$  is any arrow of  $C$ , and  $(B, \Phi)$  (in uncurried form) is an  $F$ -algebra (equivalently, we can take a  $\Phi'$  in curried form, an arrow of  $C$  of type  $B \rightarrow RB$ , making  $B$  into an  $R$ -coalgebra).

There is a total category  $\mathbf{iMRS}_{[F]}$  having the tuples  $(A, B; \langle f, \Phi \rangle_F)$  of the above type as objects, and where a morphism  $(A, B; f, \Phi) \rightarrow (A', B'; g, \Psi)$  consists of the obvious choice of arrows  $u : A' \rightarrow A, v : B \rightarrow B'$  such that  $v \circ f \circ u = g$  and in addition,  $v$  is an  $F$ -algebra homomorphism.

**3.2. Cartesian incoherent  $(M, R)$ -systems.** We now specialise to the case of a Cartesian category  $C$ . The additional structure supplies canonical ‘trivial’ systems and comparison functors that are unavailable in the general monoidal setting.

**Definition 3.46** (Cartesian incoherent  $(M, R)$ -system). When  $C$  is a Cartesian monoidal category, i.e.  $\otimes = \times$  is the categorical product and  $[A, B]$  is the actual exponential object  $B^A$ , we refer to an incoherent  $(M, R)$ -system of Definition 3.5 simply as a *Cartesian incoherent  $(M, R)$ -system*; unlike the general monoidal case,  $C$  is automatically symmetric and equipped with diagonal maps  $\Delta_X : X \rightarrow X \times X$  and projections to the terminal object  $1$ , both of which we will use below to construct the ‘trivial’  $(M, R)$ -systems.

**Definition 3.47** (Trivial and identity  $(M, R)$ -systems). Let  $C$  be a Cartesian category, and let  $A$  be an object of  $C$ . The *terminal  $(M, R)$ -system* based at  $A$ , denoted  $\mathcal{M}_A^\top$  (or just  $\mathcal{M}^\top$  for short), is the incoherent  $(M, R)$ -system  $\langle !_A, \Phi_A \rangle$  where  $!_A : A \rightarrow 1$  is the unique arrow to the terminal object, and  $\Phi_A : 1 \rightarrow [A, 1] \cong 1$  is the unique arrow to the internal hom.

By a similar token, the *identity  $(M, R)$ -system* based at  $A$ , denoted  $\mathcal{M}_A^0$ , is the incoherent  $(M, R)$ -system  $\langle 1, \Phi^0 \rangle$  where  $\Phi^0 : A \rightarrow [A, A]$  is the curried projection (which means  $\Phi^0 = \lambda xy.x$  in lambda notation).

**Lemma 3.48.** For every object  $A \in C$ , the terminal  $(M, R)$ -system  $\mathcal{M}_A^\top$  based at  $A$  is the terminal object of the fibre  $\mathbf{iMRS}^A$  of Definition 3.21.

*Proof.* Let  $\langle g, \Psi \rangle$  be any incoherent  $(M, R)$ -system based at  $(A, B)$ . The unique arrow  $B \rightarrow 1$  to the terminal object of  $C$  is automatically the unique candidate arrow  $v : (B; \langle g, \Psi \rangle) \rightarrow (1; \langle !_A, \Phi_A \rangle)$  in  $\mathbf{iMRS}^A$ : the first commutativity condition  $v \circ g = !_A$  holds because  $1$  is terminal in  $C$ , so any two arrows  $A \rightarrow 1$  coincide; the second condition  $[A, v] \circ \Psi = \Phi_A \circ v$  holds for the same reason, since  $[A, 1]$  is terminal too (being isomorphic to  $1$ ), so any two arrows into it agree. Hence  $\mathcal{M}^\top$  receives a unique arrow from every object of  $\mathbf{iMRS}^A$ , i.e. it is terminal.  $\square$

The terminal system  $\mathcal{M}_A^\top$  has output  $1$ . The identity system  $\mathcal{M}_A^0$  instead has process  $1_A$  and repair  $\Phi^0 = \lambda xy.x$ , whose uncurried map is the first projection  $A \times A \rightarrow A$ . Extending this constant repair family to arbitrary output objects gives the comparison functors below.

The presence of the trivial  $(M, R)$ -systems allows one to prove the existence of functors that we do not have in the purely monoidal case, and that are adjoint to the ‘destructors’ defined in (3.30), where we have established the existence of a span

$$C \xleftarrow{[-]_A} \mathbf{iMRS} \xrightarrow{[-]_B} C$$

**Theorem 3.49** ( $\hookrightarrow$ ). There exists a functor  $L : C^{[0 \rightarrow 1]} \rightarrow \mathbf{iMRS}$  defined as follows:

- on objects,  $L \left[ \begin{smallmatrix} A \\ \downarrow \\ B \end{smallmatrix} \right] f$  is the  $(M, R)$ -system  $\langle f, \Phi_{A,B}^0 \rangle$ , where  $\Phi_{A,B}^0 : B \rightarrow [A, B]$  is the *constant repair map*, i.e. the transpose of the projection  $B \times A \rightarrow B$ ; in lambda notation  $\Phi_{A,B}^0 = \lambda xy.x$ . This is the evident extension to a variable codomain of the identity  $(M, R)$ -system’s repair map  $\Phi^0$  of Definition 3.47, which is the case  $B = A$ ;

- on arrows,  $L$  sends a morphism  $f \rightarrow g$  in  $C^{[0 \rightarrow 1]}$ ,

$$\begin{array}{ccc} A & \xrightarrow{f} & B \\ h \downarrow & & \downarrow k \\ X & \xrightarrow{g} & Y \end{array} \quad (3.72)$$

to the same square, carrying the constant repair maps along: the compatibility square for  $\Phi^0$  commutes because both composites discard their  $A$ -argument.

**Theorem 3.50** ( $\mathcal{C}$ ). There exists a functor  $L' : \mathbf{Tw}(C) \rightarrow \mathcal{E}(\mathbf{iMRS})$  defined as follows:

- on objects,  $L' \left[ \begin{smallmatrix} A \\ \downarrow \\ B \end{smallmatrix} \right] f$  is the  $(M, R)$ -system  $\langle f, \Phi_{A,B}^0 \rangle$  of Theorem 3.49;
- on arrows,  $L'$  sends a twisted morphism  $f \rightarrow g$  in  $\mathbf{Tw}(C)$ ,

$$\begin{array}{ccc} A & \xrightarrow{f} & B \\ h \uparrow & & \downarrow k \\ X & \xrightarrow{g} & Y \end{array} \quad (3.73)$$

to the same twisted square, again carrying the constant repair maps along.

Observe that then we are in the situation depicted by the diagram

$$C^{[0 \rightarrow 1]} \begin{array}{c} \xrightarrow{L} \\ \xleftarrow{[-]_f} \end{array} \mathbf{iMRS} \quad \mathbf{Tw}(C) \begin{array}{c} \xrightarrow{L'} \\ \xleftarrow{[-]_f} \end{array} \mathcal{E}(\mathbf{iMRS}) \quad (3.74)$$

It is now natural to conjecture that these two pairs form adjunctions (in fact, coreflections, since  $L, L'$  are fully faithful and their definition suggests they are left adjoints). However, one can check that  $L$  and  $L'$  fall just short of satisfying the adjunction property.<sup>7</sup>

**Proposition 3.51** (Semisimplicial structure on  $\mathbf{iMRS}^{(n)}$ ,  $\mathcal{C}$ ). The assignment that sends a finite nonempty ordinal  $[n]$  to the category  $\mathbf{iMRS}^{(n)}$  of Remark 3.26 has a canonical semisimplicial structure, at least in the low degrees

$$C \begin{array}{c} \xleftarrow{\quad} \\ \xrightarrow{\quad} \end{array} \mathbf{iMRS} \begin{array}{c} \xleftarrow{\quad} \\ \xrightarrow{\quad} \end{array} \mathbf{iMRS}^{(2)} \quad \dots \quad (3.75)$$

More precisely, let  $\Delta_{\text{inj}}$  be the category of finite nonempty total orders, and *injective* monotone maps; the categories  $\mathbf{iMRS}^{(n)}$  for  $n = 0, 1, 2$  (under the convention that  $\mathbf{iMRS}^{(0)} = C$ ), together with suitably defined face and degeneracy functors, form a *2-truncated semisimplicial object*  $\Delta_{\text{inj}}^{\text{op}} \rightarrow \mathbf{Cat}$ .

The natural choice of degeneracies does not, however, make this into a full simplicial object in the category of all incoherent  $(M, R)$ -systems: there seems to be no way to extend the above diagram by filling in the dotted arrow with functors satisfying all simplicial identities.

*Proof.* Represent an incoherent  $(M, R)$ -system in the usual way as

$$A \xrightarrow{f} B \xrightarrow{\Phi} [A, B] \quad (3.76)$$

A 2-simplex  $A \xrightarrow{\xi} B \xrightarrow{\eta} C$  in  $\mathbf{iMRS}^{(2)}$  consists of a composable pair that we can depict as in (3.44).

<sup>7</sup>The obstruction is easy to locate: the counit at an  $(M, R)$ -system  $\langle f, \Phi \rangle$  would be a morphism  $\langle f, \Phi_{A,B}^0 \rangle \rightarrow \langle f, \Phi \rangle$  over the identities, and this forces  $\Phi = \Phi_{A,B}^0$ . Passing to *coherent*  $(M, R)$ -systems does repair it, but only at a price: the same computation then requires  $\text{Nat}(\text{id}, [A, -])$  to be a singleton for every  $A$ , i.e. that the Bernstein centre of  $C$  be a terminal object (Proposition 4.19); in the Cartesian case this is exactly the condition that  $\mathfrak{z}C \cong 1$ , a condition guaranteed by well-pointedness, though strictly weaker than it, cf. Remark 4.6. That is a severe restriction, and it is exactly the hypothesis that Proposition 4.23 shows one must abandon in order to have any nontrivial coherent  $(M, R)$ -system at all. The adjunction is therefore available only in the setting where the objects it relates are all trivial, and we do not pursue it.



The two outer faces select  $\xi$  and  $\eta$ , while the middle face is given by

$$\xi \circ \eta = (g \circ f, [f, 1_C]_1 \circ \Psi). \quad (3.77)$$

The three face–face identities assert only that this composite has source  $A$  and target  $C$ , and that the two original edges have the common endpoint  $B$ . These identities hold naturally. Composition is also associative up to the canonical equality supplied by functoriality of the internal hom, since

$$[f, 1]_1 \circ [g, 1]_1 \circ \Omega = [g \circ f, 1]_1 \circ \Omega. \quad (3.78)$$

The current Agda formalisation verifies this through degree 2.

In the Cartesian case, let  $\kappa_{A,B}$  be the transpose of the projection, i.e.

$$\kappa_{A,B}: B \longrightarrow [A, B], \quad \kappa_{A,B}(b)(a) = b, \quad (3.79)$$

and let  $\mathbf{1}_A = (1_A, \kappa_{A,A})$  be the evident degenerate edge. One unit identity holds:  $\mathbf{1}_A \circ \xi \cong \xi$ , as is immediate to check.

The other identity does not. Indeed,

$$\xi \circ \mathbf{1}_B = (f, [f, 1_B]_1 \circ \kappa_{B,B}) = (f, \kappa_{A,B}), \quad (3.80)$$

which is independent of the original repair map  $\Phi$ . The simplicial identity  $d_1^2 s_0^1 \cong \text{id}_{\mathbf{MRS}}$  would require every incoherent repair map to satisfy

$$\Phi(b)(a) = b. \quad (3.81)$$

This is false in general; we will see in Lemma 4.2 that for coherent  $(\mathbf{M}, \mathbf{R})$ -systems in a Cartesian category this is, however, the only possible choice.<sup>8</sup>

To see a concrete counterexample, take  $A = \{*\}$ ,  $B = \{0, 1\}$  in the category **Set** of sets and functions; let  $f(*) = 0$ , and define  $\Phi(b)(*) = 0$ . The normalised repair satisfies  $\kappa_{A,B}(1)(*) = 1$ , so the original system and its right-unit composite are not equal.  $\square$

**Remark 3.52** (Possible corrections). One may simply omit the degeneracies and use the semi-simplicial object described above. There are several ways to remove the obstruction, depending on which information one wishes to retain.

SE1) If one studies  $(\mathbf{M}, \mathbf{R})$ -systems with trivial repair function, where  $\Phi = \kappa_{A,B}$ , one essentially obtains the ordinary nerve of the arrow category  $C^{[0 \rightarrow 1]}$ . We will see in Lemma 4.2 that for Cartesian coherent  $(\mathbf{M}, \mathbf{R})$ -systems in the category of sets this is the only possible choice.

SE2) One may enlarge the notion of simplex. For example, a 2-simplex may consist of a complete triangle

$$(\xi_{01}, \xi_{12}, \xi_{02}), \quad U\xi_{02} = U\xi_{12} \circ U\xi_{01}, \quad (3.82)$$

where  $U$  forgets the repair map and the repair carried by  $\xi_{02}$  is stored independently. The three faces are then the three projections, and the degeneracies are  $s_0(\xi) = (\xi, \mathbf{1}, \xi)$ ,  $s_1(\xi) = (\mathbf{1}, \xi, \xi)$ . In higher degrees one stores an independently decorated edge  $\xi_{ij}$  for every  $i < j$ , and faces and degeneracies act by reindexing. This produces a natural candidate for a genuine simplicial object retaining arbitrary repairs, at the cost of no longer requiring the diagonal repair to be computed from the two adjacent repairs.

**Proposition 3.53** ( $(\mathbf{M}, \mathbf{R})$ -systems as dynamical systems in a coKleisli category). In a Cartesian category, uncurrying gives  $f: A \rightarrow B$  and  $\hat{\Phi}: B \times A \rightarrow B$ . Since  $S_A = \_ \times A$  is a comonad, these are coKleisli maps  $S_A 1 \rightarrow B$  and  $S_A B \rightarrow B$ , respectively, i.e. arrows

$$1 \xrightarrow{f} B \quad B \xrightarrow{\Phi} B \quad (3.83)$$

in  $\mathbf{coKl}(S_A)$ ; thus,  $\Phi$  is a dynamical system, one orbit of which is the sequence  $\{f, \Phi \cdot f, \Phi \cdot (\Phi \cdot f), \dots\}$  (here  $\cdot$  denotes coKleisli composition); we can think of  $(\mathbf{M}, \mathbf{R})$ -systems as *parametric algebras* of some sort, in the sense that follows: first, observe that one can lift the “Maybe”

<sup>8</sup>It is worth recording how narrowly the construction misses: of the thirteen identities of a truncated simplicial object, twelve hold, and  $d_1^2 s_0^1 \cong 1$  is the only casualty. This is not an impression but a checked statement — the Agda development builds the record carrying all twelve,  $\mathcal{U}$ , alongside the semisimplicial object of Proposition 3.51 that omits the degeneracies altogether.

endofunctor  $1 + \_$  to  $\mathbf{coKl}(S_A)$ , because for every  $A$  there exists a natural transformation  $S_A(1 + \_) \rightarrow 1 + S_A \_$  whose components are

$$A \times (1 + X) \cong A + A \times X \xrightarrow{!+A \times X} 1 + A \times X, \quad (3.84)$$

and which is compatible with the comonad structure. Thus, we get a lifting  $1 +_{[A]} \_$  filling the square<sup>9</sup>

$$\begin{array}{ccc} \mathbf{coKl}(S_A) & \xrightarrow{1+_{[A]} \_} & \mathbf{coKl}(S_A) \\ \uparrow & & \uparrow \\ C & \xrightarrow{1+ \_} & C \end{array} \quad (3.85)$$

into a commutative one (the vertical arrows being the cofree functors), together with an identification  $\mathbf{iMRS}^A \cong \mathbf{Alg}(1 +_{[A]} \_)$ .

Before turning to coherent systems in section 4, we give a computational reading of the Cartesian case. An  $(M, R)$ -system can be represented by the datatype

```
data MRS a b = MRS
  { f    :: a -> b
  , phi  :: b -> (a -> b)
  }
```

Repair can then be read as an ‘unfold’ seeded by a function. The connection with machines, developed by Rosen [Ros64a] and Arbib [Arb66], is treated in section 6. Here we use Haskell notation [Has, Ell17] for intuition: although unrestricted Haskell does not form a Cartesian closed category [Bau16], the categorical reasoning applies to its total fragment [DHJG06].

**Definition 3.54.** We read an *MRS record*

$$\text{MRS } A \ B = (f : A \rightarrow B, \phi : B \rightarrow (A \rightarrow B)) \quad (3.86)$$

as a *two-stage, feedback-sensitive* computation. The field  $f$  produces a provisional output  $x = f(a)$ ; the post-processor  $\phi$  then returns a revised process  $\phi(x) : A \rightarrow B$ . We also write  $\phi(b, a)$  for  $\phi(b)(a)$ .

Iterating this update defines maps  $k^{(n)} : A \rightarrow B$ :

$$k^{(0)} := f \quad k^{(n+1)} = \lambda a. \phi(k^{(n)}(a))(a) \quad (3.87)$$

Each output thus revises the next process.

Uncurrying–recurring (Definition 2.19) gives an  $A$ -indexed family of endomorphisms

$$\phi_a : B \rightarrow B, \quad \phi_a(b) = \phi(b, a), \quad (3.88)$$

For a fixed  $b_0$ , the same map gives the process  $a \mapsto \phi(b_0, a)$ .

This viewpoint can be coupled with the assumption that  $A$  or  $B$  carry more structure, such as that of a monoid or semigroup in **Hask**, to obtain more specific patterns of behaviour. In general,  $\phi$  equips  $B$  with the structure of a right  $A$ -set, i.e. there is a right action of the free monoid  $A^*$  on  $B$ . If such an action has a nontrivial kernel, or if  $A$  is not free, and we assume that  $\phi$  uncurries to an action  $B \times A \rightarrow B$ , the situation becomes more complex and interesting, as the dynamics of the system is constrained by algebraic relations.

**Example 3.55** (Assuming  $B$  is a semigroup / monoid). Let  $*$  be the multiplication of a semigroup  $B$ , and write  $u_B$  for its unit when  $B$  is a monoid. These operations give two simple post-processors:

- A ‘semigroup augment’ pattern: one uses the monoid operation of  $B$  and sets

$$\text{MRS}_h = \begin{cases} f : A \rightarrow B \\ \phi : B \times A \rightarrow B : (x, a) \mapsto x * h(a). \end{cases} \quad (3.89)$$

for some function  $h : A \rightarrow B$  (possibly constant at an element of  $B$ ).

<sup>9</sup>Note that the object 1 remains terminal in  $\mathbf{coKl}(S_A)$ ; this motivates the intuition of the lifting in question as a lifted Maybe monad.

```
semigroupAugment :: Semigroup b =>
  (a -> b) -> (a -> b) -> MRS a b
semigroupAugment base extra =
  MRS
  { f    = base
  , phi = \x a -> x <> extra a
  }
```

Closure requires a single seed  $x_0$  with

$$f(a) = x_0 * \text{extra}(a) \quad \text{for all } a. \quad (3.90)$$

- Similarly, one can define a ‘monoid gate/override’: let  $p : A \rightarrow \mathbf{Bool}$  be a predicate on  $A$ , and set

$$\text{MRS}_p = \begin{cases} f : A \rightarrow B \\ \phi : B \times A \rightarrow B : (x, a) \mapsto \begin{cases} x & \text{if } p(a), \\ u_B & \text{otherwise.} \end{cases} \end{cases} \quad (3.91)$$

Closure forces  $f$  to be constant on the  $p$ -true locus and equal to  $u_B$  on its complement.

Note that in nondegenerate cases ( $f$  nonconstant) the image of the seed map  $\lambda a. \phi(b_0, a)$  is small inside the set of functions  $A \rightarrow B$ , so closedness is a ‘non-generic’ property.

**Example 3.56** (Assuming  $B$  is of coproduct / partial type  $\mathbf{1} + T$ ). The category of sets and partial functions has products and coproducts, so upon uncurrying  $\phi$  into a map  $B \times A \rightarrow B$ , we can study MR systems whose components  $f$  and  $\phi$  are partially defined.

If  $f$  is undefined on an input, repair can supply a fixed fallback value in  $T$ , regarded as a point  $1 \rightarrow T$ . This replaces failure by a default output:

```
repair :: (a -> Maybe t) -> t -> MRS a (Maybe t)
repair base fallback =
  MRS
  { f    = base
  , phi = \mx a -> case mx of
    Just _ -> mx
    Nothing -> Just fallback
  }
```

- Repair:

$$\phi(\text{Nothing}, a) = \text{Just}(\text{fallback}). \quad (3.92)$$

- Validate:

$$\phi(mx, a) = \begin{cases} mx & \text{if } \text{ok}(a), \\ \text{Nothing} & \text{otherwise.} \end{cases} \quad (3.93)$$

```
validate :: (a -> Maybe t) -> (a -> Bool) -> MRS a (Maybe t)
validate base test =
  MRS
  { f    = base
  , phi = \mx a -> if test a then mx else Nothing
  }
```

- Enrich:

$$\phi(\text{Lefte}, a) = \text{Left}(e * \text{ctx}(a)). \quad (3.94)$$

```
enrichLeft :: Semigroup e =>
  (a -> Either e t) -> (a -> e) -> MRS a (Either e t)
enrichLeft base ctx =
  MRS
  { f    = base
  , phi = \ex a -> case ex of
    Left e -> Left (e <> ctx a)
```

**Right**  $y \rightarrow$  **Right**  $y$   
 $\}$

#### 4. COHERENT (M,R)-SYSTEMS

We now ask how to make repair natural while retaining nontrivial examples and gaining a functorial dependence on both the parameters  $A, B$ , something which allows us to use the powerful of fibrations to attach invariants to (M,R)-systems.

There are indeed at least two reasons to seek a more coherent presentation for (M,R)-systems.

- The first motivation is purely mathematical. In Remark 3.20 we constructed a span

$$C \xleftarrow{[-]_A} \mathbf{iMRS} \xrightarrow{[-]_B} C \quad (4.1)$$

whose left leg is a fibration, and whose fibres over  $(A, B)$  are the discrete sets  $\mathbf{iMRS}(A, B)$  of Remark 3.22. However, we then observed that  $[-]_B$  is not an opfibration: the assignment  $B \mapsto \mathbf{iMRS}_B$  is not a covariant functor. It would be desirable to obtain, instead, a *two-sided discrete fibration* of (M,R)-systems—or equivalently (given the equivalence in Theorem 2.21), a profunctor

$$\mathbf{MRS}(-, -) : C^{\text{op}} \times C \longrightarrow \mathbf{Set} : (A, B) \longmapsto \mathbf{MRS}(A, B) \quad (4.2)$$

which depends functorially not only on  $A$ , but also on  $B$ . (Note that we lose the ‘i’ of the  $\mathbf{iMRS}$  in Definition 3.5). If we denote as  $(A, B) \xrightleftharpoons[u]{u} (A', B')$  a morphism of the category  $C^{\text{op}} \times C$ , by which we mean a pair  $u : A' \rightarrow A, v : B \rightarrow B'$ , the fact that  $\mathbf{MRS}$  is a profunctor means that there exists an induced *function*

$$\mathbf{MRS}(A, B) \xrightarrow{\mathbf{MRS}(u,v)} \mathbf{MRS}(A', B') \quad (4.3)$$

between sets. This makes the tools of fibred category theory [Gra66, Jac99, Bén85] available for constructing invariants. Naturality can also supply comparison functors unavailable for incoherent systems and adjunctions that their incoherent counterparts, such as Theorem 3.49 and Theorem 3.50, fail to satisfy.

- The semantic motivation is to replace a single repair map by a *polymorphic family*

$$\forall X. \Phi_X : X \rightarrow [A, X], \quad (4.4)$$

varying naturally with  $X$ . Such a family should be seen as a *blueprint for repair*: its component at  $B = \text{cod} f$  recovers the original map, while its other components prescribe repair at other types.

Evaluating at the output object should recover an incoherent system, ideally through a functor

$$\mathbf{MRS} \xrightarrow{(-)^b} \mathbf{iMRS} \quad (4.5)$$

that ‘flattens’ a coherent (M,R)-system into an incoherent one. We compare several ways to impose naturality. Some are highly restrictive (Lemma 4.2); others fail to produce the desired two-sided fibration. These constraints reflect the connection between naturality, polymorphism, and the Yoneda lemma [Yon60, Str74].

The alternatives displayed in subsection 4.1–subsubsection 4.3.5 below balance these requirements differently.

Some, like the ‘coslice-natural’ systems in ?? have a direct semantic interpretation but do not yield the profunctor we are after; **cod**-natural systems of ?? do, but are as restrictive as **id**-coherent (or just ‘natural’) in Cartesian categories. Profunctorial systems offer a more abstract alternative.

Our purpose here is foundational: to make the alternatives and their consequences explicit. The appropriate choice depends on what a model is intended to describe.

Most of the theory uses the following two notions of coherence.

**Definition 4.1** (Coherent and **cod**-coherent (M,R)-systems). Let  $A, B$  be objects of  $C$ .

- An *id-coherent*  $(M, R)$ -system (or simply a *coherent* one) is a pair  $\langle f, \Phi \rangle$  where  $f : A \rightarrow B$  and  $\Phi : \text{id} \Rightarrow [A, -]$  is a natural transformation,  $\mathcal{U}$ ; thus, one can recover the incoherent  $(M, R)$ -system  $\langle f, \Phi_B \rangle$  by evaluating  $\Phi$  at the component  $B$ ;
- a *cod-coherent*  $(M, R)$ -system is a pair  $\langle f, \Phi \rangle$  where  $f : A \rightarrow B$  and  $\Phi : \text{cod} \Rightarrow [A, \text{cod}]$  is a natural transformation,  $\mathcal{U}$ ; thus, one can recover the incoherent  $(M, R)$ -system  $\langle f, \Phi_B \rangle$  by evaluating  $\Phi$  at the component  $f$  (which is, obviously, an object of  $C^{[0 \rightarrow 1]}$ ).

These notions are equivalent because  $\text{cod} : C^{[0 \rightarrow 1]} \rightarrow C$  is a *lax epimorphism* [LNS22]; see Lemma 4.33. We generally use the object-indexed, id-coherent presentation.

Whenever a coherence condition yields a profunctor, we can apply the tabulator construction of Definition 2.24. This culminates in Definition 4.10, at the end of the first half of the section, establishing that there exists a *fibration of  $(M, R)$ -systems* giving the title to this paper.

In the next section, section 5, we study closure and its unit-retraction criterion. We then turn, in subsection 5.3, to a question persistently pursued by Rosen himself and by several other authors [Ros12, LSAGA<sup>+</sup>06, Lou17]: extending the notion of  $(M, R)$ -system to a sequence of the form

$$A \longrightarrow B \longrightarrow [A, B] \longrightarrow [B, [A, B]] \longrightarrow \dots \quad (4.6)$$

This extension was studied especially in connection with *closure under efficient causation*, a major theme in Rosen's work, to which we intend to apply the language developed in this first paper. We define a system that 'exhibits metabolic closure' in Definition 5.1 and show that having a 'universal closure point' is a sufficient condition for it (Definition 5.3; the converse does not hold in general, since a closure point witnesses the condition at the single component  $\Phi_B$ , while a universal one witnesses it uniformly for processes with the given input object). In Proposition 5.4 we show that reindexing preserves metabolic closure. In subsection 5.4 we distinguish levelwise closure from compatibility of the repair families across a longer system, and prove precise conditions for transporting universal closure points along its connecting maps.

Notably, the extension (4.6) can also be carried out for incoherent  $(M, R)$ -systems, and is thus of independent value even for those who want to work exclusively with incoherent  $(M, R)$ -systems. We focus on the construction of a chain

$$(i)\mathbf{MRS} =: (i)\mathbf{MRS}_2 \longleftarrow (i)\mathbf{MRS}_3 \longleftarrow (i)\mathbf{MRS}_4 \longleftarrow (i)\mathbf{MRS}_5 \longleftarrow \dots \quad (4.7)$$

Both the coherent and incoherent towers carry links to the Agda formalisation.

**4.1. id-coherent  $(M, R)$ -systems.** The simplest possible way to obtain natural repair maps is to regard  $\Phi : B \rightarrow [A, B]$  as a single component, at  $B \in C_0$ , of a natural transformation of type

$$\Phi : \text{id} \Rightarrow [A, -] : C \rightarrow C \quad \text{or in cell form,} \quad C \begin{array}{c} \xrightarrow{\text{id}} \\ \Downarrow \Phi \\ \xrightarrow{[A, -]} \end{array} C. \quad (4.8)$$

An *id-coherent*  $(M, R)$ -system is then a pair  $\langle f, \Phi \rangle$  as above. We denote by  $\text{id-MRS}(A, B)$ , or simply  $\mathbf{MRS}(A, B)$ , the class of id-coherent  $(M, R)$ -systems.

The naturality requirement makes this quite a rigid object; in short, a natural transformation of this type amounts to specifying an  $(- \otimes A)$ -algebra structure on each object  $B \in C_0$ . As a consequence of the following observations, *there are no nontrivial examples of id-coherent  $(M, R)$ -systems in the category of sets.*

**Lemma 4.2** ( $\mathcal{U}$ ). Let  $C = \mathbf{Set}$ , with its usual Cartesian closed structure, and let  $A$  be a set; then the unique natural transformation  $\text{id} \Rightarrow (-)^A$  is the one having components the constant maps  $X \rightarrow X^A : x \mapsto \lambda a. x$ .

*Proof.* For an element  $x \in X$ , regarded as a map  $x : 1 \rightarrow X$  with  $1 = \{\bullet\}$ , naturality of  $\alpha : \text{id} \Rightarrow (-)^A$  gives

$$\begin{array}{ccc} 1 & \xrightarrow{\alpha_1} & [A, 1] \\ x \downarrow & & \downarrow x \circ - \\ X & \xrightarrow{\alpha_X} & [A, X] \end{array} \quad (4.9)$$

that is,  $\alpha_X(x)a = x \circ \alpha_1(\bullet)$  as desired.  $\square$

In a monoidal closed category, evaluation at the unit defines a projection

$$p : [C, C](\text{id}, [A, -]) \rightarrow C(A, I) \quad (\text{U})$$

by uncurrying  $\alpha_I$  and composing with the inverse left unitor:

$$A \xrightarrow{\lambda_A^{-1}} I \otimes A \xrightarrow{\alpha_I \otimes 1_A} [A, I] \otimes A \xrightarrow{\epsilon} I. \quad (4.10)$$

Conversely, an arrow  $a : A \rightarrow I$  defines a natural family  $\iota(a)$  whose component at  $X$  is

$$X \xrightarrow{\eta_X} [I, X \otimes I] \xrightarrow{[I, \rho_X]} [I, X] \xrightarrow{[a, X]} [A, X]. \quad (\text{U})$$

**Lemma 4.3**  $(\text{U})$ . The composite  $p \circ \iota$  is the identity of  $C(A, I)$ .

*Proof.* The composite  $p \circ \iota$  sends an arrow  $a : A \rightarrow I$  of  $C$  to the composite arrow marked in red in the diagram

$$\begin{array}{ccccccc}
 & & & & I \otimes I & & \\
 & & & & \nearrow \lambda_I & & \\
 I & \xrightarrow{\lambda_I^{-1}} & I \otimes I & \xrightarrow{\eta_I \otimes I} & [I, I \otimes I] \otimes I & \xrightarrow{\epsilon_{I \otimes I}} & [I, I] \otimes I \xrightarrow{\lambda_I} I \\
 \uparrow a & & \uparrow I \otimes a & & \uparrow 1 \otimes a & & \uparrow 1 \otimes a \\
 A & \xrightarrow{\lambda_A^{-1}} & I \otimes A & \xrightarrow{\eta_I^{(I)} \otimes A} & [I, I \otimes I] \otimes A & \xrightarrow{[I, \lambda_I] \otimes A} & [I, I] \otimes A \xrightarrow{[a, I] \otimes A} [A, I] \otimes A \xrightarrow{\epsilon_I^{(A)}} I
 \end{array} \quad (4.11)$$

Each square commutes, because of naturality, bifactoriality, or the cowedge condition of  $\epsilon$ ; thus,  $(p \circ \iota)(a) = \lambda_I \circ \lambda_I^{-1} \circ a = a$ , as desired.  $\square$

**Remark 4.4** (Representable-coherence). As an immediate consequence of Lemma 4.3,  $\iota$  is injective (a split monomorphism) and  $p$  is surjective (a split epimorphism); we call a coherent  $(M, R)$ -system *representable* if it lies in the image of  $\iota$ .

**Remark 4.5**  $(\text{U})$ . On the other hand, the composite  $\iota \circ p$  is *not* the identity; we can build a monoidal (even Cartesian) category  $C$  such that  $\iota$  is the inclusion of a singleton in a two-element set. We defer the human-readable proof to Proposition 4.23 below.

An important observation is now in order: despite its size, the literature on  $(M, R)$ -systems that we have been able to consult works, without exception, in a category of sets, where an  $(M, R)$ -system is made of functions; in that setting Lemma 4.2 shows that no nontrivial coherent  $(M, R)$ -system can exist, an obstruction which appears to have gone unremarked.

**Remark 4.6** (On  $(M, R)$ -systems and classical set theory). The key property that allows us to reproduce the argument in Lemma 4.2 is that the terminal object of **Set** is also a *generator*: parallel maps that agree on all global elements are equal, or in other words the functor  $\Gamma = \mathbf{Set}(1, -) : \mathbf{Set} \rightarrow \mathbf{Set}$  is faithful. Fortunately, this condition—called *well-pointedness* in [MLM92, p. 275]—is a strong property for a Cartesian closed category, one we cannot expect in full generality and that rules out interesting examples of  $(M, R)$ -systems.

Indeed, in a (nondegenerate) well-pointed topos  $\mathcal{E}$  the global sections functor  $\Gamma_{\mathcal{E}} = \mathcal{E}(1, -)$  is faithful, so that element-theoretic reasoning is legitimate; every object  $X \neq 0$  admits a global element, since otherwise the two coprojections  $X \rightrightarrows 1 + 1$  would agree vacuously on points; and the topos is necessarily two-valued [MLM92, p. 274] and Boolean [MLM92, VI, Proposition 1]. The constraint is severe enough that, classically, **Set** is the only Grothendieck topos that is well-pointed.<sup>10</sup>

<sup>10</sup>A note for the logicians: this is not a purely elementary phenomenon: well-pointedness together with an NNO and choice axiomatises Lawvere's ETCS, which has many inequivalent models, so a well-pointed topos is better thought of as a *model of a certain classical set theory* than as **Set** itself. It is the non-elementary hypothesis of cocompleteness that collapses the distinction.

Examples of non-well-pointed toposes are correspondingly abundant. In the topos  $\mathbf{Set}^G$  of  $G$ -sets for a nontrivial group  $G$ , the global elements of an object are its fixed points, so any free action gives an object with no points at all. In Proposition 4.23 we will indeed use a category of the form  $\mathbf{Set}^G$  (with  $G$  the cyclic group of order 2) to build the counterexample of an  $(M, R)$ -system alluded to in Remark 4.5.<sup>11</sup> As a rather informal aside, there is a line of research in the foundations of quantum mechanics that relates the absence of global points to the failure of a quantum system to admit a classical presentation, and that finds a natural home for such systems in a universe whose internal logic is intuitionistic.<sup>12</sup>

By contrast, coherent  $(M, R)$ -systems are relatively abundant in linear settings: let's consider a toy example in the category of vector spaces.

**Corollary 4.7.** Fix a field  $k$ ; in the category of vector spaces (so, allowing infinite-dimensional ones) over  $k$ , the set of coherent  $(M, R)$ -systems  $\langle f, - \rangle$  such that  $[f, -]_f = f$  can be big, and in particular it can be bigger than  $A$  in a way that can be estimated; for example, consider the  $(M, R)$ -system

$$A \xrightarrow{f} k \quad (-) \xRightarrow{\Phi} [A, -] \quad (4.12)$$

of vector spaces, where  $A = k[T]$  is the polynomial ring in one indeterminate; then,  $\mathbf{Vect}(A, k) \cong k[[T]]$  is the ring of formal power series over  $k$ . There is then a unique  $(M, R)$ -system for every formal power series  $q(T) = \sum a_n T^n$ , determined as the natural transformation  $\Phi^q : \text{id} \Rightarrow [A, -]$  with components  $\Phi_X^q(x) := (p \mapsto (\sum_n a_n p_n)x) \in [A, X]$  for  $x \in X$ , using the coefficients  $p = \sum_n p_n T^n$  of a polynomial  $p \in A$ ; naturality in  $X$  is immediate, since for every linear map  $g : X \rightarrow Y$ ,  $g(\Phi_X^q(x)(p)) = (\sum_n a_n p_n)g(x) = \Phi_Y^q(g(x))(p)$ . Conversely, every  $\text{id}$ -coherent  $\Phi : \text{id} \Rightarrow [A, -]$  arises this way: evaluating the component  $\Phi_k$  at  $X = k$  recovers a functional  $q := \Phi_k(1) \in \mathbf{Vect}(A, k) \cong k[[T]]$ , and naturality (applied to the map  $k \rightarrow X$ ,  $1 \mapsto x$ ) forces  $\Phi_X(x) = \Phi_k(1) \cdot x$  for every  $X$  and  $x \in X$ , an instance of the general fact that, for  $F$  representable, natural transformations  $\text{id} \Rightarrow F$  correspond bijectively to elements of  $F(k)$ .

(The remark above, sketching an example in  $\mathbf{Chu}(\mathbf{Vect}_k, k)$ , is *not* an instance of the present notion: as stated in its own text it produces an *incoherent*  $(M, R)$ -system in the different category  $\mathbf{Chu}(\mathbf{Vect}_k, k)$ , fixed at a single pair rather than natural in a varying target as required here. It is left commented out, and would belong instead in section 3 as a further Chu-space example alongside the Hilbert-space one, should it be developed further.)

<sup>11</sup>A second note for the logicians: more generally, no presheaf topos  $[C^{\text{op}}, \mathbf{Set}]$  over a category with more than one object is well-pointed—witness the Sierpiński topos  $\mathbf{Set}^{\rightarrow}$ , simplicial sets, or the topos of graphs. Neither is a topos of sheaves  $\mathbf{Sh}(X)$  for all but the most degenerate spaces  $X$ : already  $X$  a two-point discrete space gives  $\mathbf{Sh}(X) \simeq \mathbf{Set} \times \mathbf{Set}$ , which fails to be two-valued. Realisability toposes such as the effective topos  $\mathbf{Eff}$  (see [vO08]) fail for a different reason, being neither Boolean nor two-valued.

<sup>12</sup>More precisely, here's a footnote for the physicists: in topos-theoretic approaches to basic quantum mechanics like [IB98, DI08, HLS09], one replaces the noncommutative algebra  $\mathcal{A}$  of observables of a quantum system with the poset  $\mathcal{V}(\mathcal{A})$  of its commutative subalgebras, thought of as the 'classical contexts', i.e. those families of observables that admit a joint measurement. With this, it attaches to  $\mathcal{A}$  the *spectral presheaf*  $\underline{\Sigma} : \mathcal{V}(\mathcal{A})^{\text{op}} \rightarrow \mathbf{Set}$  assigning to each context its *Gel'fand spectrum*. The Kochen–Specker theorem [KS67] then reads, in this language, as the assertion that for  $\mathcal{A} = B(\mathcal{H})$  with  $\mathcal{H}$  a Hilbert space of dimension at least three, the object  $\underline{\Sigma}$  admits no global sections  $1 \rightarrow \underline{\Sigma}$ . Thus, a quantum system carries no consistent global assignment of values to all its observables, only a coherent family of local, contextual ones. Our Lemma 4.2 records the same tension read from the other side, so that both settings are pushed outside  $\mathbf{Set}$  for what is formally the same reason. A bridge between Rosen and this tradition has in fact been attempted by [Zaf12], who transports the apparatus onto Rosen's *modelling relation* rather than onto  $(M, R)$ -systems (no metabolism, repair map, or closure occurs there): *op. cit.* introduces a category  $\mathcal{Z}$  of 'complex information structures' having objects partially commutative unital rings of observables, which is shown to be equivalent to a topos of sheaves over a small base  $\mathcal{Y}$  of commutative rings. The mathematics underlying that equivalence is the well-known *nerve–realisation adjunction*: for any  $M : \mathcal{Y} \rightarrow \mathcal{Z}$  with  $\mathcal{Y}$  small, the nerve  $N_M = \mathcal{Z}(M \_, =) : \mathcal{Z} \rightarrow [\mathcal{Y}^{\text{op}}, \mathbf{Set}]$  exists unconditionally, and acquires a left adjoint  $\text{Lan}_{\mathcal{Y}} M$  precisely when  $\mathcal{Z}$  is cocomplete—a hypothesis *op. cit.* imposes on  $\mathcal{Z}$  explicitly, citing [MLM92], where the adjunction itself is proved, while the restriction to an equivalence onto sheaves is the density of  $M$  for the chosen topology. Given this, the approach in *op. cit.* seems extendable to other settings: no step of that construction uses the ring structure, let alone its commutativity; the choice is motivated by quantum theory, where observables attached to a system form algebras, rather than by anything in Rosen. In any case, Zafiris' work seems to have no relation with  $(M, R)$ -systems, so we take this merely as an interesting aside.

**Remark 4.8** (id-coherent  $(M, R)$ -systems and ends). Assume  $C$  is symmetric, or at least biclosed.

Under the uncurrying-recurring operation of Definition 2.19 we can transform the *natural* transformation  $\Phi^\natural : \text{id} \Rightarrow [A, -]$  into a *dinatural* transformation with components

$$\Phi_X^\natural : A \longrightarrow [X, X]. \quad (4.13)$$

This is a *wedge* in the sense of the dual of (2.21). But then, any id-coherent  $(M, R)$ -system can be uncurried-recurring and turned into a single ‘classifying map’

$$\bar{\Phi}^\sharp : A \longrightarrow \int_X [X, X] \quad (4.14)$$

with codomain the *end* of  $[-, -] : C^{\text{op}} \times C \rightarrow C$  (of course, provided the limit that defines  $\int_X [X, X]$  exists in  $C$ ). Note that  $\int_X [X, X]$  is a  $\otimes$ -monoid in  $C$ , where the multiplication is the map

$$\left( \int_X [X, X] \right) \otimes \left( \int_Y [Y, Y] \right) \longrightarrow \int_{XY} [X, X] \otimes [Y, Y] \xrightarrow{(\star)} \int_Z [Z, Z] \quad (4.15)$$

where  $(\star)$  is induced by  $\int_{XY} [X, X] \otimes [Y, Y] \xrightarrow{\pi_{ZZ}} [Z, Z] \otimes [Z, Z] \xrightarrow{\circ} [Z, Z]$  and the unit  $I \rightarrow \int_X [X, X]$  is induced by uncurrying and recurring the unitors; but then, again assuming the colimit that defines  $A^* = \sum_n A^{\otimes n}$  exists in  $C$ , we get that

Any id-coherent  $(M, R)$ -system on a category  $C$  that admits the end  $\int_X [X, X]$  and countable coproducts induces a monoid homomorphism

$$(\bar{\Phi}^\sharp)^* : A^* \longrightarrow \int_X [X, X]. \quad (4.16)$$

**Definition 4.9** (The profunctor of coherent  $(M, R)$ -systems,  $\mathcal{C}^{\text{co}}$ ). The assignment  $(A, B) \mapsto \mathbf{MRS}(A, B)$  defines a profunctor

$$\mathbf{MRS} : C^{\text{op}} \times C \longrightarrow \mathbf{Set} \quad (4.17)$$

that we call the *profunctor of (id-coherent)  $(M, R)$ -systems*.

To verify functoriality, let  $u : A' \rightarrow A$  and  $v : B \rightarrow B'$ . Define

$$\mathbf{MRS}(u, v) \langle f, \Phi \rangle = \langle v \circ f \circ u, u^* \Phi \rangle, \quad (u^* \Phi)_X = [u, X] \circ \Phi_X. \quad (4.18)$$

Naturality of  $\Phi$  and bifactoriality of the internal hom make  $u^* \Phi$  natural. The same bifactoriality shows that reindexing preserves identities and composition.

**Definition 4.10** (The fibration of  $(M, R)$ -systems,  $\mathcal{C}^{\text{fib}}$ ). To the profunctor of id-coherent  $(M, R)$ -systems one associates the *tabulator* (see Definition 2.24)  $\otimes \mathbf{MRS}$  i.e. the category having

- objects are triples  $(A, B; \xi)$  where  $A, B \in C_0$  and  $\xi = \langle f, \Phi \rangle \in \mathbf{MRS}(A, B)$  is a coherent  $(M, R)$ -system based at  $(A, B)$ ;
- arrows  $(A, B; \langle f, \Phi \rangle) \rightarrow (A', B'; \langle g, \Psi \rangle)$  are pairs  $(u, v)$  with  $u : A' \rightarrow A$ ,  $v : B \rightarrow B'$  in  $C$  (i.e. exactly a morphism  $(A, B) \rightarrow (A', B')$  of  $C^{\text{op}} \times C$ ) such that  $\mathbf{MRS}(u, v)(\langle f, \Phi \rangle) = \langle g, \Psi \rangle$ , i.e.  $v \circ f \circ u = g$  and  $u^* \Phi = \Psi$  (Definition 4.9).

By a similar token, we can define

**Definition 4.11** ( $\mathcal{C}^{\text{tw}}$ ). The category of twisted elements of the profunctor in Definition 4.9 is the category  $\mathcal{E}(\mathbf{MRS})$  having

- objects are the same triples  $(A, B; \langle f, \Phi \rangle)$  as above;
- arrows  $(A, B; \langle f, \Phi \rangle) \rightarrow (A', B'; \langle g, \Psi \rangle)$  are pairs  $(u, v)$  with  $u : A \rightarrow A'$  (reversed relative to the total category above, mirroring Definition 3.41) and  $v : B \rightarrow B'$ , such that  $v \circ f \circ u = g$  and  $[u, -] \circ \Phi = \Psi$  as natural transformations (which means, of course, componentwise equal).

Similarly to what happens for incoherent  $(M, R)$ -systems, there are equivalent presentations for Definition 4.1, useful to have at hand while working.



**Remark 4.12.** Given that  $f : A \rightarrow B$  is an object of  $C^{[0 \rightarrow 1]}$ , a coherent  $(M, R)$ -system can be seen as a *pointed* natural transformation  $\Phi : \text{id} \Rightarrow [A, \text{id}]$ ; thus, one can equivalently present the data of Definition 4.1 above as just a natural transformation  $\Phi$  of the type above, with a distinguished component  $\Phi_f : B \rightarrow [A, B]$  as another arrow of  $C$ .

Evaluation at the output object gives the ‘flattening’ functor

$$(-)^b : \mathbb{O}\mathbf{MRS} \longrightarrow \mathbf{iMRS}. \quad (4.19)$$

**Remark 4.13.** With the type given in Definition 4.1, and by the universal property of  $\mathbf{coAlg}[A, -]$  as an inserter (cf. Definition 2.7),  $\Phi$  corresponds in turn to a unique functor

$$C \longrightarrow \mathbf{coAlg}[A, -]. \quad (4.20)$$

Clearly,  $b\Phi = \Phi_f$  is a  $[A, -]$ -coalgebra. One may therefore regard  $\mathbf{MRS}(A, B)$  as the set of ‘basepoint-preserving’ functors

$$(C, f : A \rightarrow B) \longrightarrow (\mathbf{coAlg}[A, -], \Phi_B : B \rightarrow [A, B]). \quad (4.21)$$

Equally clearly,  $\Phi$  may be regarded as an  $(A \otimes -)$ -algebra, and dual considerations apply. We shall freely use either characterisation, according to which is more convenient in the discussion.

As soon as  $C$  is symmetric monoidal closed, Remark 4.8 absorbs the entire repair datum of an  $(M, R)$ -system into a single arrow with a fixed codomain. That codomain deserves a name.

**Notation 4.14** (Bernstein centre of a category). We write

$$\mathfrak{z}C := \int_X [X, X] \quad (4.22)$$

for the end of Remark 4.8, we call it the *Bernstein centre* of  $C$ , and we say that  $C$  is *Bernstein-complete* when this end exists. The terminology is borrowed from representation theory, where Bernstein [Ber84] studies  $\mathbf{End}(\text{id}_{\mathbf{Rep}(G)})$  for  $G$  a reductive  $p$ -adic group (and himself writes ‘centre’ in scare quotes); Remark 4.17 records what carries over and what does not. The notion is unrelated to the *Drinfeld–Joyal–Street* centre  $\mathcal{Z}(C)$ , which is a braided category rather than an object of  $C$ , and to the internal (Huq) centre of an internal group.

**Lemma 4.15** (Representability lemma). Assume  $C$  is symmetric monoidal closed, complete, and Bernstein-complete. Then the profunctor of  $(M, R)$ -systems of Definition 4.9 is representable; more precisely, it is isomorphic to the functor  $C(-, - \times \mathfrak{z}C)$ .

*Proof.* Send a coherent  $(M, R)$ -system  $\langle f, \Phi \rangle$  to the pair  $(f, \bar{\Phi}^\#)$ , where  $\bar{\Phi}^\# : A \rightarrow \mathfrak{z}C$  is the classifying map of  $\Phi$  obtained by uncurrying and recurrying as in (4.14). By Remark 4.8 the assignment  $\Phi \mapsto \bar{\Phi}^\#$  is a bijection between repair families based at  $A$  and arrows  $A \rightarrow \mathfrak{z}C$ , whence a bijection

$$\mathbf{MRS}(A, B) \cong C(A, B) \times C(A, \mathfrak{z}C) \cong C(A, B \times \mathfrak{z}C), \quad (4.23)$$

natural in  $A$  and in  $B$ .  $\square$

**Corollary 4.16** (Tabulator as a comma object). Under the same hypotheses, the category  $\mathbb{O}\mathbf{MRS}$  of coherent  $(M, R)$ -systems is isomorphic to the comma category  $(C / - \times \mathfrak{z}C)$ .

We use this to characterise the coherent versions of the  $(M, R)$ -systems sketched in Example 3.9–3.19, and to prove that  $\mathbb{O}\mathbf{MRS}$  is complete and cocomplete whenever  $C$  is; the latter mirrors the incoherent Proposition 3.36.

**Remark 4.17** (Elementary properties of the Bernstein centre). The Bernstein centre ‘classifies repair maps’ in the sense of Lemma 4.15. By Remark 4.8 it is moreover an internal monoid under composition of families, with unit  $\eta : I \rightarrow \mathfrak{z}C$  picking out the identity family; the projection  $\mathfrak{z}C \xrightarrow{\pi_I} [I, I] \cong I$  retracts  $\eta$ , so the unit is always a retract of  $\mathfrak{z}C$ , and the only question is how much larger the latter is.

Its global points recover the classical centre of a category,  $C(I, \mathfrak{z}C) \cong \mathbf{End}(\text{id}_C)$ , a commutative monoid by Eckmann–Hilton; this is the point of contact with [Ber84], and the analogy stops there. Commutativity is not inherited by  $\mathfrak{z}C$ : for a small groupoid  $\mathcal{G}$  one has

$$\mathfrak{z}([\mathcal{G}^{\text{op}}, \mathbf{Set}]) : x \longmapsto \text{Aut}_{\mathcal{G}}(x), \quad (4.24)$$

with restriction along  $u : x' \rightarrow x$  given by conjugation  $\gamma \mapsto u^{-1}\gamma u$ , and global points  $\prod_{x \in \pi_0 \mathcal{G}} Z(\text{Aut}(x))$ . Taking points is thus lossy: for  $\mathcal{G} = BG$  the centre is  $G$  acting on itself by conjugation, while  $\mathbf{End}(\text{id}) \cong Z(G)$  is a singleton as soon as  $G$  has trivial centre. It is the internal object that governs the theory, a coherent  $(M, R)$ -system carrying an *arrow*  $A \rightarrow {}_3C$  and not a point of it. Note finally that the definition of  $\mathbf{End}(\text{id}_C)$  requires no monoidal structure, whereas the definition of  ${}_3C$  requires closedness.

**Remark 4.18** (A category without a Bernstein centre). Bernstein-completeness does not follow from completeness of  $C$ : the end is indexed by the whole of  $C$ , and may fail to exist for size reasons already when  $C$  is essentially small and rigid, so that  $\mathbf{Nat}(A \otimes -, \text{id})$  is unproblematically a set.

Let  $k$  be a field and  $C = \mathbf{Vect}_{\mathbf{Z}\text{-gr}}^{\text{fd}}$  the category of finite-dimensional  $\mathbf{Z}$ -graded  $k$ -vector spaces and degree-preserving maps, with the convolution structure

$$(V \otimes W)_n = \bigoplus_{p+q=n} V_p \otimes W_q, \quad [V, W]_n = \prod_{m \in \mathbf{Z}} \text{Hom}_k(V_m, W_{m+n}), \quad (4.25)$$

and unit  $I = k(0)$ , where  $k(n)$  denotes  $k$  placed in degree  $n$ . Both constructions stay finite-dimensional, so  $C$  is symmetric monoidal closed, and it is rigid, with  $(V^\vee)_n = (V_{-n})^*$ . Every object is a finite direct sum of copies of the  $k(n)$ , with  $C(k(m), k(n)) = 0$  for  $m \neq n$  and  $C(k(n), k(n)) = k$ . If  $\alpha_{k(n)} = c_n \cdot 1_{k(n)}$  for  $\alpha \in \mathbf{End}(\text{id}_C)$ , naturality along  $k(n) \rightarrow V$  forces  $\alpha_V$  to act on  $V_n$  by  $c_n$ . Conversely every family  $(c_n)_{n \in \mathbf{Z}}$  defines a natural transformation, since morphisms preserve degrees. Hence  $\mathbf{End}(\text{id}_C) \cong k^{\mathbf{Z}}$ . If some  $E \in C$  satisfied  $C(A, E) \cong \mathbf{Nat}(A \otimes -, \text{id}_C)$ , taking  $A = I$  would give  $E_0 \cong k^{\mathbf{Z}}$ , contradicting  $\dim_k E < \infty$ .

Lemma 4.15 is not a triviality result, and the reduction to pairs  $(A \rightarrow B, A \rightarrow I)$  that it seems to promise must be read with care. What it shows is that the repair datum is concentrated in the single arrow  $\bar{\Phi}^\# : A \rightarrow {}_3C$ ; the theory therefore degenerates exactly when  ${}_3C$  is *terminal*, which by the retraction of Remark 4.17 happens if and only if  ${}_3C \cong I$  and  $I$  is itself terminal. (If  ${}_3C$  is terminal, every  $C(X, I)$  is nonempty,  $\pi_I$  being composable with the unique  $X \rightarrow {}_3C$ , and injects into  $C(X, {}_3C)$  along  $\eta$ .) Outside the Cartesian world a trivial centre thus still leaves the datum  $A \rightarrow I$ , which can be substantial:  ${}_3\mathbf{Vect}_k \cong k$ , and the repair families based at  $A$  are the functionals  $A \rightarrow k$  of Corollary 4.7. There remains to determine when  ${}_3C$  collapses onto  $I$ .

**Proposition 4.19** (Trivialisation criteria). Let  $C$  be monoidal closed. Each of the following implies that  $C$  is Bernstein-complete with  ${}_3C \cong I$ :

- (i) the unit  $I$  is a generator of  $C$ ;
- (ii)  $C$  is Cartesian closed and its subterminal objects form a generating family (for a topos, this says exactly that it is *localic*).

Whenever  ${}_3C \cong I$  one has  $\mathbf{Nat}(\text{id}, [A, -]) \cong C(A, I)$  for every  $A$ , the pair  $(p, \iota)$  of Remark 4.4 consists of mutually inverse bijections, and every coherent  $(M, R)$ -system is representable in the sense of Remark 4.4.

*Proof.* Both cases rest on the same remark: if a family  $u : U \rightarrow X$  is jointly epimorphic, so is  $(A \otimes u)_u$ , since  $f \circ (A \otimes u) = g \circ (A \otimes u)$  for all  $u$  transposes to  $f^\# \circ u = g^\# \circ u$  for all  $u$ , whence  $f^\# = g^\#$  and  $f = g$ .

(i) The generalised elements  $x : I \rightarrow X$  are jointly epimorphic, hence so are the  $A \otimes x$ ; naturality of  $\theta : A \otimes - \Rightarrow \text{id}$  along  $x$  gives  $\theta_X \circ (A \otimes x) = x \circ \theta_I$  modulo the unitor  $A \otimes I \cong A$ , so  $\theta \mapsto \theta_I$  is injective. Modulo the transposition  $\mathbf{Nat}(A \otimes -, \text{id}) \cong \mathbf{Nat}(\text{id}, [A, -])$  this map is the  $p$  of Lemma 4.3, surjective because  $\iota$  is a section of it; hence  $\mathbf{Nat}(A \otimes -, \text{id}) \cong C(A, I)$  naturally in  $A$ , which is the universal property of  ${}_3C$ , and  ${}_3C \cong I$  by Yoneda.

(ii) Here  $I = 1$ , and the generating objects need not have points: it suffices that they be too rigid to carry an operation. If  $U$  is subterminal then  $U \rightarrow 1$  is monic, so any two arrows with codomain  $U$  and common domain agree; in particular  $\theta_U = \pi_U$  for every natural  $\theta : A \times - \Rightarrow \text{id}$ . Naturality along  $u : U \rightarrow X$  gives  $\theta_X \circ (A \times u) = u \circ \pi_U = \pi_X \circ (A \times u)$ , and the  $A \times u$  are jointly epimorphic, so  $\theta_X = \pi_X$ . Thus  $\mathbf{Nat}(A \times -, \text{id})$  is a singleton naturally in  $A$ , as is  $C(A, 1)$ , and  $1 = I$  has the universal property of  ${}_3C$ .

In either case  $\mathbf{Nat}(\mathrm{id}, [A, \_]) \cong C(A, \mathfrak{z}C) \cong C(A, I)$  by Lemma 4.15, and under this identification  $p$  becomes a bijection with inverse  $\iota$ .  $\square$

**Remark 4.20** (Neither criterion is necessary).  $\mathfrak{z}C \cong I$  is not a disguised way of saying that  $I$  generates. Criterion (ii) refutes the converse of (i): in the Sierpiński topos  $\mathbf{Set}^\rightarrow$ , the smallest non-well-pointed presheaf topos (Remark 4.6), the object  $P = (\emptyset \rightarrow 1)$  has no global points while admitting two distinct maps to  $(\emptyset \rightarrow \{0, 1\})$ , so  $1$  does not generate; but the representables  $(1 \rightarrow 1)$  and  $P$  are subterminal and do, whence  $\mathfrak{z}(\mathbf{Set}^\rightarrow) \cong 1$ . Nor is (ii) necessary: in  $\mathbf{Cat}$  the only subterminals are  $0$  and  $\mathbf{1}$ , while  $\mathbf{N}$ -graded sets under the Cauchy product and  $\mathbf{Chu}(\mathbf{Set}, K)$  are not Cartesian at all, and the centre is trivial in all three (Remark 4.22). What survives is one implication: a nontrivial centre *requires* the unit to fail to generate. So it should be, since the centre measures the isotropy of  $C$  and not its supply of points:  $\mathbf{Set}^{C_2}$  is nontrivial because its group acts on every object at once, by a natural automorphism of the identity, whereas  $\mathbf{Set}^\rightarrow$  has as few points and no such automorphism.

**Remark 4.21** (The enriched end always collapses). All of the above concerns the *ordinary* end, indexed by the underlying category of  $C$ . The self-enriched end is always  $I$ , by the enriched Yoneda lemma:

$$\int_X^C [X, X] \cong \int_X [[I, X], X] \cong I. \quad (4.26)$$

The construction is non-vacuous only in its unenriched reading, and a reader who enriches it silently will conclude that the whole notion collapses.

**Remark 4.22** (Where the Bernstein centre is trivial, and where it is not). The centre trivialises in

- $\mathbf{Set}$ ,  $\mathbf{Vect}_k$  and  $\mathbf{Mod}(R)$  for  $R$  commutative, where it is  $1$ ,  $k$  and  $R$ : criterion (i);
- localic toposes (sheaves  $\mathbf{Sh}(X)$  on a space,  $\mathbf{Sh}(L)$  on a locale, presheaves on a poset), by criterion (ii);
- $\mathbf{Chu}(\mathbf{Set}, K)$ , where the unit does not generate and neither criterion applies: points  $I \rightarrow X$  fix the first component of a natural  $\theta : A \otimes \_ \Rightarrow \mathrm{id}$  to be the projection, and, every state of  $X$  being a copoint  $X \rightarrow I^\perp$ , naturality along copoints determines the second component from the single component  $\theta_{I^\perp}$ ; the resulting bijection  $\mathbf{Nat}(A \otimes \_, \mathrm{id}) \cong \mathbf{Chu}(A, I)$  gives  $\mathfrak{z}C \cong I = (1, K, \mathrm{ev})$ ;
- $\mathbf{N}$ -graded sets under the Cauchy product, and graded objects over any cancellative monoid, where the unit, a singleton in degree  $0$ , is far from generating, and triviality is read off the base as in the last item below;
- $\mathbf{Cat}$  under the Cartesian and under the funny tensor product, where  $\mathbf{1}$  and  $\mathbf{2}$  generate strongly and the end is  $\mathbf{1}$ ; and  $\mathbf{Set}^{\mathbf{N}}$  under the degreewise Cartesian product, where it is terminal.

It does not trivialise in

- $\mathbf{Set}^G$ , where  $\int_X X^X \cong G$  with the conjugation action, the family being  $\alpha_X(z, x) = z \cdot x$ : our test case, since the leg  $A \rightarrow \mathfrak{z}C$  of Lemma 4.15 is then substantial data, and the internal monoid is noncommutative although its global points  $Z(G)$  commute;
- $\mathbf{Set}^{B^{\mathbf{N}}}$ , sets with an endomorphism, where the end is the constant object  $\underline{\mathbf{N}}$  with  $\alpha_X(n, x) = f_X^n(x)$ ;
- Joyal species under the Cauchy product, i.e. Day convolution along  $(\mathbf{Bij}, +)$ , where the end is  $I \amalg I$ , the extra element acting on  $F[S]$  by the transposition for  $|S| = 2$  and trivially elsewhere;  $k$ -linear species give  $k[\mathbf{Z}/2]$ . Under the Cartesian structure one gets instead the groupoid formula of Remark 4.17,  $S \mapsto \mathrm{Aut}(S)$ : the two tensors have different internal homs, hence different centres;
- more generally  $[\mathcal{D}^{\mathrm{op}}, \mathcal{V}]$  with Day convolution, for  $\mathcal{D}$  monoidal with nontrivial centre: the criterion is read off the base, since  $(\mathfrak{z}[\mathcal{D}^{\mathrm{op}}, \mathcal{V}])(d) \cong \mathbf{Nat}(d \otimes \_, \mathrm{id}_{\mathcal{D}})$ , so the end is  $I$  exactly when  $\mathcal{D}$  has no nontrivial central endomorphisms, as for  $\mathcal{D} = (\mathbf{N}, +)$  discrete.

The canonical nontrivial example thus remains  $\mathbf{Set}^G$ , or  $\mathbf{Set}^{\mathcal{G}}$  for a groupoid  $\mathcal{G}$ . More generally still,  $\mathbf{Set}^{\mathcal{D}}$  is often Bernstein-complete, with  $\mathfrak{z}(\mathbf{Set}^{\mathcal{D}}) : \mathcal{D} \rightarrow \mathbf{Set}$  sending  $B$  to the endomorphisms of  $\mathrm{dom} : \mathcal{D}/B \rightarrow \mathcal{D}$  in  $\mathbf{Cat}/\mathcal{D}$ ; this is nontrivial as soon as some object of  $\mathcal{D}$  has nontrivial endomorphisms, each  $k : B \rightarrow B$  inducing  $k_* : \mathcal{D}/B \rightarrow \mathcal{D}/B$  by postcomposition.

In a topos the two criteria have familiar names: (i) is well-pointedness (Remark 4.6), and (ii) is locality. A well-pointed topos is generated by 1, which is subterminal, so (i) implies (ii), and the triviality that Lemma 4.2 established under a hypothesis that classically pins down **Set** itself in fact holds over an arbitrary locale. Triviality is therefore not an artefact of set-theoretic foundations but the normal state of affairs in every space-like topos, and a nontrivial coherent  $(M, R)$ -system requires isotropy: one must leave the localic world, as the toposes  $\mathbf{Set}^M$  of monoid and group actions do, their only subterminals being 0 and 1 and their global points the fixed points. The smallest such topos already suffices, and settles the debt contracted in Remark 4.5.

**Proposition 4.23** ( $\mathcal{U}$ ). In the topos  $\mathbf{Set}^{C_2}$  of  $C_2$ -sets, there is an object  $A$  (namely  $A := I$ , the monoidal unit/terminal object) admitting strictly more than one natural transformation  $\text{id} \Rightarrow [A, -]$ ; consequently  $\iota \circ p \neq 1$  on  $\text{Nat}(\text{id}, [A, -])$ , as promised in Remark 4.5.

*Proof.* Write  $C_2 = \{e, \sigma\}$  and  $\mathcal{E} = \mathbf{Set}^{C_2}$ , the Cartesian closed topos of sets with an involution; its unit  $I$  is the terminal object, a fixed point. Take  $A := I$ . Every morphism of  $\mathcal{E}$  is equivariant, so the family  $\tau_X := \sigma_X$  is a natural transformation  $\tau : \text{id} \Rightarrow \text{id}$ ; composing with the isomorphism  $i : \text{id} \xrightarrow{\cong} [I, -]$  of Definition 2.15 gives  $i \circ \tau : \text{id} \Rightarrow [I, -]$ . It differs from  $i$ : at the regular  $C_2$ -set  $A_{\text{reg}} := C_2$  and the base point  $e$ , the first returns  $\sigma \cdot e = \sigma$  and the second  $e$ , the translation action being free.

Now  $\mathcal{E}(A, I) = \mathcal{E}(I, I)$  is a singleton, so  $p(i) = p(i \circ \tau) = 1_I$ , whence  $\#\text{Nat}(\text{id}, [A, -]) > \#\mathcal{E}(A, I) = 1$ ; and  $\iota(p(i \circ \tau)) = \iota(1_I) = i \neq i \circ \tau$ , so  $\iota \circ p \neq 1$ .  $\square$

This is consistent with Proposition 4.19 in the only way it could be: 1 does not generate  $\mathbf{Set}^{C_2}$ , a free action having no global points,  $\mathfrak{z}(\mathbf{Set}^{C_2}) \cong C_2$ , and  $i \circ \tau$  is the second global point of the centre.

**Remark 4.24.** Note that the rather heavy-handed machinery of topos theory plays no essential role here; a topos is just a category that is as close as possible to **Set** without having a terminal generator.

It is an interesting observation that in the category **Cat** of categories and functors, which is not a topos, the terminal category is *not* a generator either (the two-element subcategory  $\{\bullet, [0 \rightarrow 1]\}$  is, see [AR94]), and yet an id-coherent  $(M, R)$ -system doesn't recover a truly new notion:

 An id-coherent  $(M, R)$ -system in **Cat** boils down to an incoherent  $(M, R)$ -system in funny categories, as described in Example 3.18.

This is not in tension with Proposition 4.19: as recorded in Remark 4.22, **1** and **2** generate **Cat** strongly, and the end is **1**: failure of the terminal object alone to generate is necessary, but not sufficient, for a nontrivial centre.

**4.2. Invariants of coherent  $(M, R)$ -systems.** The categories we defined in Definition 4.10–4.11 come equipped with canonical forgetful functors, that we define at once.

**Definition 4.25** (The first edge functor,  $\mathcal{U}$ ). The *first edge functor*

$$[-]_f : \mathbb{O}\mathbf{MRS} \longrightarrow C^{[0 \rightarrow 1]} \quad (4.27)$$

sends a coherent  $(M, R)$ -system  $\langle f, \Phi \rangle$  based at  $(A, B)$  to the first edge of its defining diagram, i.e. to the process map  $f : A \rightarrow B$ , regarded as an object of the arrow category. On a morphism  $(l, r)$  of  $\mathbb{O}\mathbf{MRS}$ , with  $l : A \rightarrow A'$  and  $r : B \rightarrow B'$ , it returns the commutative square

$$\begin{array}{ccc} A & \xrightarrow{f} & B \\ l \downarrow & & \downarrow r \\ A' & \xrightarrow{g} & B' \end{array} \quad (4.28)$$

whose commutativity is the first of the two conditions defining a morphism of  $(M, R)$ -systems. Note that this is the very functor  $[-]_f$  that Remark 3.1 wishes for and shows to be unavailable for incoherent  $(M, R)$ -systems: passing to the tabulator resolves the variance obstruction, and the wished-for process selector exists after all.

**Definition 4.26** (The last edge functor,  $\mathcal{U}$ ). The *last edge functor*

$$\zeta_{-}\mathcal{J}_{\Phi} : \mathcal{E}(\mathbf{MRS}) \longrightarrow C^{[0 \rightarrow 1]} \quad (4.29)$$

sends a coherent (M,R)-system  $\langle f, \Phi \rangle$  based at  $(A, B)$ , regarded as an object of the category of twisted elements, to the last edge of its defining diagram, i.e. to the component  $\Phi_B : B \rightarrow [A, B]$  of its repair family, again as an object of the arrow category. A morphism of  $\mathcal{E}(\mathbf{MRS})$  consists of  $l : A' \rightarrow A$  and  $r : B \rightarrow B'$ , twisted as in Definition 2.26, and  $\zeta_{-}\mathcal{J}_{\Phi}$  returns the square

$$\begin{array}{ccc} B & \xrightarrow{\Phi_B} & [A, B] \\ r \downarrow & & \downarrow [l, r] \\ B' & \xrightarrow{\Psi_{B'}} & [A', B'] \end{array} \quad (4.30)$$

obtained by acting on the internal hom with  $l$  contravariantly and  $r$  covariantly. It is the coherent counterpart of the right leg of the span of Proposition 3.44, and the notation is deliberately the paper's own: square brackets  $[-]_f$  mark a functor out of a total category, bags  $\zeta_{-}\mathcal{J}_{\Phi}$  one out of twisted elements, and the asymmetry between the two edge functors is exactly that the process map is already functorial on  $\odot\mathbf{MRS}$  while the repair map is only so after twisting.

**Definition 4.27** (The twisted edge functor,  $\mathcal{U}$ ). The *twisted edge functor*

$$\zeta_{-}\mathcal{J}_f : \mathcal{E}(\mathbf{MRS}) \longrightarrow \mathbf{Tw}(C) \quad (4.31)$$

sends a coherent (M,R)-system  $\langle f, \Phi \rangle$  based at  $(A, B)$ , regarded as an object of the category of twisted elements, to its process map  $f : A \rightarrow B$  — like  $[-]_f$  of Definition 4.25, but landing in the *twisted* arrow category rather than in  $C^{[0 \rightarrow 1]}$ . On a morphism of  $\mathcal{E}(\mathbf{MRS})$ , given by  $l : A' \rightarrow A$  and  $r : B \rightarrow B'$ , it returns the twisted square

$$\begin{array}{ccc} A & \xrightarrow{f} & B \\ l \uparrow & & \downarrow r \\ A' & \xrightarrow{g} & B' \end{array} \quad (4.32)$$

which is the shape of a morphism of  $\mathbf{Tw}(C)$ .

Taken together, the three functors just defined are the coherent counterparts of the incoherent ones, and the notation records which is which:  $[-]_f$  is the process selector out of the total category — the one Remark 3.1 wishes for and shows to be unavailable incoherently — while  $\zeta_{-}\mathcal{J}_f$  and  $\zeta_{-}\mathcal{J}_{\Phi}$  are the two legs of the span of Proposition 3.44, out of twisted elements.

**Definition 4.28** (The fibration of repairs,  $\mathcal{U}$ ). The *fibration of repairs* **Rep** is the category of elements of the assignment  $A \mapsto \text{Nat}(\text{id}, [A, -])$ ; equivalently, it is what remains of an (M,R)-system once the process map is forgotten and only the repair family is kept. Explicitly:

- objects are pairs  $(A, \Phi)$  with  $A \in C_0$  and  $\Phi : \text{id} \Rightarrow [A, -]$  a natural transformation, i.e. the repair datum of Definition 4.1 on its own;
- an arrow  $(A, \Phi) \rightarrow (A', \Phi')$  is a map  $u : A \rightarrow A'$  of  $C$  such that  $[u, -] \circ \Phi' = \Phi$ , where  $[u, -] : [A', -] \Rightarrow [A, -]$  is the natural transformation induced by  $u$  on internal homs;
- identities and composition are those of  $C$ .

The condition on arrows is precisely the second of the two conditions defining a morphism of (M,R)-systems, the first — the one constraining process maps — being simply absent. Proposition 4.30 makes that observation exact.

**Remark 4.29.** It is evident that when a repair map is presented in uncurried form, **Rep** is just another name for the category  $C \ltimes_{\odot} C$  of Definition 3.34, and the functor  $K$  of Proposition 4.30 is precisely the comparison that forgets the process map and keeps only the repair datum. [how does it fit in the square of \(3.61\)?](#)

**Proposition 4.30** (The repairs coreflect into the total category,  $\mathcal{U}$ ). The fibration of repairs **Rep** of Definition 4.28 is a coreflective subcategory of the total category  $\mathbb{MRS}$  of Definition 4.10: there is an adjunction

$$\mathbf{Rep} \begin{array}{c} \xrightarrow{J} \\ \xleftarrow{K} \end{array} \mathbb{MRS} \quad (4.33)$$

in which  $J$  sends a repair system  $(A, \Phi)$  to the  $(M, R)$ -system  $\langle 1_A, \Phi \rangle$  based at  $(A, A)$ , and  $K$  forgets the process map, retaining only the repair datum. The unit of  $J \dashv K$  is an identity; equivalently  $J$  is fully faithful (which has already been observed), and thus  $K$  is a coreflector.

**Remark 4.31.** The content of the proposition is that a repair blueprint carries no less information than an  $(M, R)$ -system whose process map is an identity: choosing  $\Phi$  and choosing  $\langle 1_A, \Phi \rangle$  are the same choice. Contrast this with the situation for the functors  $L, L'$  of Theorem 3.49 and Theorem 3.50, where the analogous comparison fails for incoherent systems.

The representability result Lemma 4.15 also provides a route to completeness and cocompleteness of  $\mathbb{MRS}$ .

**4.3. Related notions of coherence.** Here we collect several variations on the notion of id-naturality introduced above, each offering a different balance between semantic transparency and the goal of producing a genuine profunctor of  $(M, R)$ -systems: **cod-coherent** systems (§4.3.1), which we show to be equivalent to **id-coherent** ones;  **$A/C$ -coherent** systems (§4.3.2), a semantically finer variant restricted to processes sharing the domain of  $f$ ; **fully polymorphic, dinatural** systems (§4.3.3);  **$U$ -coherent** systems (§4.3.4) for a generic functor  $U$ ; and, at the top of the abstraction ladder, **fully profunctorial** systems (§4.3.5).

**4.3.1. cod-coherent  $(M, R)$ -systems.** In a ‘cod-coherent’  $(M, R)$ -system based at  $A$ , there is a process  $f : A \rightarrow B$ , and the repair function has type

$$\Phi : \text{cod} \Rightarrow [A, \text{cod}]. \quad (4.34)$$

We denote  $\text{cod-MRS}(A, B)$  the class of the  $(M, R)$ -systems in this sense.

It is evident how any **id-coherent**  $(M, R)$ -system  $\langle f, \Phi \rangle$  gives rise to a **cod-coherent** one by whiskering:  $\langle f, \Phi * \text{cod} \rangle$ . One could then suspect that **cod-coherent**  $(M, R)$ -systems constitute a more general concept than **id-coherent** ones, but this is not true. We prove it in a moment, in Lemma 4.33 below.

The **cod-coherent** presentation has a useful semantic interpretation.

In the intuition on  $(M, R)$ -systems that regards them as process-repair pairs, it is natural to think that  $\Phi$  is a ‘blueprint for reconstruction’ modelled on processes more than on objects:  $\Phi$  has a component  $\Phi_h : Y \rightarrow [A, Y]$  for every *arrow*  $h : X \rightarrow Y$ , not just for any *object*  $Y$  of  $C$ . Evidently this recovers an **incoherent**  $(M, R)$ -system as the component at  $f$  of  $\Phi$ :

$$\begin{array}{ccc} A & \xrightarrow{f} & B \\ \parallel & & \parallel \\ \text{cod } f & \xrightarrow{\Phi_f} & [A, \text{cod } f] \end{array} \quad (4.35)$$

Naturality in this case takes the following form:

For every morphism  $(p, q) : h \rightarrow h'$  of  $C^{[0 \rightarrow 1]}$ , i.e. every commutative square

$$\begin{array}{ccc} X & \xrightarrow{h} & Y \\ p \downarrow & & \downarrow q \\ X' & \xrightarrow{h'} & Y' \end{array} \quad (4.36)$$

in  $C$ , the diagram

$$\begin{array}{ccc} Y & \xrightarrow{\Phi_h} & [A, Y] \\ q \downarrow & & \downarrow [A, q] \\ Y' & \xrightarrow{\Phi_{h'}} & [A, Y'] \end{array} \quad (4.37)$$

commutes, i.e.  $[A, q] \circ \Phi_h = \Phi_{h'} \circ q$ .

Now, despite the intuitive idea behind this choice having a solid foundation, we are in the following situation:

**Definition 4.32** (Lax epimorphism [LNS22]). A functor  $f : A \rightarrow B$  is a *lax epimorphism* if, for every category  $C$ , the precomposition functor

$$_ \circ f : \mathbf{Cat}(B, C) \longrightarrow \mathbf{Cat}(A, C) \quad (4.38)$$

is fully faithful.

The relevant property of  $\text{cod}$  follows by inspecting a few elementary squares.

**Lemma 4.33** ( $\mathcal{C}^{[0 \rightarrow 1]}$ ). The codomain functor  $\text{cod} : C^{[0 \rightarrow 1]} \rightarrow C$  is a lax epimorphism. As a consequence, there is a bijection between  $\text{id}$ -coherent  $(M, R)$ -systems and  $\text{cod}$ -coherent ones.

**Remark 4.34.** We can cast the naturality condition of  $\Phi$  on various ‘degenerate’ morphisms of  $C^{[0 \rightarrow 1]}$  to obtain the following equations; the last one will entail the claim that whiskering with  $\text{cod}$  induces a bijection between classes of natural transformations

$$\mathbf{Cat}(C, C)(\text{id}, [A, _]) \cong \mathbf{Cat}(C^{[0 \rightarrow 1]}, C)(\text{cod}, [A, \text{cod}]). \quad (4.39)$$

- On degenerate squares of the form  $\begin{array}{ccc} X & \xrightarrow{ku} & Z \\ u \downarrow & & \parallel \\ Y & \xrightarrow{k} & Z \end{array}$ , for composable arrows  $k, u$ , naturality of  $\Phi$

means that the diagram

$$\begin{array}{ccc} Y & \xrightarrow{k} & Z \\ \Phi_u \downarrow & & \downarrow \Phi_{1_Z} \\ [A, Y] & \xrightarrow{k \circ _} & [A, Z] \end{array} \quad (4.40)$$

is commutative, giving  $[A, k] \circ \Phi_u = \Phi_{1_Z} \circ k$ . In particular, if  $k = f$ , then  $[A, f] \circ \Phi_u = \Phi_{1_B} \circ f$  for every  $u : X \rightarrow A$ ; the right-hand side is independent of  $u$ .

- So, when  $k = 1_Y$ , and  $u : X \rightarrow Y$ , naturality of  $\Phi$  means that the diagram

$$\begin{array}{ccc} Y & \xlongequal{\quad} & Y \\ \Phi_u \downarrow & & \downarrow \Phi_{1_Y} \\ [A, Y] & \xlongequal{\quad} & [A, Y] \end{array} \quad (4.41)$$

is commutative, giving  $\Phi_u = \Phi_{1_Y}$ .

Thus  $\Phi_u = \Phi_{1_{\text{cod}u}}$ : components depend only on the codomain. The assignment  $X \mapsto \Phi_{1_X}$  defines a natural transformation  $\Phi^\natural : \text{id} \Rightarrow [A, _]$ , and  $\Phi = \Phi^\natural * \text{cod}$ .

As a corollary of Lemma 4.33 one associates to  $\text{cod}$ -coherent  $(M, R)$ -systems the same tabulator of Definition 4.10.

**4.3.2.  $A/C$ -coherent  $(M, R)$ -systems.** In a ‘coslice-coherent’  $(M, R)$ -system, the repair function has type

$$\Phi : \text{cod}^A \Rightarrow [A, \text{cod}^A] \quad (4.42)$$

where  $\text{cod}^A : A/C \rightarrow C$  is the codomain functor restricted to the subcategory  $A/C \subseteq C^{[0 \rightarrow 1]}$  of arrows with domain  $A$ . A pair  $\langle f, \Phi \rangle$  of this type, where  $f : A \rightarrow B$ , is a  $\text{cod}^A$ -coherent  $(M, R)$ -system; we denote by  $\text{cod}^A\text{-MRS}(A, B)$  the class of all such systems.

The semantic interpretation for such a structure is even more straightforward:  $\Phi$  is defined and natural on a narrower class of processes, those sharing the same domain with  $f$ ; it is then able to regenerate terms of type  $f$  if it can access the shape of the domain  $A$  of  $f$ .

Naturality for  $\text{cod}^A$ -coherent systems takes the following form:

For every morphism  $q : h \rightarrow h'$  of the coslice category  $A/C$ , i.e. for every triangle

$$\begin{array}{ccc} & A & \\ h \swarrow & & \searrow h' \\ X & \xrightarrow{q} & X' \end{array} \quad (4.43)$$

in  $C$ , the diagram

$$\begin{array}{ccc} X & \xrightarrow{\Phi_h} & [A, X] \\ q \downarrow & & \downarrow [A, q] \\ X' & \xrightarrow{\Phi_{h'}} & [A, X'] \end{array} \quad (4.44)$$

commutes, i.e.  $[A, q] \circ \Phi_h = \Phi_{h'} \circ q$ .

Evidently this recovers an incoherent  $(M, R)$ -system as the component at  $f$  of  $\Phi$ :

$$A \xrightarrow{f} B \xrightarrow{\Phi_f} [A, B] \quad (4.45)$$

Yet, under this notion of naturality there seems to be no way to turn the assignment  $(A, B) \mapsto \text{cod}^A\text{-MRS}(A, B)$  into a profunctor

$$\text{cod}^A\text{-MRS} : C^{\text{op}} \times C \longrightarrow \mathbf{Set}. \quad (4.46)$$

Being able to ‘reindex’ along a morphism  $(A, B) \xrightleftharpoons[u]{u} (A', B')$  means that one is able to take an  $(M, R)$ -system  $\Phi : \text{cod}^A \Rightarrow [A, \text{cod}^A]$  and send it to one of type

$$u^* \Phi : \text{cod}^{A'} \Rightarrow [A', \text{cod}^{A'}], \quad (4.47)$$

which amounts to being able to fill the question marks in this diagram

$$\begin{array}{ccccc} A'/C & \xleftarrow{\quad ? \quad} & A/C & \xleftarrow{\quad ? \quad} & A'/C \\ \text{cod}^{A'} \swarrow & & \text{cod}^A \swarrow & & \text{cod}^{A'} \swarrow \\ & C & \xrightarrow{\quad \Phi \quad} & C & \\ & \downarrow [u, -] & & \downarrow [u, -] & \\ & [A', -] & & & \end{array} \quad (4.48)$$

The dotted arrow on the left poses no difficulty: it is filled by the functor  $u^* : A/C \rightarrow A'/C$  sending an object  $h : A \rightarrow X$  to  $h \circ u : A' \rightarrow X$ , so that the left-hand triangle commutes even strictly, since  $\text{cod}^{A'}(h \circ u) = X = \text{cod}^A(h)$  by definition.

The dashed arrow is the problem. Filling it so that the whole diagram pastes correctly would require a comparison 2-cell running in the direction opposite to the one furnished by  $u^*$  itself. Indeed, if  $C$  admits finite colimits,  $u^*$  does have a left adjoint  $u_!$ , given by pushout along  $u$ ; but the canonical arrow this adjunction supplies points the wrong way to be pasted into the diagram above. What would be needed to correct this is a *right* adjoint  $P_u$  to  $u^*$ , which exists if and only if  $u$  is an isomorphism: assuming  $u^*$  is a left adjoint, it preserves the initial object of  $A/C$ , namely  $1_A$ ; but  $u^*(1_A) = u$ , regarded as an object of  $A'/C$ , and for this to be initial a necessary and sufficient condition is that  $u$  is an isomorphism in  $C$ . So, except in the degenerate case where  $u$  is invertible, no functorial reindexing along  $u$  exists, confirming that  $\text{cod}^A$ -coherent  $(M, R)$ -systems do not assemble into a profunctor.

The best we can do is assemble the  $\text{cod}^A$ -coherent  $(M, R)$ -systems into a *displayed* category, much like in Remark 3.24 above.



**Definition 4.35** (The displayed category of  $\mathbf{cod}^A$ -coherent  $(M,R)$ -systems). Define a category  $\mathbf{MRS}_C$  as follows:

- objects are the ‘ $\mathbf{cod}^A$ -coherent’  $(M,R)$ -systems as defined above, i.e. pairs

$$f : A \longrightarrow B \quad \Phi : \mathbf{cod}^A \Longrightarrow [A, \mathbf{cod}^A]; \quad (4.49)$$

- arrows of type  $\langle\langle f, \Phi \rangle\rangle_{\mathbf{cod}^A} \rightarrow \langle\langle g, \Psi \rangle\rangle_{\mathbf{cod}^{A'}}$  consist of pairs  $\ell : A \rightarrow A', r : B \rightarrow B'$  of arrows in  $C$ , such that the following equations hold:

$$\begin{array}{ccc} A & \xrightarrow{f} & B \\ \ell \downarrow & & \downarrow r \\ A' & \xrightarrow{g} & B' \end{array} \quad \begin{array}{ccc} A/C & \xleftarrow{\ell^*} & A'/C \\ & \searrow & \swarrow \\ & C & \xrightarrow{[A', -]} C \\ & \swarrow & \searrow \\ & C & \xrightarrow{[A, -]} C \end{array} \quad \begin{array}{ccc} A/C & \xleftarrow{\ell^*} & A'/C \\ & \searrow & \swarrow \\ & C & \xrightarrow{[A, -]} C \\ & \swarrow & \searrow \\ & C & \xrightarrow{[A, -]} C \end{array} \quad (4.50)$$

Thinking of  $\Phi$  as a pointed natural transformation, one can clearly recover an incoherent  $(M,R)$ -system by ‘evaluation at  $f$ ’.

**4.3.3. Fully polymorphic  $(M,R)$ -systems.** In a ‘fully polymorphic’  $(M,R)$ -system, the repair function has type

$$\Phi : \mathbf{cod} \Rightarrow [\mathbf{dom}, \mathbf{cod}] \quad (4.51)$$

where  $\mathbf{dom}, \mathbf{cod} : C^{[0 \rightarrow 1]} \rightarrow C$  are the obvious domain and codomain functors. Semantically, here one is abstracting the shape of  $\Phi$  so that evaluation at  $f : A \rightarrow B$  yields  $\Phi_f : \mathbf{cod} f \rightarrow [\mathbf{dom} f, \mathbf{cod} f]$  i.e.  $\Phi_f : B \rightarrow [A, B]$ .

The first caveat is that the repair cell  $\Phi$  is now a *dinatural* transformation filling the cell

$$\begin{array}{ccc} (C^{[0 \rightarrow 1]})^{\text{op}} \times (C^{[0 \rightarrow 1]}) & \xrightarrow{\mathbf{dom}^{\text{op}} \times \mathbf{cod}} & C^{\text{op}} \times C \\ \pi^r \downarrow & \Phi \nearrow & \downarrow [-, -] \\ C^{[0 \rightarrow 1]} & & C \\ & \searrow \text{cod} & \end{array} \quad (4.52)$$

Dinaturality takes the following form:

For every morphism  $(p, q) : f \rightarrow f'$  of  $C^{[0 \rightarrow 1]}$ , i.e. every commutative square  $q \circ f = f' \circ p$  with  $f : A \rightarrow B, f' : A' \rightarrow B'$ , the diagram

$$\begin{array}{ccc} B & \xrightarrow{\Phi_f} & [A, B] \\ q \downarrow & & \downarrow [A, q] \\ B' & & \\ \Phi_{f'} \downarrow & & \\ [A', B'] & \xrightarrow{[p, B']} & [A, B'] \end{array} \quad (4.53)$$

commutes, i.e.  $[A, q] \circ \Phi_f = [p, B'] \circ \Phi_{f'} \circ q$ ; this is the classical hexagon condition for a dinatural transformation, here collapsing to a square because  $\mathbf{cod} \circ \pi^r$  does not depend on the contravariant argument.

This yields a fibration, but for a very trivial reason: it does not involve the parameters  $A$  or  $B$ , so it just results in a ‘trivial bundle’ over  $C^{\text{op}} \times C$ .

**Remark 4.36.** To a fully polymorphic  $(M,R)$ -system we can associate the tabulator of Definition 2.24, and armed with the definition the reader will easily verify that, denoting by  $\mathbf{N}(\mathbf{dom}, \mathbf{cod})$  the class of (di)natural transformations  $\Phi : \mathbf{cod} \Rightarrow [\mathbf{dom}, \mathbf{cod}]$  as in (4.52) above, the tabulator universal functor is isomorphic to the span

$$C \xleftarrow{\mathbf{dom} \circ \pi^l} C^{[0 \rightarrow 1]} \times \mathbf{N}(\mathbf{dom}, \mathbf{cod}) \xrightarrow{\mathbf{cod} \circ \pi^l} C \quad (4.54)$$

**Remark 4.37** (Fully polymorphic  $(M, R)$ -systems and ends). Concretely, since  $\Phi$  does not depend on  $(A, B)$ , a ‘fully polymorphic’  $(M, R)$ -system is really just a single dinatural transformation  $\Phi : \text{cod} \rightarrow [\text{dom}, \text{cod}]$ , i.e. (via the classical correspondence between dinatural transformations and ends) an element of the end  $\int_{f \in C^{[0 \rightarrow 1]}} [\text{dom} f, \text{cod} f]$ , chosen once and for all, independently of any particular process  $f : A \rightarrow B$  under consideration. Every incoherent  $(M, R)$ -system  $\langle f, \Phi_f \rangle$  recovered by evaluation therefore shares literally the same underlying  $\Phi$ , so this notion discards exactly the freedom (choosing a repair map tailored to the specific pair  $(A, B)$ ) that made incoherent  $(M, R)$ -systems interesting in the first place; the ‘trivial bundle’ is trivial precisely because it carries no information about  $(A, B)$  beyond what is already fixed once, globally, by the single chosen element of the end.

4.3.4. *U-coherent  $(M, R)$ -systems.* In a  $U$ -coherent  $(M, R)$ -system, the repair function has type

$$\Phi : U \Rightarrow [A, U] \quad (4.55)$$

where  $U : \mathcal{E} \rightarrow C$  is a generic functor, or a fibration over  $C$ .

This is, in our view, a reasonable notion more general than id-coherent systems on which one might settle to obtain meaningful nontrivial invariants, with a small caveat that we explain below. One issue to overcome is the excessive freedom in choosing a natural canonical candidate for  $U$ ; how many ‘obvious’ functors of the type of  $U$  can be attached to a monoidal closed category like  $C$ ? One can require  $U$  to be a monoidal fibration in the sense of [Shu08], or a functor naturally induced by the monoidal structure; for example, if  $M$  is a  $\otimes$ -monoid in  $C$ , we can consider the forgetful functor  $U_M : C^M \rightarrow C$  from the category of objects equipped with an action of  $M$ ; or we can consider a monoidal closed category of modules over a von Neumann regular ring  $R$ , and leveraging the main result of [Agu97] we can consider a notion of ‘linear’ internal category  $\mathbf{Cat}_\otimes(C)$  and the associated forgetful functor  $U_{\mathbf{Cat}, R} : \mathbf{Cat}_\otimes(C) \rightarrow C$ . The choice is virtually unlimited: we can also consider  $C$ -shaped graphs and their forgetful functor,  $U_{\mathbf{Gph}} : C^{[0 \rightarrow 1]} \rightarrow C$ ;  $C$ -valued (co)simplicial objects  $U_\Delta : C^\Delta \rightarrow C$ , with their evaluation at a given dimension, their realisation, or totalisation (assuming  $C$  has enough limits and colimits); we can consider comma categories, or strict pullbacks, obtained from either of the previous choices.

For a fixed  $U$  as in (4.55), naturality takes the following form:

For every morphism  $h : X \rightarrow Y$  of  $\mathcal{E}$ , the diagram

$$\begin{array}{ccc} UX & \xrightarrow{\Phi_X} & [A, UX] \\ Uh \downarrow & & \downarrow [A, Uh] \\ UY & \xrightarrow{\Phi_Y} & [A, UY] \end{array} \quad (4.56)$$

commutes, i.e.  $[A, Uh] \circ \Phi_X = \Phi_Y \circ Uh$ .

This definition clearly yields a profunctor  $U\text{-MRS} : C^{\text{op}} \times C \rightarrow \mathbf{Set}$ , where the reindexing of the repair maps is simply given by composition:  $\ell^* \Phi : U \Rightarrow [A, U] \Rightarrow [A', U]$  if  $(A, B) \xrightarrow[r]{\ell} (A', B')$  is an arrow in  $C^{\text{op}} \times C$ . Then we can attach a tabulator to  $U$ -coherent systems, specialising Definition 2.24.

However, the notion is a bit ‘slippery’ as it requires ad hoc constraints to recover incoherent systems.

**Remark 4.38.** As is evident, one way in which we can recover incoherent  $(M, R)$ -systems is to fix the domain of  $U$  to be terminal, i.e. to require that  $U : \{*\} \rightarrow C$  picks the object  $B$ ; in greater generality some sort of compatibility between  $U$  and  $f$  has to be hard-coded to recover an incoherent  $(M, R)$ -system: something along the lines of ‘the fibre over  $f$  of  $U$  must be a singleton’, or at least pointed by a distinguished element  $h : X \rightarrow Y$  such that

$$\begin{array}{ccc} UX & \xrightarrow{Uh=f} & UY \\ \parallel & & \parallel \\ A & & B \end{array} \quad (4.57)$$

Such conditions are difficult to ensure in practice given their vagueness; even more problematic is the fact that the compatibility with the pair  $(h, Uh = f)$  has to be hard-coded as well in all proofs and constructions. So, this flavour of  $(M, R)$ -system is only a convenient one if the user abandons the idea of always being able to recover an incoherent system from a  $U$ -coherent one.

**4.3.5. Profunctorial  $(M, R)$ -systems.** This is the most abstract approach we have found so far (and it relies heavily on techniques from bi- and double category theory, for which we refer the reader to [Shu08, GP99a]).

Let  $\mathcal{X}$  be a category with a terminal object; we call a profunctor  $p : \mathcal{X} \times \mathcal{Y}^{\text{op}} \rightarrow \mathbf{Set}$  *reduced* if the curried functor  $p^\dagger : \mathcal{X} \rightarrow [\mathcal{Y}^{\text{op}}, \mathbf{Set}]$  preserves the terminal object.

We can consider  $(M, R)$ -systems that are ‘fully profunctorial’: in those,  $\Phi$  is a natural transformation of type  $p \Rightarrow [A, \text{cod}]_*$ , where  $p$  is reduced and  $[A, \text{cod}]_*$  is the *companion* of the functor  $[A, \text{cod}]$ , defined as the profunctor

$$[A, \text{cod}]_* : C^{\text{op}} \times A/C \longrightarrow \mathbf{Set} : (X, h : A \rightarrow Y) \longmapsto C(X, [A, Y]) \cong C(X \otimes A, Y). \quad (4.58)$$

In other words, in a language that is closer to double categories [GP99a], we consider a 2-cell

$$\begin{array}{ccc} C/B & \xrightarrow{p} & A/C \\ \text{dom} \downarrow & \not\Downarrow \Phi & \downarrow \text{cod} \\ C & \xrightarrow{[A, -]_*} & C \end{array} \quad (4.59)$$

in the double category  $\mathbb{Cat}$  of categories and profunctors.

Note that since  $p$  is reduced, the curried functor  $p^\dagger : C/B \rightarrow [(A/C)^{\text{op}}, \mathbf{Set}]$  maps the terminal object of  $C/B$ , namely  $1_B$ , to the terminal object of  $[A/C, \mathbf{Set}]$ , namely the constant functor at a terminal object of  $\mathbf{Set}$ ; in other words,  $p(1_B, v)$  is a singleton for every  $v \in A/C$ . It is then immediate to see that this notion recovers the incoherent  $(M, R)$ -systems of Definition 3.5 by evaluating  $\Phi$  at the component  $(1_B, f : A \rightarrow B)$ , since

$$\Phi_{1_B, f} : p(1_B, f) = \{*\} \longrightarrow C(\text{dom}(1_B), [A, \text{cod}(f)]) = C(B, [A, B]). \quad (4.60)$$

**Definition 4.39** (Profunctorial  $(M, R)$ -system,  $\langle\langle f, p, \Phi \rangle\rangle$ ). Let  $C$  be a monoidal closed category; let  $A, B$  be objects of  $C$ .

A *profunctorial  $(M, R)$ -system* based at  $(A, B)$  consists of a triple  $\langle\langle f, p, \Phi \rangle\rangle$  where

- $f : A \rightarrow B$  is an arrow of  $C$ ;
- $p : (C/B) \times (A/C)^{\text{op}} \rightarrow \mathbf{Set}$  is a profunctor;
- $\Phi : p \Rightarrow [A, \text{cod}]_*$  is a cell as in (4.59).

Call  $\mathbf{\Pi\text{-MRS}}(A, B)$  the collection of all profunctorial  $(M, R)$ -systems based at  $(A, B)$ .

Evidently, the profunctor  $p$  varies with the fibre because it is (more precisely, its domain and codomain are) indexed over  $A, B$ . Resorting to the well-known graphical representation of cells in the double category of profunctors, we can depict a given profunctorial  $(M, R)$ -system based at  $(A, B)$  as a pair

$$\begin{array}{ccc} A & & C/B \xrightarrow{p} A/C \\ f \downarrow & \text{dom} \downarrow & \not\Downarrow \Phi \quad \downarrow \text{cod} \\ B & & C \xrightarrow{[A, -]_*} C \end{array} \quad (4.61)$$

We now aim to prove that

**Proposition 4.40.** The assignment  $(A, B) \mapsto \mathbf{\Pi\text{-MRS}}(A, B)$  is a profunctor

$$\mathbf{\Pi\text{-MRS}} : C^{\text{op}} \times C \longrightarrow \mathbf{Set}. \quad (4.62)$$

**Remark 4.41.** This proposition carries no Agda marker. The file `Variants/Profunctorial.agda` formalises the profunctorial  $(M, R)$ -systems of Definition 4.39 and proves that they form a se-toid, but it does not build the infrastructure for the conjoint of  $v$ .

To this end, given an arrow  $(A, B) \xrightleftharpoons[v]{u} (A', B')$  in  $C^{\text{op}} \times C$ , we have to exhibit reindexings

$$\mathbf{\Pi}\text{-MRS}(u, v) : \mathbf{\Pi}\text{-MRS}(A, B) \longrightarrow \mathbf{\Pi}\text{-MRS}(A', B'). \quad (4.63)$$

Let's argue graphically; given a system like in (4.61), and  $(A, B) \xrightleftharpoons[v]{u} (A', B')$ , we define the image of  $\langle\langle f, p, \Phi \rangle\rangle$  under  $\mathbf{\Pi}\text{-MRS}(u, v)$  as the cell

$$\begin{array}{ccccc} A' & \xrightarrow{u} & A & & C/B' & \xrightarrow{v^\dagger} & C/B & \xrightarrow{p} & A/C & \xrightarrow{u_\dagger} & A'/C \\ & & \downarrow f & & \downarrow \text{dom} & & \downarrow \text{dom} & & \downarrow \text{cod} & & \downarrow \text{cod} \\ B' & \xleftarrow{v} & B & & C & \xrightarrow{=} & C & \xrightarrow{[A, -]_*} & C & \xrightarrow{=} & C \\ & & & & & & \downarrow u^* & & & & \\ & & & & & & [A', -] & & & & \end{array} \quad (4.64)$$

where  $u_\dagger$  and  $v^\dagger$  are the profunctors respectively defined as follows; ‘companion’ and ‘conjoint’ have here the meaning they have in double category theory [Shu08], and the notation for the former is the one already used for  $[A, \text{cod}]_*$  above.

- $u_\dagger : A/C \rightarrowtail A'/C$  is the *companion* of the functor

$$u/C : A/C \longrightarrow A'/C : (a : A \rightarrow X) \longmapsto (a \circ u : A' \rightarrow X) \quad (4.65)$$

induced by  $u : A' \rightarrow A$  between the coslices: its value at  $a : A \rightarrow X$  and  $a' : A' \rightarrow X'$  is the hom-set  $(A'/C)(a', (u/C)a)$ , i.e. the set of arrows  $h : X \rightarrow X'$  of  $C$  such that  $h \circ a' = a \circ u$ .

- $v^\dagger : C/B' \rightarrowtail C/B$  is the *conjoint* of the functor

$$C/v : C/B \longrightarrow C/B' : (x : X \rightarrow B) \longmapsto (v \circ x : X \rightarrow B') \quad (4.66)$$

induced by  $v : B \rightarrow B'$  between the slices: its value at  $x' : X' \rightarrow B'$  and  $x : X \rightarrow B$  is the hom-set  $(C/B')((C/v)x, x')$ , i.e. the set of arrows  $h : X \rightarrow X'$  of  $C$  such that  $v \circ x = x' \circ h$ .

Both exist for every functor  $u, v$ , and this specific choice is forced by the direction of the reindexing function; the composite profunctor  $u_\dagger \odot p \odot v^\dagger$  is the coend

$$\begin{aligned} (v^\dagger \odot p \odot u_\dagger)(x', a') &= \int^{x \in C/B, a \in A/C} v^\dagger(x, x') \times p(a, x) \times u_\dagger(a', a) \\ &\cong \int^{x \in C/B} \int^{a \in A/C} (C/B')((C/v)x, x') \times p(x, a) \times (A'/C)(a', (u/C)a). \end{aligned}$$

## 5. CLOSURE UNDER EFFICIENT CAUSATION

Closure under efficient causation is the claim that a system contains, within its own organisation, the causes responsible for its activity. An “efficient cause” here is a function responsible for a transformation, not merely a material component. Rosen’s relational formulation therefore concerns organisational rather than material closure: every such cause is “entailed within the system” [Lou09, pp. 215–220]. For an  $(M, R)$  system this is expressed by a metabolism  $f : A \rightarrow B$ , a repair/replacement map  $\Phi : B \rightarrow [A, B]$ , and a product  $b \in B$  such that  $\Phi(b) = f$ . The product therefore provides the value at which repair reconstructs the metabolism, so no external efficient cause is required. This is the Set-level formulation of metabolic closure. “Organisational invariance” means that this functional organisation is maintained even while the material components participating in it change; Letelier et al. identify it with metabolic closure [LSAGA<sup>+</sup>04, pp. 450–455]; [LSAGA<sup>+</sup>06, pp. 949–958]. Louie also gives its diagrammatic counterpart: each connected component must contain a closed path through all efficient-cause arrows [Lou09, pp. 219–220].

Several authors have tried to make this proposal mathematically precise. In Rosen’s extended diagram, the higher-order map  $\beta$  is the “repair-of-repair” map which assigns to a metabolism  $f$  the repair map  $\Phi$  that realises it. Letelier et al. emphasise that  $\beta$  must recover repair from  $f$ , so closure involves evaluation and uniqueness/invertibility, not mere feedback [LSAGA<sup>+</sup>06, pp. 953–961]. More explicitly, with

$$\text{ev} : [A, B] \times A \longrightarrow B, \quad (g, a) \longmapsto g(a), \quad (5.1)$$

closure asks for a section of the relevant evaluation map in the following pointwise sense: the metabolism  $f$  must lie in the image of evaluation at a product  $b$ , namely  $\Phi(b) = f$ ;  $\beta$  records the corresponding inverse choice. Cárdenas et al. similarly insist that the internal witness must not be confused with the product object  $B$ , and argue that this inverse-evaluation structure is missing from a simple lambda-calculus encoding [CLG<sup>+</sup>10, pp. 80–91]. By contrast, Mossio, Longo, and Stewart translate Rosen’s three equations into untyped  $\lambda$ -calculus:  $B = f(A)$ ,  $f = \Phi(B)$ , and  $\Phi = B(f)$ . Substitution reduces these to a single self-application equation, which they solve explicitly with the fixed-point combinator  $Y$ : for a suitable term  $G$ , take  $f = YG$ , so that  $f = Gf$ ;  $B$  and  $\Phi$  are then obtained by substitution [MLS09, pp. 489–498]. Since untyped  $\lambda$ -calculus is a canonical formalism for computation, they conclude that this expression of closure is computable. Cárdenas et al. reply that the construction does not represent the inverse-evaluation requirement, so the dispute concerns what a faithful formalisation must preserve: self-reference alone, or self-reference together with repair’s internal inverse structure [CLG<sup>+</sup>10, pp. 80–91].

The definition below transports the Set-level witness into a monoidal closed category:  $b \in B$  becomes a generalised element  $b_0 : I \rightarrow B$ , and  $\Phi(b) = f$  becomes the displayed evaluation square. A closure point means that seeding repair at  $b_0$  regenerates the whole process  $f$ . This generalised-element formulation, and the equivalence of its two closed-category presentations, are extensions proved here rather than results attributed to Rosen or Letelier. Naturality can strengthen closure for one process to a universal point for all processes with input  $A$ , as explained below.

We use *metabolic closure* for this specific  $(M, R)$  condition. Related proposals should not be identified with it without further hypotheses: Montévil and Mossio’s account concerns mutual dependence among constraints at different scales [MM15, pp. 179–191], while Brown and Vitadello’s account defines network closure through cycles of mutual process enablement [BV25]. Both are relevant neighbouring notions, but neither is equivalent to the single-witness condition  $\exists b \Phi(b) = f$  by itself.

**Definition 5.1** ( $\hookrightarrow$ ). A coherent  $(M, R)$ -system *allows metabolic closure* if there exists a generalised element  $b_0 : I \rightarrow B$  such that one of these two equivalent conditions holds:

- the square

$$\begin{array}{ccc} I \otimes A & \xrightarrow{b_0 \otimes A} & B \otimes A \\ \downarrow \lambda_A & & \downarrow \hat{\Phi} \\ A & \xrightarrow{f} & B \end{array} \quad (5.2)$$

is commutative, where  $\hat{\Phi} : B \otimes A \xrightarrow{\Phi_B \otimes 1_A} [A, B] \otimes A \xrightarrow{\epsilon_B} B$  is the mate of the repair component  $\Phi_B$ ;

- the square

$$\begin{array}{ccc} I & \xrightarrow{b_0} & B \\ j_A \downarrow & & \downarrow \Phi_B \\ [A, A] & \xrightarrow{[A, f]} & [A, B] \end{array} \quad (5.3)$$

is commutative, where  $j_A : I \rightarrow [A, A]$  is the structure map of Definition 2.15. Thus  $[A, f] \circ j_A = \bar{f} : I \rightarrow [A, B]$  is the Eilenberg–Kelly name of  $f$ . This expresses the first condition using only the closed structure.

Such a  $b_0$  is called a *closure point*. The system is ‘self-sustained’ in the specific sense that its repair map, seeded at  $b_0$ , reconstructs the entire process  $f$ . The witness is internal:  $b_0$  is a point of the output object  $B$ .

**Remark 5.2.** In categorical semantics, the identity is a ‘universal term’, representing the judgement  $\{A\} \vdash A$ . Its role in the Yoneda lemma suggests testing closure against the identity process.

The natural repair family includes the component

$$\Phi_A : A \longrightarrow [A, A], \quad (5.4)$$

so we can ask for a point  $a_0 : I \rightarrow A$  making the square

$$\begin{array}{ccc} I \otimes A & \xrightarrow{a_0 \otimes A} & A \otimes A \\ \lambda_A^{-1} \uparrow & & \downarrow \hat{\Phi} \\ A & \xlongequal{\quad} & A \end{array} \quad (5.5)$$

commute. (In the category of sets, where  $A$  has genuine points and functions can be extensionally equivalent, this question becomes: is there  $a_0 \in A$  such that  $\Phi(a_0) \equiv 1_A : A \rightarrow A$ —extensional equality of functions?)

Naturality then makes  $f \circ a_0$  a closure point for every process  $f : A \rightarrow B$   $\mathcal{U}$ :

$$\Phi_B \circ f \circ a_0 = [A, f] \circ \Phi_A \circ a_0 = [A, f] \circ j_A = \bar{f}. \quad (5.6)$$

Thus

A point recovering  $1_A$  under  $\Phi_A$  gives metabolic closure for every process with input  $A$ , using the same repair family.

**Definition 5.3** (Universal closure point,  $\mathcal{U}$ ). A *universal closure point* for the coherent  $(M, R)$ -system  $\langle f, \Phi \rangle$  is a point  $u : I \rightarrow A$  such that  $\Phi_A \circ u = j_A$ , the name of  $1_A$ .

First, metabolic closure is preserved by reindexing.

**Proposition 5.4** (Reindexing is compatible with closure,  $\mathcal{U}$ ). Let  $(u, v) : (A, B) \rightarrow (A', B')$  be a morphism in the indexing category, so  $u : A' \rightarrow A$  and  $v : B \rightarrow B'$ . If  $\langle f, \Phi \rangle$  has closure point  $b_0 : I \rightarrow B$ , then its reindexing has closure point

$$b'_0 := v \circ b_0 : I \rightarrow B'. \quad (5.7)$$

*Proof.* The reindexed system has  $f' = v \circ f \circ u$  and  $\Phi'_X = [u, X] \circ \Phi_X$ . Naturality of  $\Phi$ , followed by the closure equation, gives

$$\begin{aligned} \Phi'_{B'} \circ b'_0 &= [u, B'] \circ \Phi_{B'} \circ v \circ b_0 \\ &= [u, B'] \circ [A, v] \circ \Phi_B \circ b_0 \\ &= [u, v] \circ \bar{f} = \overline{v \circ f \circ u} = \bar{f}'. \end{aligned}$$

The last equality is naturality of the name construction in both variables. This is the closure equation for  $b'_0$ .  $\square$

**5.1. Universal closure and retracts of the unit.** For a fixed repair family  $\Phi : \text{id} \Rightarrow [A, -]$ , the universal closure points form the fibre of

$$C(I, \Phi_A) : C(I, A) \rightarrow C(I, [A, A]) \cong C(A, A) \quad (5.8)$$

over  $1_A$ . Thus they depend on  $\Phi$ , independently of the process  $f$ . If the pullback

$$\begin{array}{ccc} P & \xrightarrow{\quad} & A \\ \downarrow & \lrcorner & \downarrow \Phi_A \\ I & \xrightarrow{j_A} & [A, A] \end{array} \quad (5.9)$$

along  $j_A$  exists, they correspond to *sections of the projection*  $P \rightarrow I$ .

**Proposition 5.5** (The unit-retraction criterion,  $\mathcal{U}$ ). Let  $C$  be monoidal closed and  $\Phi : \text{id} \Rightarrow [A, -]$  be natural. Write  $q := p(\Phi) : A \rightarrow I$  in the notation of ??, so that  $\Phi_I : I \rightarrow [A, I]$  is the name of  $q$ . A point  $u : I \rightarrow A$  is a universal closure point if and only if  $u \circ q = 1_A$ .

Consequently  $A$  is a retract of the monoidal unit whenever such a point exists, and the point is unique  $\mathcal{U}$ .

*Proof.* Naturality at the arrow  $u : I \rightarrow A$  gives

$$\Phi_A \circ u = [A, u] \circ \Phi_I = \overline{u \circ q} \quad (\mathcal{U})$$

where  $\overline{u \circ q}$  is the mate of  $u \circ q$ . Since  $j_A = \overline{1_A}$  and names determine arrows, this proves the equivalence. For uniqueness, suppose  $u \circ q = v \circ q = 1_A$ , and put  $e = q \circ u$ ,  $d = q \circ v$ . Then

$$e \circ d = d, \quad d \circ e = e. \quad (5.10)$$

Endomorphisms of the monoidal unit commute: composition and tensor product, transported along  $I \otimes I \cong I$ , have the same unit and satisfy the interchange law, so the Eckmann–Hilton argument applies  $\curvearrowright$ . Hence  $e = d$ , and  $v = u \circ q \circ v = u \circ d = u \circ e = u \circ q \circ u = u$ .  $\square$

Combining Proposition 5.5 and Lemma 4.3 we also obtain the following.

**Corollary 5.6** (Inputs admitting universal closure,  $\curvearrowright$ ). In a monoidal closed category, an object  $A$  admits a natural repair family with a universal closure point if and only if  $A$  is a retract of  $I$ . More precisely, every splitting

$$A \xrightarrow{q} I \xrightarrow{u} A, \quad u \circ q = 1_A, \quad (5.11)$$

yields the representable repair family  $\iota(q)$  with universal point  $u$ . It can be paired with any process  $f : A \rightarrow B$ .

The corollary allows us to *choose* the repair family; it does not assert that every family on a retract of  $I$  has a universal point. For example,  $A = k$  is the unit of  $\mathbf{Vect}_k$ , but the zero repair family has  $p(\Phi) = 0$ . Then  $u \circ p(\Phi) = 0 \neq 1_k$  for every  $u : k \rightarrow k$ , so this family has no universal point.

**Corollary 5.7** (The Cartesian case). Let  $C$  be Cartesian closed. A coherent  $(M, R)$ -system based at  $(A, B)$  admits a universal closure point if and only if  $A \cong 1$ , and the point is then unique.

*Proof.* Here  $p(\Phi) = !_A : A \rightarrow 1$ . A universal point  $u$  satisfies  $u \circ !_A = 1_A$ , while  $!_A \circ u = 1_1$  by terminality, so  $A \cong 1$ . Conversely, if  $A \cong 1$ , its unique point retracts  $!_A$ , and Proposition 5.5 applies.  $\square$

**Remark 5.8** (The role of the Bernstein centre). The Bernstein centre classifies *whole repair families*, as in Lemma 4.15; universal closure points are the left inverses of  $p(\Phi) : A \rightarrow I$ .

These objects answer different questions. The unit-retraction criterion requires neither symmetry nor existence of the Bernstein centre, and does not require  $\Phi$  to be representable. Nor does the existence of universal closure points force representability: in  $\mathbf{Set}^{C^2}$ , both families on  $A = 1$  from Proposition 4.23 have the unique universal point, although only one is representable. The process  $1 \rightarrow B$  is still arbitrary.

**5.2. Idempotent scalars and examples.** The possible inputs in Corollary 5.6 are controlled by idempotents in the (commutative, by Eckmann–Hilton) monoid of *scalars*  $\mathbf{End}_C(I)$ . This gives a uniform way to find categories with many such inputs. In particular, a universal closure point  $u$  determines the idempotent scalar  $q \circ u$ , where  $q = p(\Phi) \curvearrowright$ : the identity  $u \circ q = 1_A$  gives  $(q \circ u)^2 = q \circ (u \circ q) \circ u = q \circ u$ .

**Proposition 5.9** (Retracts of the unit and scalars). Isomorphism classes of retracts of  $I$  correspond bijectively to split idempotents in  $\mathbf{End}_C(I)$ , hence to all idempotent scalars when  $C$  is idempotent-complete. Writing  $I_e$  for a splitting object of  $e$ , one has

$$I_e \otimes I_d \cong I_{ed}, \quad I_e \otimes I_e \cong I_e. \quad (5.12)$$

Thus, when  $C$  is also monoidal closed, these split idempotents classify the input objects admitting some universally closed repair family.

*Proof.* A retraction  $u \circ q = 1_A$  gives  $e = q \circ u$  with  $e^2 = e$ , and a splitting of  $e$  is precisely such a pair. It remains to check that the idempotent depends only on the isomorphism class of  $A$ , not on its chosen retraction data. Suppose  $u \circ q = 1_A$ ,  $v \circ r = 1_B$ , and  $h : A \xrightarrow{\cong} B$ . The scalars

$$a = q \circ h^{-1} \circ v, \quad b = r \circ h \circ u \quad (5.13)$$

satisfy  $ab = q \circ u = e$  and  $ba = r \circ v = d$ . Scalars commute by the Eckmann–Hilton argument in Proposition 5.5, so  $e = d$ . Conversely two splittings of the same idempotent are isomorphic:  $v \circ q : A \rightarrow B$  and  $u \circ r : B \rightarrow A$  are inverse, since

$$(u \circ r) \circ (v \circ q) = u \circ e \circ q = 1_A, \quad (5.14)$$

and similarly in the other order. Finally tensoring splittings of  $e$  and  $d$  gives a splitting of  $e \otimes d$  on  $I \otimes I$ . Under  $I \otimes I \cong I$ , this scalar is  $ed$ , again by interchange and the unit laws. This proves the tensor formulas, and Corollary 5.6 proves the last assertion.  $\square$

Multiplication makes the idempotent scalars a meet-semilattice, with  $e \leq d$  defined by  $ed = e$ . In an additive idempotent-complete category,  $1 - e$  supplies a complementary summand,  $I \cong I_e \oplus I_{1-e}$ . In general no such complement is available, as the locale-module example below shows.

**Example 5.10** (Modules over a commutative ring). Let  $C = \mathbf{Mod}(R)$  for a commutative unital ring  $R$ , with its usual tensor product and unit  $R$ . An endomorphism of  $R$  is multiplication by its value at 1, so the scalar ring is  $R$ . Every idempotent  $e \in R$  splits as

$$eR \xrightarrow{\text{incl}} R \xrightarrow{r \mapsto er} eR. \quad (5.15)$$

These are all the unit retracts, by Proposition 5.9. For a specified coherent family, naturality along the maps  $R \rightarrow X$  gives  $\Phi_X(x)(a) = q(a)x$ , with  $q = p(\Phi) : A \rightarrow R$ . Its universal point is an element  $a_0 \in A$  satisfying

$$q(a)a_0 = a \quad (a \in A). \quad (5.16)$$

Then  $q$  identifies  $A$  with  $eR$ , where  $e = q(a_0)$ : its left inverse is  $r \mapsto ra_0$ , and the associated endomorphism of  $R$  is multiplication by  $e$ . Conversely, the inclusion label on  $eR$  gives universal point  $e$ . Uniqueness follows from Proposition 5.5. In particular, over a field the only possible input objects are 0 and  $k$ ; on 0 the unique family is universally closed, and on a one-dimensional input exactly the nonzero labels are.

**Example 5.11** (Families of vector spaces). For a set  $S$ , take  $C = \prod_{s \in S} \mathbf{Vect}_k$  with componentwise tensor and internal hom. The unit is  $I = (k)_{s \in S}$ , its scalar ring is  $\prod_{s \in S} k$ , and its idempotents are characteristic functions of subsets  $U \subseteq S$ . Their splitting objects are

$$(I_U)_s = \begin{cases} k, & s \in U, \\ 0, & s \notin U. \end{cases} \quad (5.17)$$

The inclusion and projection are componentwise the identity on  $U$  and zero off  $U$ . Thus there are exactly  $2^{|S|}$  isomorphism classes of unit retracts, and  $I_U \otimes I_V \cong I_{U \cap V}$ . The inclusion label gives a universally closed repair family on every  $I_U$ .

**Example 5.12** (Quasi-coherent sheaves). Let  $X$  be a quasi-compact quasi-separated scheme and take the symmetric monoidal closed category  $\mathbf{QCoh}(X)$  with tensor product over  $\mathcal{O}_X$ . The unit is  $\mathcal{O}_X$ , and

$$\mathbf{End}_{\mathcal{O}_X}(\mathcal{O}_X) = \Gamma(X, \mathcal{O}_X). \quad (5.18)$$

Indeed, an endomorphism is multiplication by the image of the global section 1. Its idempotents correspond to clopen subsets  $U \subseteq X$ : an idempotent has value 0 or 1 in every local ring, and the loci of these two values are complementary open sets; conversely the sections 1 on  $U$  and 0 on its complement glue to an idempotent. The corresponding summand  $e\mathcal{O}_X$  equals  $\mathcal{O}_X$  on  $U$  and vanishes off  $U$ , and inclusion followed by multiplication by  $e$  splits it. Every unit retract has this form. Thus a nonempty connected scheme has only the zero and whole-unit retracts, while a disjoint union of  $m$  nonempty connected schemes has  $2^m$ . This is the geometric version of Example 5.10.

**Example 5.13** (Bimodules and central idempotents). For a unital, possibly noncommutative ring  $R$ , the category of  $R$ -bimodules is monoidal biclosed under  $\otimes_R$ , with unit  $R$ . A bimodule endomorphism  $h : R \rightarrow R$  satisfies

$$h(r) = rh(1) = h(1)r, \quad (5.19)$$

so its scalar ring is the ordinary ring-theoretic centre  $Z(R)$ . Hence unit retracts are precisely the bimodules  $eR$  for *central* idempotents  $e$ , with the same inclusion and multiplication splitting as above. Noncentral idempotents need not define bimodule retracts. For  $R = \prod_{j=1}^m M_{n_j}(k)$ , the centre is  $k^m$ , giving exactly  $2^m$  unit retracts, indexed by subsets of the matrix blocks.



**Example 5.14** (Modules over a locale). Let  $L$  be a frame, regarded as a commutative unital quantale with multiplication  $a \wedge b$  and unit  $\top$ . Consider complete  $L$ -modules: sup-lattices  $M$  with a unital associative action  $L \times M \rightarrow M$  preserving arbitrary joins in both variables, and join-preserving  $L$ -linear maps. Their relative tensor product represents maps that preserve joins separately and satisfy

$$b(a \cdot m, n) = a \cdot b(m, n) = b(m, a \cdot n). \quad (5.20)$$

It can be constructed from the free sup-lattice on  $M \times N$  by imposing these relations and the separate-join relations. Its unit is  $L$ , and the internal hom is the sup-lattice of  $L$ -linear join-preserving maps, with pointwise joins and action. Currying the displayed bilinear maps gives the closed adjunction. Every endomorphism of the unit has the form

$$h(x) = h(x \wedge \top) = x \wedge h(\top), \quad (5.21)$$

so the scalar monoid is  $(L, \wedge, \top)$ . Every scalar  $a$  is idempotent, with splitting

$$\downarrow a \xrightarrow{\text{incl}} L \xrightarrow{x \mapsto a \wedge x} \downarrow a. \quad (5.22)$$

Both maps preserve joins and the  $L$ -action, and their composite on  $\downarrow a$  is the identity. Hence the frame  $L$  indexes all unit retracts. For  $L = [0, 1]$ , with joins given by suprema and multiplication by minimum, these are the pairwise nonisomorphic modules  $[0, a]$  for all  $a \in [0, 1]$ . If  $0 < a < 1$ , there is no scalar complement  $b$ :  $\min(a, b) = 0$  forces  $b = 0$ , whence  $\max(a, b) \neq 1$ . Here we use the *category of modules over  $L$* . The posetal monoidal category  $L$  itself, as in Example 3.11, has only identity endomorphisms and therefore only the unit as a unit retract.

**Remark 5.15** (Scarcity and abundance). The number of unit retracts is governed by split idempotent scalars, rather than by the size of the category or the number of its repair families. In  $\mathbf{Vect}_k$ , in representations of a group over  $k$ , and in finite-dimensional Hilbert spaces, the scalar ring is a field and the only unit retracts are 0 and  $I$ . In every Cartesian category there is only  $I$ , even when the Bernstein centre is nontrivial. The product, geometric, and equivariant examples above supply complementary pieces, while locale modules supply retracts without complements. In each monoidal closed example, Corollary 5.6 turns the explicit retraction into a repair family with its unique universal point.

**5.3. Extended (M,R)-systems, or “*quis reficiet ipsos refectores?*”.** The repair component of an (M,R)-system is a piece of machinery like any other, and machinery wears out. In [Cas88, p. 145] Casti puts the question this subsection is about in the bluntest possible form: ‘who repairs the repairers? It would lead to a useless infinite regress to introduce another level of repair mechanisms, but what is the alternative?’.

The same difficulty is stated by Letelier et al. as the observation that ‘the same wear-and-tear argument that was applied to  $M$  applies equally well to the  $R$ ’, so that invoking a second-level repairer ‘just raises the new question of how to replace the second-level repairers, and is thus no solution at all’ [LSAGA<sup>+</sup>04, p. 451].

Rosen’s own answer is a *replication* map. Already in [Ros59, Ros66b] he observes that a set  $X$  embeds into  $H(H(X, Y), Y)$  by sending  $x$  to the evaluation  $\hat{x} : f \mapsto f(x)$  (‘the abstract version of the familiar embedding of a vector space into its second dual space’, as he puts it in [Ros72]), and applies this with  $X = B$ ,  $Y = H(A, B)$ : an element  $b \in B$  then yields  $\hat{b} \in H(H(B, H(A, B)), H(A, B))$ , and when  $\hat{b}$  is injective its inverse is a map

$$\beta = \hat{b}^{-1} : H(A, B) \longrightarrow H(B, H(A, B)) \quad (5.23)$$

sending the single value  $\Phi(b) = f$  to the whole repair map  $\Phi$ . Louie’s account [Lou09, §§11.11–11.17] isolates what matters in this construction. First, that the *form* of  $\beta$  is incidental and only its codomain is forced: a replication map is anything whose values are repair maps, so  $\beta : Z \rightarrow H(B, H(A, B))$  for some  $Z$ , and the choice  $Z = H(A, B)$  is made only ‘for the convenience of iterative combination’. Second, that with this choice the pair  $(\Phi, \beta)$  is again an (M,R)-system,

$$B \xrightarrow{\Phi} H(A, B) \xrightarrow{\beta} H(B, H(A, B)), \quad (5.24)$$

whose metabolism is the repair of the old one, and that ‘one may sequentially extend this formalism ad infinitum’. Third, and crucially, that the injectivity hypothesis on  $\hat{b}$  (two repair maps agreeing at  $b$  agree everywhere, the abstract form of the one-gene-one-enzyme hypothesis) is what allows the sequence to be *truncated*: ‘the apparent infinite sequence of maps that may arise with the iteration of “repair the repair” is truncated by a mathematical argument that turns it into a hierarchical cycle of maps instead’. Rosen already noted the obstruction posed by categories that are too rich in maps [Ros66b, p. 150]. Our universal-point criterion in Proposition 5.5 instead concerns a retraction of the unit; the tower results below use that criterion rather than assuming inverse evaluation.

The literature is divided on exactly how much of this is essential. Letelier et al. read  $\beta$  as an inverse evaluation  $\text{ev}_b^{-1}$  and regard its invertibility, unusual for an evaluation map, as the substance of Rosen’s central result [LSAGA<sup>+</sup>04, pp. 451–453]; Cárdenas et al. insist that the internal witness must not be confused with the product object  $B$ , and that a formalisation dropping the inverse-evaluation requirement has not formalised closure [CLG<sup>+</sup>10, pp. 86–91]; Mossio, Longo and Stewart, on the other hand, hold that the regress is ‘perfectly handled by recursion’, the tower being the Böhm tree of a fixed-point combinator [MLS09, pp. 489–498]. We have already declined to arbitrate this dispute in the discussion preceding Definition 5.1, and we decline again here; it suffices to say that one reason why little attention has been given to this circle of ideas is that to any trained mathematician the existence of an injective function  $(B^A)^B \rightarrow B^A$  sounds extremely fishy.

Yet, it is worth saying what our formalism does with it. Rather than truncating the tower by fiat, we *build* it, as a genuine cofiltered limit of categories (Construction 5.16 and Definition 5.17), and only then ask which towers are governed by a single global datum. That question is not the same as closure at one level: metabolic closure in the sense of Definition 5.1 asks one level to reconstruct its own process from within, and there is no a priori reason for countably many such reconstructions to cohere. A truly self-sustaining process is one where each closure point at level  $n$  can be fed to the repair map one level higher, still being a closure point.

Isolating the extra compatibility is the point of *tower-coherence* (Remark 5.21 and Definition 5.22), which Proposition 5.23 identifies, under its stated hypotheses, with a single transformation defined on the colimit of the tower. Finite-stage determination of a tower-coherent family requires an additional condition: Proposition 5.27 gives one in terms of compatible universal closure points. An *extended* (M,R)-system consists of a chain

$$F_0 \longrightarrow F_1 \longrightarrow F_2 \longrightarrow F_3 \longrightarrow \dots \quad (5.25)$$

that follows the ‘Fibonacci style’ inductive rule

$$\begin{cases} F_0 = A \\ F_1 = B \\ F_{k+2} = [F_k, F_{k+1}] \end{cases} \quad (5.26)$$

So, in particular, an (M,R)-system of length 3 looks like

$$A \longrightarrow B \longrightarrow [A, B] \longrightarrow [B, [A, B]] \quad (5.27)$$

and one of length 4 looks like

$$A \longrightarrow B \longrightarrow [A, B] \longrightarrow [B, [A, B]] \longrightarrow [[A, B], [B, [A, B]]]. \quad (5.28)$$

One would like to study these longer (M,R)-systems with the intuition that they provide repair-of-repair functions  $\Phi_n$  for the process  $\Phi_{n-1}$ . However, one faces a problem analogous to the one that led us to introduce coherent (M,R)-systems: the dependence on  $B$  of a two-step system  $A \rightarrow B \rightarrow [A, B]$  is not bifunctorial.

Here we face a similar problem, with one additional layer of complexity in the dependence of an extended (M,R)-system on the pair  $(A, B)$ ; even assuming a simple description for (5.27) above, what then with sequences of length four, five, and so on?

The inductive nature of this idea—already a half-baked definition, but requiring refinement—suggests a similarly inductive mechanism for constructing a family of total categories  $\mathbf{MRS}(n)$

containing sequences like (5.25) up to length  $n$ , each equipped with a canonical “first edge” functor, as follows:

**Construction 5.16** ( $(M, R)$ -systems of arbitrary length,  $\mathcal{U}$ ).

- Start the induction by defining  $(M, R)$ -systems of length 2 as the total category of Definition 4.10, equipped with the functor that chooses  $f$ :

$$[-]_f : \mathbb{M}\mathbf{RS} \longrightarrow C^{[0 \rightarrow 1]} \quad (5.29)$$

- Note that by Proposition 3.44 above, one has another functor into  $C^{[0 \rightarrow 1]}$  that chooses  $\Phi$ , but defined from the *category of elements* of the profunctor  $\mathbf{MRS}$ :

$$\wr_{\Phi} : \mathcal{E}(\mathbf{MRS}) \longrightarrow C^{[0 \rightarrow 1]} \quad (5.30)$$

The category  $\mathbf{MRS}(3)$  is then defined as the pullback

$$\begin{array}{ccc} \mathbf{MRS}(3) & \longrightarrow & \mathbb{M}\mathbf{RS} \\ \downarrow & \lrcorner & \downarrow [-]_f \\ \mathcal{E}(\mathbf{MRS}) & \xrightarrow{[-]_{\Phi}} & C^{[0 \rightarrow 1]} \end{array} \quad (5.31)$$

Inspecting the universal property of such a pullback, one easily sees that an object of  $\mathbf{MRS}(3)$  consists of the following data

- an  $(M, R)$ -system of length 2 in  $\mathbb{M}\mathbf{RS}$ , denoted  $(f, \Phi)$ , so that  $\Phi_f = B \rightarrow [A, B]$ ;
- an  $(M, R)$ -system of length 2 in  $\mathcal{E}(\mathbf{MRS})$ , denoted  $(g, \Psi)$ , so that  $\Psi_g = Y \rightarrow [X, Y]$ ;
- subject to the condition that  $\Phi_f = g$ :

$$\begin{array}{ccccc} A & \longrightarrow & B & \longrightarrow & [A, B] \\ & & \parallel & & \parallel \\ X & \longrightarrow & Y & \longrightarrow & [X, Y] \end{array} \quad (5.32)$$

- To proceed with the induction, observe that there exists a ‘first edge’ functor

$$[-]_f^3 : \mathbf{MRS}(3) \longrightarrow C^{[0 \rightarrow 1]} \quad (5.33)$$

such that  $[A \xrightarrow{f} B \rightarrow [A, B] \rightarrow [B, [A, B]]] = f$ . Then, one can define

$$\begin{array}{ccc} \mathbf{MRS}(n+1) & \longrightarrow & \mathbf{MRS}(n) \\ \downarrow & \lrcorner & \downarrow [-]_f^n \\ \mathcal{E}(\mathbf{MRS}) & \xrightarrow{[-]_{\Phi}} & C^{[0 \rightarrow 1]} \end{array} \quad (5.34)$$

which yields ‘gluing data’ for a pair of an  $n$ -step and a 2-step  $(M, R)$ -system to identify the  $\Phi$  part of one with the  $f$  part of the other.

This provides a reasonable, self-contained interpretation of what a sequence like (5.25) might mean formally, without entangling oneself in explicit reasoning about the parametric dependence of the construction. However, we can go further: the canonical projections from the pullbacks assemble into a cofiltered sequence of categories

$$\mathbf{MRS}(2) \longleftarrow \mathbf{MRS}(3) \longleftarrow \mathbf{MRS}(4) \longleftarrow \mathbf{MRS}(5) \longleftarrow \dots \quad (5.35)$$

**Definition 5.17** ( $\mathcal{U}$ ). The category of infinitely extended  $(M, R)$ -systems is defined as the limit (in  $\mathbf{Cat}$ ) of the cofiltered chain

$$\mathbf{MRS}_{\infty} := \lim \left( \mathbf{MRS}(2) \longleftarrow \mathbf{MRS}(3) \longleftarrow \mathbf{MRS}(4) \longleftarrow \mathbf{MRS}(5) \longleftarrow \dots \right) \quad (5.36)$$

In fact, the chain above lives entirely in the slice  $\mathbf{Cat}/C^{[0 \rightarrow 1]}$ .

**Remark 5.18.** There exists a cofiltered sequence of *sliced* categories

$$\begin{array}{c} \text{MRS}(2) \xleftarrow{\Pi_2} \text{MRS}(3) \xleftarrow{\Pi_3} \text{MRS}(4) \xleftarrow{\Pi_4} \text{MRS}(5) \xleftarrow{\Pi_5} \dots \\ \downarrow V_2 \quad \downarrow V_3 \quad \downarrow V_4 \quad \downarrow V_5 \\ \text{C}^{[0 \rightarrow 1]} \end{array} \quad (5.37)$$

where the functor  $V_n$  is defined as follows:

$$\dots \quad (5.38)$$

**Definition 5.19** ( $\mathcal{U}$ ). A similar construction can be performed for incoherent systems: we sketch the construction but we refer the reader to the Agda codebase  $\mathcal{U}$ . Given that metabolic closure (in the sense of Definition 5.1) only makes sense for coherent systems, however, we mainly work in the setting of Definition 5.17 above.

**Remark 5.20.** A coherent extended system contains more data than the chain (5.25). For example, a system of length four consists of a process  $f : A \rightarrow B$  and natural families  $\Phi_1 : \text{id} \Rightarrow [A, -]$ ,  $\Phi_2 : \text{id} \Rightarrow [B, -]$ , and  $\Phi_3 : \text{id} \Rightarrow [[A, B], -]$ . Evaluating at the successive output objects gives the chain

$$A \xrightarrow{f} B \xrightarrow{\Phi_{1,B}} [A, B] \xrightarrow{\Phi_{2,[A,B]}} [B, [A, B]] \xrightarrow{\Phi_3} [[A, B], [B, [A, B]]]. \quad (5.39)$$

All components of these families are available, not merely those shown in the chain. Closure can fail either because a required nontrivial family is unavailable, or because its selected component is unable to reconstruct the preceding process. To put it simply, in two (M,R)-systems ‘spliced’ together according to the rule (??), one can ask for a coherence condition stronger than just ‘the first and second system both admit a closure point’.

**Remark 5.21.** More precisely, levelwise metabolic closure gives each process its own reconstruction witness, and does not relate the repair families at different levels. We introduce *tower-coherence* to express this additional separate compatibility, and then examine how it interacts with closure points.

**5.4. Tower-coherence.** To start the discussion, take the case  $n = 4$  above and apply the functor  $[-, X]$  to (5.39), obtaining

$$[A, X] \xleftarrow{[f,X]} [B, X] \xleftarrow{[\Phi_{1,B},X]} [[A, B], X] \xleftarrow{[\Phi_{2,[A,B]},X]} [[B, [A, B]], X] \quad (5.40)$$

Tower-coherence is then the requirement that the components of the successive repair families, probed at one and the same object  $X$ , assemble into a cone over this diagram: each  $\Phi_{n,X} : X \rightarrow [F_{n-1}, X]$  must be the restriction of  $\Phi_{n+1,X}$  along the connecting map, so that the whole tower is determined by probing it as deep as one likes.

We first formulate this compatibility independently of the existence of a colimit, and then relate it to the classification of repair families and to universal closure points.

**Definition 5.22** (Tower-coherence).

$$\begin{array}{c} X \\ \downarrow \Phi_{1,X} \quad \searrow \Phi_{2,X} \quad \searrow \Phi_{3,X} \quad \searrow \Phi_{4,X} \\ [A, X] \xleftarrow{[f,X]} [B, X] \xleftarrow{[\Phi_{1,B},X]} [[A, B], X] \xleftarrow{[\Phi_{2,[A,B]},X]} [[B, [A, B]], X] \end{array} \quad (5.41)$$

Let  $\mathcal{C}$  be monoidal closed. Fix an infinitely extended (M,R)-system with  $F_0 = A$ ,  $F_1 = B$ ,  $F_{n+1} = [F_n, F_n]$ , and natural transformations  $\Phi_n : \text{id} \Rightarrow [F_{n-1}, -]$  for  $n \geq 1$ . Set

$$c_0 := f : F_0 \rightarrow F_1, \quad c_n := \Phi_{n,F_n} : F_n \rightarrow F_{n+1} \quad (n \geq 1). \quad (5.42)$$

The level with input  $F_{n-1}$  is the pair  $(c_{n-1}, \Phi_n)$ . The gluing in Construction 5.16 specifies its repair component as the next process; it imposes no equation between the two natural transformations

$\Phi_n$  and  $\Phi_{n+1}$ . We say the tower is *tower-coherent*, or *self-coherent*, if, for every  $n \geq 1$  and every  $X \in C_0$ ,

$$\Phi_{n,X} = [c_{n-1}, X] \circ \Phi_{n+1,X}, \quad (5.43)$$

i.e. the triangles pictured above commute for every  $X$ . For a finite family  $\Phi_1, \dots, \Phi_N$ , the same definition uses  $1 \leq n < N$ . Iteration gives

$$\Phi_{n,X} = [c_{m-2} \circ \dots \circ c_{n-1}, X] \circ \Phi_{m,X} \quad (1 \leq n < m). \quad (5.44)$$

Thus each finite compatible family is determined by its top transformation and the intervening connecting maps.

**Proposition 5.23** (The colimit description). Suppose  $C$  is symmetric monoidal closed and the colimit  $F_\infty := \text{colim}_{n \geq 0} F_n$  exists, with cocone maps  $\kappa_n : F_n \rightarrow F_\infty$ . The given extended system is tower-coherent if and only if there exists a natural transformation  $\Phi_\infty : \text{id} \Rightarrow [F_\infty, -]$  such that

$$\Phi_n = [\kappa_{n-1}, -] \circ \Phi_\infty \quad (n \geq 1). \quad (5.45)$$

This transformation is unique. More generally, symmetry can be replaced by the assumption that each functor  $T \otimes -$  preserves this colimit; in particular, biclosedness suffices.

*Proof.* ( $\Leftarrow$ ) Given such a  $\Phi_\infty$ , contravariant bifunctionality of  $[-, X]$  together with the cocone identity  $\kappa_{n-1} = \kappa_n \circ c_{n-1}$  give, for every  $X$ ,

$$[c_{n-1}, X] \circ \Phi_{n+1,X} = [c_{n-1}, X] \circ [\kappa_n, X] \circ \Phi_{\infty,X} = [\kappa_n \circ c_{n-1}, X] \circ \Phi_{\infty,X} = [\kappa_{n-1}, X] \circ \Phi_{\infty,X} = \Phi_{n,X}, \quad (5.46)$$

so  $(\Phi_n)$  is tower-coherent.

( $\Rightarrow$ ) In the symmetric monoidal closed case  $T \otimes -$  is a left adjoint, so it preserves the given colimit. For any objects  $T, X$ , the adjunction and the colimit property give natural bijections

$$\begin{aligned} C(T, [F_\infty, X]) &\cong C(T \otimes F_\infty, X) \\ &\cong \lim_{n \geq 0} C(T \otimes F_n, X) \cong \lim_{n \geq 0} C(T, [F_n, X]). \end{aligned}$$

Thus  $[F_\infty, X]$ , with projections  $[\kappa_n, X]$ , is the limit of the inverse chain of internal homs. No separate completeness assumption is needed. Tower-coherence makes the maps  $\Phi_{n+1,X} : X \rightarrow [F_n, X]$  a cone, so there is a unique *arrow*  $\Phi_{\infty,X} : X \rightarrow [F_\infty, X]$  with the required restrictions. To check naturality, let  $g : X \rightarrow Y$ . Postcomposing  $[F_\infty, g] \circ \Phi_{\infty,X}$  and  $\Phi_{\infty,Y} \circ g$  with  $[\kappa_n, Y]$  gives, respectively,

$$[F_n, g] \circ \Phi_{n+1,X} \quad \text{and} \quad \Phi_{n+1,Y} \circ g, \quad (5.47)$$

which agree by naturality of  $\Phi_{n+1}$ . The limit projections are jointly monic, proving naturality of  $\Phi_\infty$ . They also prove uniqueness componentwise. The argument uses precisely the preservation hypothesis stated above.  $\square$

**Lemma 5.24** (Restriction of labels). For  $h : C \rightarrow D$ ,  $a : D \rightarrow I$ , and  $\Psi : \text{id} \Rightarrow [D, -]$ ,

$$[h, -] \circ \iota(a) = \iota(a \circ h), \quad p([h, -] \circ \Psi) = p(\Psi) \circ h. \quad (5.48)$$

*Proof.* Writing  $i_X : X \xrightarrow{\cong} [I, X]$  for the closed-unit isomorphism, the definition of  $\iota$  gives

$$[h, X] \circ \iota(a)_X = [h, X] \circ [a, X] \circ i_X = [a \circ h, X] \circ i_X = \iota(a \circ h)_X. \quad (5.49)$$

For the second identity,  $\Psi_I$  is the name of  $p(\Psi)$ , and  $[h, I] \circ \Psi_I$  is the name of  $p(\Psi) \circ h$ .  $\square$

**Proposition 5.25** (Compatibility of labels). Let  $C$  be symmetric monoidal closed with Bernstein centre  $\mathfrak{z}C$ , and let  $z_n : F_{n-1} \rightarrow \mathfrak{z}C$  classify  $\Phi_n$ . Tower-coherence is equivalent to

$$z_n = z_{n+1} \circ c_{n-1} \quad (n \geq 1). \quad (5.50)$$

If  $F_\infty$  exists, these equations are equivalent to a unique  $z_\infty : F_\infty \rightarrow \mathfrak{z}C$  satisfying  $z_n = z_\infty \circ \kappa_{n-1}$ . When every  $\Phi_n$  is representable, write  $\Phi_n = \iota(q_n)$  with  $q_n : F_{n-1} \rightarrow I$ . Then tower-coherence is equivalent to

$$q_n = q_{n+1} \circ c_{n-1} \quad (n \geq 1), \quad (5.51)$$

and, when the colimit exists, to a unique global label  $q_\infty : F_\infty \rightarrow I$  with  $q_n = q_\infty \circ \kappa_{n-1}$ . This representable case requires only monoidal closedness. In particular it applies to every tower when  $I$  is a generator, as in Proposition 4.19.

*Proof.* The classification of repair families is natural in their input object: uncurrying the restriction  $[h, -] \circ \Psi$  precomposes each uncurried component by  $1_X \otimes h$ , so its classifying map is  $z_\Psi \circ h$ . The classifying bijection therefore turns the tower-coherence equation into the displayed equation for  $z_n$ . The universal property of the colimit gives  $z_\infty$ . In the representable case, Lemma 5.24 and injectivity of  $\iota$  give

$$\iota(q_n) = [c_{n-1}, -] \circ \iota(q_{n+1}) \iff q_n = q_{n+1} \circ c_{n-1}. \quad (5.52)$$

The colimit property again gives the unique  $q_\infty$ , and  $\iota(q_\infty)$  restricts to all the  $\Phi_n$ . For arbitrary repair families, the equations for  $q_n = p(\Phi_n)$  remain necessary by Lemma 5.24; their sufficiency here uses representability.  $\square$

**Proposition 5.26** (Universal closure gives retractions along the tower). *Let a tower in a monoidal closed category be tower-coherent, and put  $q_n = p(\Phi_n)$ . If  $u : I \rightarrow F_{n-1}$  is a universal closure point for  $\Phi_n$ , then*

$$r_n := u \circ q_{n+1} : F_n \rightarrow F_{n-1} \quad \text{satisfies} \quad r_n \circ c_{n-1} = 1_{F_{n-1}}. \quad (5.53)$$

Moreover,  $c_{n-1} \circ u$  is a universal closure point for  $\Phi_{n+1}$  if and only if  $c_{n-1}$  is an isomorphism. It is always a metabolic closure point for the level  $(c_{n-1}, \Phi_n)$ . Under the hypotheses of Proposition 5.23, write  $q_\infty = p(\Phi_\infty)$ . Then  $u \circ q_\infty$  retracts  $\kappa_{n-1}$ , and  $\kappa_{n-1} \circ u$  is a universal closure point for  $\Phi_\infty$  if and only if  $\kappa_{n-1}$  is an isomorphism.

*Proof.* By Lemma 5.24,  $q_n = q_{n+1} \circ c_{n-1}$ , and Proposition 5.5 gives  $u \circ q_n = 1_{F_{n-1}}$ . Hence

$$r_n \circ c_{n-1} = u \circ q_{n+1} \circ c_{n-1} = u \circ q_n = 1_{F_{n-1}}. \quad (5.54)$$

The same proposition says that  $c_{n-1} \circ u$  is universal at the next level exactly when

$$c_{n-1} \circ u \circ q_{n+1} = c_{n-1} \circ r_n = 1_{F_n}. \quad (5.55)$$

Together with the first identity this is precisely invertibility of  $c_{n-1}$ ; if that map is invertible, its left inverse  $r_n$  is its inverse. For metabolic closure, naturality gives

$$\Phi_{n,F_n} \circ c_{n-1} \circ u = [F_{n-1}, c_{n-1}] \circ \Phi_{n,F_{n-1}} \circ u = \overline{c_{n-1}}, \quad (5.56)$$

the required closure equation. Finally restriction along  $\kappa_{n-1}$  gives  $q_n = q_\infty \circ \kappa_{n-1}$ . Replacing  $c_{n-1}, q_{n+1}, r_n$  by  $\kappa_{n-1}, q_\infty, u \circ q_\infty$  in the same calculation proves both colimit assertions.  $\square$

**Proposition 5.27** (Compatible universal points and stabilisation). *For a tower-coherent extended system in a monoidal closed category, the following are equivalent:*

- (1) There are points  $u_n : I \rightarrow F_n$ , universal for  $\Phi_{n+1}$ , with  $u_{n+1} = c_n \circ u_n$  for every  $n \geq 0$ .
- (2)  $\Phi_1$  has a universal closure point and every  $c_n$  is an isomorphism.

When these conditions hold, the family of points is unique. The chain has a colimit isomorphic to  $F_0$ , and the induced global family has a unique universal closure point. For fixed  $F_0, F_1$  and initial system  $(f, \Phi_1)$ , there is at most one such tower.

*Proof.* The equivalence follows by applying Proposition 5.26 successively, and uniqueness follows from Proposition 5.5. Put  $d_0 = 1_{F_0}$  and  $d_n = c_{n-1} \circ \cdots \circ c_0 : F_0 \rightarrow F_n$ . The maps  $d_n^{-1} : F_n \rightarrow F_0$  form a colimit cocone: any cocone  $h_n : F_n \rightarrow Y$  satisfies  $h_n \circ d_n = h_0$ , hence  $h_n = h_0 \circ d_n^{-1}$ , with  $h_0$  uniquely determined. Iterated tower-coherence gives

$$\Phi_1 = [d_n, -] \circ \Phi_{n+1}, \quad \Phi_{n+1} = [d_n^{-1}, -] \circ \Phi_1. \quad (5.57)$$

Thus on this colimit the global family is  $\Phi_1$  itself, with universal point  $u_0$ . This argument needs no tensor-preservation assumption, since the cocone maps are isomorphisms. Finally the object recursion fixes every  $F_n$  from  $F_0, F_1$ . Starting with  $c_0 = f$  and  $\Phi_1$ , each successive family is forced by

$$\Phi_{n+1} = [c_{n-1}^{-1}, -] \circ \Phi_n, \quad (5.58)$$

and the next connecting map is already fixed as  $c_n = \Phi_{n,F_n}$ . Induction proves uniqueness of the entire extension. This is an at-most-one assertion: an arbitrary initial system need not produce invertible connecting maps.  $\square$

**Remark 5.28** (Tower-coherence is vacuous in **Set**, but not elsewhere). In **Set**, every repair family is  $\iota(!)$ , so the label equations hold automatically. The same holds in every Cartesian closed category with terminal Bernstein centre, including localic toposes. This does not imply closure: for  $F_0 = F_1 = \{0, 1\}$  and  $f = 1_{F_0}$ , repair produces only constant functions and cannot recover  $f$ . The tower is nevertheless tower-coherent. Conversely, even universal closure at every level need not imply tower-coherence. In **Set**<sup>C<sub>2</sub></sup>, take  $F_0 = F_1 = 1$  and alternate the identity and involution families of Proposition 4.23. Every  $F_n$  is terminal, so every level has a universal point by Corollary 5.7, and all connecting maps are the unique isomorphisms. Tower-coherence would require successive families to agree under these identifications, but they do not.

**Proposition 5.29** (Six-periodic linear towers). Work in **Vect**<sub>k</sub>, take  $F_0 = F_1 = k$ , and identify each subsequent  $F_n$  with  $k$  using  $\text{Hom}_k(k, k) \cong k$ . Write  $f = t \cdot 1_k$  and let  $a_n \in k$  be the scalar label of  $\Phi_n$ , so

$$\Phi_{n,X}(x)(v) = a_n vx, \quad c_n = a_n \cdot 1_k \quad (n \geq 1). \quad (5.59)$$

The tower is tower-coherent precisely in one of the following cases:

- (1) Every  $a_n = 0$ , with  $t$  arbitrary.
- (2) There are  $s, r \in k^\times$  such that  $t = s/r$  and the labels repeat with period dividing six:

$$(a_1, a_2, a_3, a_4, a_5, a_6) = (s, r, r/s, 1/s, 1/r, s/r), \quad a_{n+6} = a_n. \quad (5.60)$$

In the second case every level has a universal closure point, these points are compatible, and the colimit is  $k$ .

*Proof.* The label criterion becomes

$$a_1 = ta_2, \quad a_n = a_{n-1}a_{n+1} \quad (n \geq 2). \quad (5.61)$$

If  $a_1 = 0$ , the equation at  $n = 2$  forces  $a_2 = 0$ , and induction forces every later label to vanish. Conversely the zero labels satisfy all equations. If  $a_1 \neq 0$ , then  $t, a_2 \neq 0$ . The equation at  $n = 2$  forces  $a_3 \neq 0$ , and induction shows that every label is nonzero. Setting  $s = a_1$ ,  $r = a_2$ , the recurrence  $a_{n+1} = a_n/a_{n-1}$  gives the displayed six values, followed by  $a_7 = s$ ,  $a_8 = r$ . Uniqueness of the recurrence proves periodicity. Substitution verifies the equations for this sequence, including  $t = s/r$ , and proves the converse. The universal point at  $F_n$  is  $u_n = 1/a_{n+1}$ , by Proposition 5.5. For  $n = 0$ ,  $t/a_1 = 1/a_2$ ; for  $n \geq 1$ , the recurrence gives  $a_n/a_{n+1} = 1/a_{n+2}$ . Thus  $c_n(u_n) = u_{n+1}$ , and Proposition 5.27 identifies the colimit.  $\square$

**Example 5.30** (Levelwise closure and tower-coherence are independent). In **Vect**<sub>Q</sub>, use the identifications of Proposition 5.29. Take  $t = 1$ ,  $a_1 = a_2 = 1$ ,  $a_3 = 2$ , and  $a_n = 1$  for  $n \geq 4$ . Every level has a universal point because every label is nonzero, but tower-coherence at  $n = 2$  would require  $1 = 1 \cdot 2$ . Conversely,  $t = 1$  and  $a_n = 0$  for all  $n$  define a tower-coherent system whose first process,  $1_k$ , cannot be recovered by zero repair. Thus levelwise universal closure does not imply tower-coherence, and tower-coherence does not imply levelwise metabolic closure.

**Remark 5.31** (Global data). The colimit theorem packages repair families on a *given* chain; it does not by itself imply finite determination or reflexivity. Finite determination follows under the additional hypotheses of Proposition 5.27. Letelier et al. consider the related recursion  $C_{n+1} = H(C_{n-1}, C_n)$  and propose a reflexive limiting object [LSAGA<sup>+</sup>06, §7, pp. 958–959]; Soto-Andrade and Varela’s construction uses embedding–retraction pairs and continuity in chain-complete posets [SAV84, §2.2, Theorem 4 and Corollary 3]. Tower-coherence alone supplies neither this structure nor an isomorphism  $F_\infty \cong [F_\infty, F_\infty]$ . Indeed, the **Set**-tower with  $F_0 = F_1 = \{0, 1\}$  and  $f = 1_{F_0}$  has injective connecting maps, given thereafter by constant-function embeddings. Its colimit has at least two points and cannot be isomorphic to its full endomorphism set, by Cantor’s theorem.

### 5.5. A comonad structure on the profunctor of coherent (M,R)-systems.

**Remark 5.32.** A *procomonad* is a comonad in profunctors: an endoprofunctor  $s : C \multimap C$ , i.e. a functor

$$s : C^{\text{op}} \times C \longrightarrow \mathbf{Set} \quad (5.62)$$

equipped with a counit  $\epsilon : s \Rightarrow H_C$  and a comultiplication  $\delta : s \Rightarrow s \circ s$ , where  $H_C = C(-, -)$  is the identity profunctor. Diagrammatically,

$$\begin{array}{ccc} C & \xrightarrow{s} & C \\ & \searrow s & \downarrow \delta \\ & & C \end{array} \quad \begin{array}{ccc} C & \xrightarrow{s} & C \\ & \searrow s & \downarrow \delta \\ & & C \end{array} \quad (5.63)$$

satisfying appropriate axioms of counitality,

$$\begin{array}{ccc} C & \xrightarrow{s} & C \\ & \searrow s & \downarrow \delta \\ & & C \end{array} \quad \begin{array}{ccc} C & \xrightarrow{s} & C \\ & \searrow s & \downarrow \delta \\ & & C \end{array} = 1_s = \begin{array}{ccc} C & \xrightarrow{s} & C \\ & \searrow s & \downarrow \delta \\ & & C \end{array} \quad \begin{array}{ccc} C & \xrightarrow{s} & C \\ & \searrow s & \downarrow \delta \\ & & C \end{array} \quad (5.64)$$

and coassociativity

$$\begin{array}{ccc} C & \xrightarrow{s} & C \\ & \searrow s & \downarrow \delta \\ & & C \end{array} \quad \begin{array}{ccc} C & \xrightarrow{s} & C \\ & \searrow s & \downarrow \delta \\ & & C \end{array} = \begin{array}{ccc} C & \xrightarrow{s} & C \\ & \searrow s & \downarrow \delta \\ & & C \end{array} \quad \begin{array}{ccc} C & \xrightarrow{s} & C \\ & \searrow s & \downarrow \delta \\ & & C \end{array} \quad (5.65)$$

These diagrams have to be interpreted in the (bi)category of profunctors, where composition  $s \circ s$  is defined as the coend

$$(s \circ s)(X, Y) := \int^C s(X, C) \times s(C, Y) \quad (5.66)$$

(see Definition 2.13).

We now aim to prove the following result.

**Theorem 5.33** (The procomonad of (M,R)-systems). Let  $C$  be a Cartesian closed category. There is a procomonad structure on the profunctor of Definition 4.9.

*Proof.* The counit  $\epsilon : \mathbf{MRS} \Rightarrow H_C$  has components

$$\mathbf{MRS}(A, B) \longrightarrow C(A, B) \quad (5.67)$$

sending an (M,R)-system  $\langle f, \Phi \rangle$  to  $f$ . Clearly this is just the fibrewise action of the functor  $[-]_f$  defined on page 30, which relates the tabulator  $\odot \mathbf{MRS}$  of the  $\mathbf{MRS}$  profunctor to the tabulator  $C^{[0 \rightarrow 1]} = \odot C(-, -)$  of the hom functor.

We also have to define the comultiplication  $\delta : \mathbf{MRS} \Rightarrow \mathbf{MRS} \circ \mathbf{MRS}$ , which intuitively expresses the idea that an (M,R)-system  $\langle f, \Phi \rangle$  based at  $(A, B)$  can be decomposed into a ‘left’ and a ‘right’ part, along a certain intermediate object  $X$ :

$$\delta_{AB} : \mathbf{MRS}(A, B) \longrightarrow \int^X \mathbf{MRS}(A, X) \times \mathbf{MRS}(X, B), \quad (5.68)$$

where the coend accounts for the choice of said intermediate object. Concretely,  $\delta_{AB}$  makes use of the ‘identity’ (M,R)-system of Definition 3.47; there are essentially two ways to decompose  $\langle f, \Phi \rangle$  that get identified in the quotient that defines the coend, a property which guarantees that the assignment is well defined.

This is not immediately clear, since one has to determine a family of maps *into* a coend –thus into a colimit, thus in the opposite direction of usual universal property arguments.



The idea is to use the fact that for any two given objects  $A, B$  the square

$$\begin{array}{ccc}
 \mathbf{MRS}(A, B) & \xrightarrow{\delta' = \mathcal{M}_A^0 \times 1} & \mathbf{MRS}(A, A) \times \mathbf{MRS}(A, B) \\
 \delta'' = 1 \times \mathcal{M}_B^0 \downarrow & & \downarrow j_A \\
 \mathbf{MRS}(A, B) \times \mathbf{MRS}(B, B) & \xrightarrow{j_B} \int^X & \mathbf{MRS}(A, X) \times \mathbf{MRS}(X, B)
 \end{array} \quad (5.69)$$

commutes (if  $\mathcal{M}^0$  is the ‘identity’ (M,R)-system defined as in Definition 3.47,  $j_A$  is the composite

$$\mathbf{MRS}(A, A) \times \mathbf{MRS}(A, B) \hookrightarrow \sum_X \mathbf{MRS}(A, X) \times \mathbf{MRS}(X, B) \twoheadrightarrow \int^X \mathbf{MRS}(A, X) \times \mathbf{MRS}(X, B),$$

and similarly for  $j_B$ ); thus there is a well-defined diagonal yielding the component at  $A, B$  of  $\delta$ ; it is then a routine check to establish naturality and the comonad equations.

Then we prove the comonad axioms:

- left and right counitality are the easiest to prove, as one can choose in each case the presentation for  $\delta$  that trivialises the counitality equation;
- coassociativity of  $\delta$  also relies on the freedom to pick the easiest representative for the equivalence class of  $\delta_{AB}\langle f, \Phi \rangle$ ; all in all, the axiom that  $\delta$  is coassociative boils down to proving that

$$\begin{array}{ccc}
 \langle f, \Phi \rangle & \xrightarrow{\quad} & (\mathcal{M}^0, \langle f, \Phi \rangle) \\
 \downarrow & & \downarrow \\
 & & ((\mathcal{M}^0, \langle f, \Phi \rangle), \mathcal{M}^0) \\
 \downarrow & & \downarrow \\
 (\langle f, \Phi \rangle, \mathcal{M}^0) & \xrightarrow{\quad} & (\mathcal{M}^0, (\langle f, \Phi \rangle, \mathcal{M}^0))
 \end{array} \quad (5.70)$$

become equivalent in the coend to the same class  $\mathcal{M}^0 \otimes \langle f, \Phi \rangle \otimes \mathcal{M}^0$ . This is a straightforward check and it concludes the proof of the comonad axioms.  $\square$

## 6. (M,R)-SYSTEMS AND MEALY AUTOMATA

A central aspect of Rosen’s intuition was to relate (M,R)-systems to ‘sequential machines’ (or ‘state machines’, ‘automata’, etc.) [Ros64a, Ros64b, Ros66a]; Arbib and others developed this connection further [Arb66, AM75, PA07]. We express this as a functor from the twisted-elements category  $\mathcal{E}l(\mathbf{iMRS})$  to a category  $\tau\mathbf{Mly}$  of Mealy automata bearing correspondingly twisted morphisms as arrows.

The choice of morphisms matters when input and output vary together. A functor over the base must fit into

$$\begin{array}{ccc}
 \mathcal{E}l(\mathbf{iMRS}) & \xrightarrow{\quad} & \tau\mathbf{Mly} \\
 & \searrow \quad \swarrow & \\
 & C^{\text{op}} \times C &
 \end{array} \quad (6.1)$$

and therefore induce fibrewise functors

$$\mathcal{A}_{(A,B)} : \mathbf{iMRS}(A, B) \longrightarrow \tau\mathbf{Mly}(A, B); \quad (6.2)$$

Since each source fibre is discrete, a fibrewise functor is simply a function

$$(\mathcal{A}_{(A,B)})_0 : \mathbf{iMRS}(A, B) \longrightarrow \tau\mathbf{Mly}(A, B)_0; \quad (6.3)$$

The compatibility with ‘interfibre’ arrows in the domain is the substantive part of the construction we propose.

We first recall the categorical definition of Mealy automata, closely following [Lor25a].

**Definition 6.1** ( $\mathcal{U}$ ). Let  $C$  be a category,  $F : C \rightarrow C$  an endofunctor, and  $B$  an object.

Define the category  $\mathbf{Mly}(F, B)$  of *Mealy automata with input  $F$  and output  $B$*  as the following strict 2-pullback in  $\mathbf{Cat}$ :

$$\begin{array}{ccc} \mathbf{Mly}(F, B) & \longrightarrow & F/B \\ \downarrow & \lrcorner & \downarrow U \\ \mathbf{Alg}(F) & \xrightarrow{V} & C \end{array} \quad (6.4)$$

where  $\mathbf{Alg}(F)$  is the category of *endofunctor algebras* of  $F$ ,  $F/B$  the comma category of arrows  $(X, FX \rightarrow B)$  (and  $U, V$  are the most obvious forgetful functors).

We observe in passing that using the techniques of [Lor25a] one can show that  $\mathbf{Mly}(F, B)$  inherits several good properties from  $C$ ; in particular, if  $C$  is complete, or cocomplete, or accessible, or locally presentable [AR94, GU71], then so is  $\mathbf{Mly}(F, B)$ .

**Remark 6.2.** When  $C$  is monoidal and  $F = \_ \otimes A$ , we write  $\mathbf{Mly}(A, B)$  for  $\mathbf{Mly}(F, B)$ . An object  $\left\| \frac{E}{d,s} \right\|$  is a span

$$\left\| \frac{E}{d,s} \right\| : E \xleftarrow{d} E \otimes A \xrightarrow{s} B. \quad (6.5)$$

Here  $E$  is the state object,  $d$  the transition, and  $s$  the output map. We place the input on the right, consistently with the adjunction  $\_ \otimes A \dashv [A, \_]$ . These machines and their categorical treatments have a long history; see [Mea55, Moo56, Net59, Gog72, Gog73, AM75].

**Remark 6.3.** Let  $F : C \rightarrow C$ ,  $G : \mathcal{H} \rightarrow \mathcal{H}$ , and  $P : C \rightarrow \mathcal{H}$ , with a natural transformation  $\pi : GP \Rightarrow PF$ . Applying  $P$  and precomposing with  $\pi$  sends  $F$ -algebras to  $G$ -algebras and maps  $FX \rightarrow B$  to  $GPX \rightarrow PB$ . The pullback property induces a functor

$$\pi^* : \mathbf{Mly}_C(F, B) \longrightarrow \mathbf{Mly}_{\mathcal{H}}(G, PB). \quad (6.6)$$

Thus the construction in Definition 6.1 is implicitly functorial in the pair  $F, B$ , an observation which allows one to study the properties of the pseudofunctor  $(F, B) \mapsto \mathbf{Mly}(F, B)$ .

**Definition 6.4.** The *total Mealy 2-category*  $\mathbf{Mly}$  is defined as follows:

- the objects are triples  $(C; F, B)$  where  $F : C \rightarrow C$  is an endofunctor of a category  $C$ , and  $B$  an object of  $C$ ;
- the morphisms  $(P, \pi, u) : (C; F, B) \rightarrow (\mathcal{H}; G, B')$  are triples where  $P : C \rightarrow \mathcal{H}$  is a functor,  $\pi : GP \Rightarrow PF$  is an *intertwiner* natural transformation between  $F$  and  $G$  and  $u : PB \rightarrow B'$  is a morphism;
- 2-cells  $\gamma : (P, \pi, u) \Rightarrow (Q, \theta, v)$  consist of natural transformations  $\gamma : P \Rightarrow Q$  compatible with the intertwiners  $\pi, \theta$  in the obvious sense, and such that  $v \circ \gamma_B = u$ .

With such a domain, the assignment sending  $(C, F, B)$  to  $\mathbf{Mly}(F, B)$  is a strict 2-functor  $\mathbf{Mly} \rightarrow \mathbf{Cat}$  ( $\mathbf{Cat}$  is the 2-category of categories, strict functors, strict natural transformations).

For a fixed ambient category  $C$ , this simplifies as follows.

**Definition 6.5** (The total categories of automata). Definition 6.1 entails at once that the correspondence  $(F, B) \mapsto \mathbf{Mly}(F, B)$  is a (pseudo)functor of type  $\mathbf{Mly} : \mathbf{Cat}(C, C)^{\text{op}} \times C \rightarrow \mathbf{Cat}$ , i.e. a pseudo-profunctor  $\mathbf{Cat}(C, C) \rightrightarrows C$  from which we can extract a two-sided fibration, i.e. a span

$$\mathbf{Cat}(C, C) \xleftarrow{p} \mathbf{Mly} \xrightarrow{q} C \quad (6.7)$$

such that  $p$  is a fibration,  $q$  is an opfibration,  $p$ -Cartesian lifts are  $q$ -vertical and  $q$ -opCartesian lifts are  $p$ -vertical. We call the tip  $\mathbf{Mly}$  of the span the *total Mealy category* constructed from  $C$ .

Similar considerations allow us to construct the total Moore category  $\mathbf{Mre}$  from the pseudo-profunctor  $(F, B) \mapsto \mathbf{Mre}(F, B)$ , and to obtain a two-sided fibration  $\mathbf{Cat}(C, C) \leftarrow \mathbf{Mre} \rightarrow C$ .

**Remark 6.6.** Unwinding the definition, it is easy to establish how reindexings of the total Mealy and Moore fibration act. In the particular case where  $\alpha : F \Rightarrow G$  is a natural transformation between left adjoints  $F \dashv R$  and  $G \dashv Q$  and  $f : B \rightarrow B'$  a morphism, the reindexing functor

$\mathbf{Mly}(\alpha, f) : \mathbf{Mly}(G, B) \rightarrow \mathbf{Mly}(F, B')$  preserves all colimits—and thus, assuming  $C$  is a locally presentable category, is a left adjoint; however, it fails to preserve limits (it already fails to preserve terminal objects).

The ‘right regular representation’  $\mathcal{T} : C \rightarrow \mathbf{Cat}(C, C)$ ,  $A \mapsto \_ \otimes A$  lets us pull back the total Mealy fibration:

$$\begin{array}{ccc} \mathbf{Mly}^\otimes & \xrightarrow{\quad} & \mathbf{Mly} \\ \downarrow & \lrcorner & \downarrow \\ C^{\text{op}} \times C & \xrightarrow[\mathcal{T}^{\text{op}} \times C]{} & \mathbf{Cat}(C, C)^{\text{op}} \times C \end{array} \quad (6.8)$$

which gives rise to the *monoidal Mealy* (two-sided) *fibration*

$$C \xleftarrow{p^\otimes} \mathbf{Mly}^\otimes \xrightarrow{q^\otimes} C \quad (6.9)$$

**Remark 6.7.** The (pseudo-)profunctor  $\mathbf{Mly}^\otimes : C^{\text{op}} \times C \rightarrow \mathbf{Cat}$  takes values in categories. Its associated two-sided fibration is (6.9), with objects as in (6.5) and morphisms  $(\ell : A \rightarrow A', r : B \rightarrow B', u : E \rightarrow E')$  such that the diagram

$$\begin{array}{ccccc} E & \xleftarrow{d} & E \otimes A & \xrightarrow{s} & B \\ u \downarrow & & u \otimes \ell \downarrow & & \downarrow r \\ E' & \xleftarrow{d'} & E' \otimes A' & \xrightarrow{s'} & B' \end{array} \quad (6.10)$$

commutes, i.e.  $s' \circ (u \otimes \ell) = r \circ s$  and  $d' \circ (u \otimes \ell) = u \circ d$ .

To the same profunctor we can also associate the category of elements, where the direction of arrows is appropriately ‘twisted’: it is this latter total category that provides the correct codomain for promoting Arbib’s construction to a functor.

**Definition 6.8** (The twisted category of Mealy automata,  $\mathcal{M}^{\ell}$ ). The *twisted category of Mealy automata*, denoted  $\tau\mathbf{Mly}$ , is defined as follows:

- the objects are the Mealy automata of (6.5), i.e. spans  $\left\| \frac{X}{d,s} \right\|_{AB} : X \leftarrow X \otimes A \rightarrow B$ ;
- the morphisms  $\left\| \frac{X}{d,s} \right\|_{AB} \rightarrow \left\| \frac{Y}{d',s'} \right\|_{A'B'}$  consist of triples  $\ell : A' \rightarrow A, r : B \rightarrow B', u : X \rightarrow Y$  such that

$$u \circ d \circ (1_X \otimes \ell) = d' \circ (u \otimes 1_{A'}) \quad r \circ s \circ (1_X \otimes \ell) = s' \circ (u \otimes 1_{A'}) \quad (6.11)$$

Note the difference with (6.10), where the direction of the arrows  $\ell$  is reversed; hence the ‘twisting’ in the name.

**Proposition 6.9** ([Arb66] reloaded,  $\mathcal{M}^{\ell}$ ). There exists a family of functions

$$(\mathcal{A}_{(A,B)})_0 : \mathbf{iMRS}(A, B) \longrightarrow \tau\mathbf{Mly}(A, B)_0 \quad (6.12)$$

which, in fact, is the fibrewise decomposition of a functor  $\mathcal{A}$  such that  $\langle p^\otimes, q^\otimes \rangle \circ \mathcal{A} = [-]_{(A,B)}$ :

$$\begin{array}{ccc} \mathcal{E}(\mathbf{iMRS}) & \xrightarrow{\mathcal{A}} & \tau\mathbf{Mly} \\ & \searrow [-]_{(A,B)} \quad \swarrow \langle p^\otimes, q^\otimes \rangle & \\ & C^{\text{op}} \times C & \end{array} \quad (6.13)$$

sending an incoherent (M,R)-system  $A \xrightarrow{f} B \xrightarrow{\Phi} [A, B]$  to the Mealy automaton

$$[A, B] \xleftarrow{\lambda(u,a).\Phi(ua)} [A, B] \otimes A \xrightarrow{\lambda(u,a).ua} B \quad (6.14)$$

where we have adopted the lambda notation for the arrows

$$\epsilon_B : [A, B] \otimes A \longrightarrow B \quad \Phi \circ \epsilon_B : [A, B] \otimes A \longrightarrow B \longrightarrow [A, B] \quad (6.15)$$

The process  $f$  supplies the distinguished initial state  $\bar{f} : I \rightarrow [A, B]$ , as in Arbib’s pointed formulation [Arb66].

Arbib then goes on to prove two fundamental results which, however, can only be stated in the Cartesian case. In the next two statements, then,  $C$  is a category with finite products, and we consider *Cartesian* Mealy automata.

**Theorem 6.10.** A Mealy automaton  $\langle d, s \rangle : X \leftarrow X \times A \rightarrow B$  lies in the essential image of  $\mathcal{A}$  in Proposition 6.9 if and only if it is of the form

$$X \xleftarrow{w \circ s} X \times A \xrightarrow{s} B \quad (6.16)$$

for some  $w : B \rightarrow X$ . These Mealy automata are called *simple*.

A convenient diagrammatic representation for a simple Mealy automaton ‘closes’ the span  $d, s$  to a commutative triangle, via  $w : B \rightarrow X$ :

$$\begin{array}{ccc} & X \times A & \\ d \swarrow & & \searrow s \\ X & \xleftarrow{w} & B \end{array} \quad (6.17)$$

More interestingly, Arbib proves that simple Mealy automata satisfy some weak form of lifting property:

**Theorem 6.11.** If  $\langle d, s \rangle : X \leftarrow X \times A \rightarrow B$  is a simple Mealy automaton representing an  $(M, R)$ -system, and

$$\begin{array}{ccccc} X & \xleftarrow{\quad} & X \times A & \xrightarrow{\quad} & B \\ f \downarrow & & f \times g \downarrow & & \parallel \\ Y & \xleftarrow{d'} & Y \times C & \xrightarrow{s'} & B \end{array} \quad (6.18)$$

is a morphism in the total fibration of Mealy automata with  $f, g$  epimorphisms and the identity on  $B$ , then  $\langle d', s' \rangle$  is simple.

Technically this looks like a ‘fibration where Cartesian lifts exist only for epimorphisms’, but the morphisms in the category **MRS** are never defined in [Arb66] except indirectly, and indeed only the correspondence on objects is defined.

**Remark 6.12.** These two results are, however, of little practical interest for us, since

- the proof of Theorem 6.10 in [Arb66] makes use of the fact that the representation  $d' : A^* \rightarrow [X, X]$  is injective (in representation-theoretic terms,  $d'$  is *faithful*); this does not seem to be a blanket assumption in modern treatments of automata theory;
- the notion of simple  $(M, R)$ -system only applies to Cartesian, incoherent  $(M, R)$ -systems.

**Proposition 6.13.** For the associated simple Mealy automaton  $\mathcal{A}\langle f, \Phi \rangle$ , metabolic closure means that the initial state  $f$  has *nontrivial isotropy* in the monoid representation of  $A^*$  on the states  $[A, B]$  [Ste16]. A universal closure point  $a_0$  (Definition 5.3) belongs to the stabiliser of  $f$ . For the moment we refrain from phrasing the results of ?? in representation-theoretic terms.

## 7. FUTURE WORK

An *autograph*, in the sense of Guitart [Gui14a, Gui14b, Gui17], is a set  $X$  equipped with two endomaps  $s, t : X \rightarrow X$ ; equivalently, an action of the free monoid on two generators, so that the category **aGph** of autographs is a presheaf topos. Its defining feature is that vertices and edges are not distinguished: every edge  $x$  has a source  $sx$  and a target  $tx$ , themselves edges, whose own sources and targets are edges again. Autographs are therefore irremediably non-well-founded, the fixpoints of the action, i.e. the joint equaliser of  $s, t$  and the identity, being loops  $x : x \rightarrow x$ ; it is exactly this feature that makes them a candidate home for the self-referential diagrams of relational biology; virtually every picture of graph-theoretic dynamics in [Lou17, Ch. 7–8] is an autograph. Guitart pushes the analogy with graphs to a square of adjunctions relating **Gph**  $\rightleftarrows$  **Cat** to **aGph**  $\rightleftarrows$  **aCat**, the category **aCat** of *autocategories* being monadic over **aGph** as **Cat** is over **Gph**.

The link with the present paper is that the *shape* of an  $(M, R)$ -system is an autograph. The 2-step shape  $Q(MR_2)$  has  $A$  and  $B$  as fixpoints and two further edges  $f : A \rightarrow B$  and  $\Phi_1 : B \rightarrow f$ ; the  $n$ -step shape of Construction 5.16 adds  $\Phi_{k+1} : \Phi_{k-1} \rightarrow \Phi_k$  for  $k \geq 1$ , with  $\Phi_0 := f$ , and its two incidence matrices  $s_n, t_n$  are obtained from  $s_{n-1}, t_{n-1}$  by an evident inductive rule. Every finite  $(M, R)$ -system shape thus becomes a representation of the free monoid on two letters, and the natural question is what its representation theory looks like. That question cannot be answered by importing the theory of quiver representations [ASS06, Sch14, Kir16] wholesale, since an autograph has no vertices to index idempotents by, and a representation theory of autographs has instead to be built from scratch, of which [Gui17] is the only precedent we know.

We intend to develop this programme elsewhere, seeking interpretations in Rosen’s theory for the following representation-theoretic statements about the autographs  $MR_n$ .

**Theorem.** The (class of morphisms of the) free autcategory on  $Q(MR_2)$  is presented by the regular language  $[AB]^*(f[AB]^*)^*\Phi^*$  over the alphabet  $X(MR_2) := \{A, B, f, \Phi\}$ , with composition the concatenation of words, a *partial* operation: precomposition with a word ending in  $\Phi$  is defined on identities only.

**Theorem.** For each  $n$ , the free autcategory on  $Q(MR_n)$  is presented by a regular language in the same way; the presentation nevertheless fails at infinity, the language  $\{\Phi_1\Phi_2\cdots\Phi_k \mid k \geq 0\}$  attached to  $MRS_\infty$  of Definition 5.17 being non-regular by a Myhill–Nerode argument.

**Theorem.** The path algebra  $\mathbb{C}\langle MR_2 \rangle$  (shorthand for  $\mathbb{C}\langle Q(MR_2) \rangle$ ) is an iterated triangular extension of  $\mathbb{C}[a]$  and  $\mathbb{C}[b]$  by free bimodules: it is isomorphic to the algebra of lower triangular matrices with diagonal  $\mathbb{C}[a], \mathbb{C}[b], \mathbb{C}$ , off-diagonal entries the bimodules  $M = \langle a^n f b^m \rangle$  and  $P = \langle b^m p \rangle \cong \mathbb{C}[b]$ , and corner  $N \cong M \otimes_{\mathbb{C}[b]} P$ . In particular the square built out of  $a, b, f$  is not required to commute.

**Theorem.** The Jacobson radical of  $\mathbb{C}\langle MR_2 \rangle$  is the two-sided ideal  $\langle f, p \rangle$  generated by the arrows lying on no oriented cycle, and the quotient  $\mathbb{C}\langle MR_2 \rangle / \text{rad} \cong \mathbb{C}[a] \oplus \mathbb{C}[b] \oplus \mathbb{C}$  is semiprimitive, though of course not semisimple.

**Theorem.** Consequently  $\text{Rep}(Q(MR_2))$  is an iterated *recollement* [BBD82, CPS89, AHKL11, PV18]: it is obtained by gluing  $\text{Mod}(\mathbb{C}[a])$  and  $\text{Mod}(\mathbb{C}[b])$  along  $M$ , and then gluing the result with  $\text{Mod}(\mathbb{C})$  along  $(N, P)$ . The simple, projective and injective representations are computed explicitly in each of the four positions  $A, B, f, \Phi$ .

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